



DEPARTMENT OF PHYSICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis in Nuclear-, Particle- and Astrophysics

**Machine learning-based optimisation of Higgs
coupling measurements in the $H \rightarrow 4l$ decay
channel with ATLAS Run 3 data**

Luca Matthias Spitzauer





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**Optimierung der Higgs-Kopplungs-Messungen
im Vier-Lepton-Kanal mit ATLAS Run3 Daten**

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I confirm that this master's thesis in nuclear-, particle- and astrophysics is my own work and I have documented all sources and material used.

A handwritten signature in dark ink, appearing to read 'Spitzauer', written in a cursive style.

Munich, 03.12.2024

Luca Matthias Spitzauer

Abstract

Cross-section measurements for the different Higgs boson production and decay processes constitute a key area in the exploration of Higgs properties, with a high sensitivity to potential physics beyond the Standard Model. Due to its exceptionally clear signal, the decay of a Higgs boson into a ZZ^* pair with a subsequent decay of each Z boson into two signature light leptons, $H \rightarrow ZZ^* \rightarrow 4l$, is one of the most important channels for the Higgs property measurements. To increase the sensitivity of the Higgs cross-section measurements, the Simplified Template Cross Section (STXS) approach has been developed, where the final state is divided into exclusive phase space regions. Optimized classification of events according to kinematic production regions is vital to improve the signal sensitivity. The previous round of STXS measurements in the $H \rightarrow 4l$ channel with the Run 2 ATLAS dataset employed an event classification using a Recurrent Neural Network (RNN) based approach. With the new Run 3 dataset at a centre-of-mass energy of 13.6 TeV, an alternative Neural Network approach based on permutation invariant Deep Sets is explored for this classification, which is motivated by permutation invariant symmetries between the $H \rightarrow ZZ^* \rightarrow 4l$ final state particles. A direct comparison between the predictive behaviour and the signal-/background separation is made between the Deep Set and RNN models in all STXS bins. Similar comparisons are made between Deep Sets trained on Run 2 and Run 3 data. The STXS classification schemes used in the RNN training during Run 2 and Run 3 are both considered for the Deep Set Neural Networks. Improved or comparable signal-to-background separations were observed for the Deep Set models compared to the RNN models in all kinematical STXS classification regions. Training on Run 3 data lead to improvements in the Deep Set performance compared to Run 2 data. The Run 2 iteration of the STXS binning scheme is in general observed to be more favourable for the Deep Set performance. The Deep Set approach can be considered as a viable alternative to the previous RNN approach.

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1. Introduction

Particle physics aims to reach a full description of the behavior of the smallest building blocks our universe consists of. An incredibly accurate description of particles and their interactions has been achieved with the *Standard Model* (SM) of particle physics, which comprises three of the four fundamental forces (electromagnetic, weak and strong interaction, not including gravity), as well as the 17 known, fundamental particles [1] [2] [3]. However, the SM does not offer solutions to several observed phenomena, such as the matter-anti-matter asymmetry in the universe [4], why the strong interaction does not violate the CP -symmetry (strong CP problem) [5] [6] or the composition of dark matter [7] [8]. Due to this, it is vital to perform highly precise tests of the SM and experimentally probe extensions of it.

One of the most important approaches to study particles in the highest energy regime are particle colliders. The Large Hadron Collider is currently the world's largest and most energetic particle accelerator with a circumference of ≈ 30 km and proton-proton collisions at a center-of-mass energy of currently $\sqrt{s} = 13.6$ TeV. In July 2012, the ATLAS and CMS experiments at the LHC both announced the discovery of the final confirmed addition to the SM, the Higgs boson, which is predicted by the Higgs mechanism in the SM providing masses to the fundamental fermions and electro-weak gauge bosons [9] [10].

Measurements of the couplings of the Higgs boson to SM particles are an important window to phenomena beyond the Standard Model. One of experimentally most favorable Higgs decay modes is $H \rightarrow ZZ^* \rightarrow 4l$ [9]. At the LHC, $H \rightarrow ZZ^* \rightarrow 4l$ cross-section measurements have been performed employing a separation of the phase space into several exclusive kinematic regions, called *Simplified Template Cross Sections* (STXS). The STXS framework was developed during Run 1 of the LHC[11]. It is vital for Higgs property measurements to distinguish the production processes of the Higgs events and discriminate them against background events by accurately classifying Higgs events in the STXS kinematic regions. To achieve this, Recurrent Neural Networks are used for the ATLAS $H \rightarrow ZZ^* \rightarrow 4l$ classification.

With the LHC in its third data-taking run, generally called *Run 3*, since July 2022 with increased center-of-mass energy of $\sqrt{s} = 13.6$ TeV, compared to the $\sqrt{s} = 13$ TeV of Run 2, new precision measurements of the Higgs properties will be conducted including new Run 3 $H \rightarrow ZZ^* \rightarrow 4l$ STXS measurements. In this context, improvements to the Neural Network classification of the Higgs production modes are currently being investigated. In

this study, an alternative to the Recurrent Neural Network approach is investigated for the Neural Network classification based on the *Deep Sets* architecture [12] which is motivated by symmetries taking into account permutations of the leptons and jets in the $H \rightarrow 4l$ final state.

The thesis is structured in the following way: An introduction to the theoretical aspects of this study is given in Chapter 2, which includes an introduction to the Standard Model, a description of the Higgs mechanism and an overview of the properties of the Higgs boson. The experimental setup of the LHC and the ATLAS detector is presented in Chapter 3. A detailed description of ATLAS cross-section measurements in the $H \rightarrow ZZ^* \rightarrow 4l$ decay channel is given in Chapter 4, which discusses the Monte Carlo Event samples used in this study, the kinematic cuts applied during the $H \rightarrow 4l$ event selection, the background contributions to the $H \rightarrow ZZ^* \rightarrow 4l$ decay channel and details of the fitting of the total and fiducial $H \rightarrow 4l$ cross-sections. An introduction to the STXS framework follows in Chapter 5 including a description of the STXS classification. In Chapter 6, a theoretical introduction to Neural Networks in general is given and a detailed discussion of the Recurrent Neural Network and Deep Sets architectures with motivation for using the Deep Sets approach. An overview of Neural Network training setup and observations made during the optimization of the Deep Set Neural Networks is given in Chapter 7. The observed predictive behaviour of the Deep Set Neural Networks is discussed and compared to the one of the Recurrent Neural Networks in Chapter 8. The performance of Deep Set Neural Networks trained with Run 2 and with Run 3 data samples are compared, as well as of the STXS classification scheme currently used during the training of Recurrent Neural Network models in Run 3 and the classification scheme previously used for Run 2 data. The importance of the selection of the input variables considered during the Deep Set training are also discussed. The thesis concludes with a summary and interpretation of the results of the Deep Set studies and a short outlook.

2. Theoretical Background

2.1. The Standard Model of Particle Physics

The *Standard Model* of particle physics [1] [2] [3] describes three of the four known fundamental forces (the strong, weak and electromagnetic interactions, leaving out gravity), in a quantum field theory and classifies all known elementary particles. A core concept of the Standard Model are invariances of the Lagrangian under local gauge transformations, referred to as *gauge symmetries*. Local transformations are defined as transformations that are parametrized by spacetime coordinates, which are realized in gauge theories via symmetry group operations. The symmetry groups that describe the gauge theory of the Standard model is

$$SU(3)_C \times SU(2)_L \times U(1)_Y. \quad (2.1)$$

The strong interaction acts on all particles carrying a color charge as described by the $SU(3)_C$ group. The united electromagnetic and weak interactions act on particles carrying weak isospin J or electrical charge Q , which are related via the weak hypercharge $Y = Q + \frac{J}{2}$, and is defined by $SU(2)_L \times U(1)_Y$. The $SU(2)_L$ interaction acts only on left-handed fermions implementing maximum parity violation of the charged weak interaction in the Standard Model.

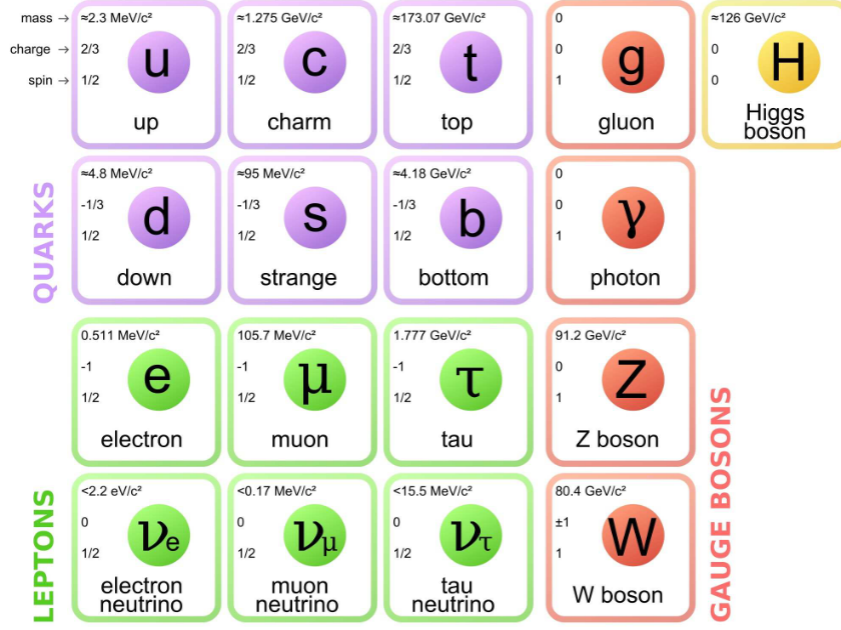


Figure 2.1.: Overview of the elementary particles of the Standard Model of particle physics. [13]

The elementary particles can be divided into two types: *bosons* and *fermions*.

Fermions are particles with half-integer spin. They can be divided into two categories, the *quarks* and the *leptons*. Three generations with increasing masses of quarks and leptons are known, where each generation consists of a pair of leptons and of quarks. The leptons pairs consist of a weak isospin-down charged lepton with electric charge $-1 e$ and a neutral weak isospin-down *neutrino*. The quark pair consist of a weak isospin-up quark with charge $\frac{2}{3} e$ and a spin-down quark with charge $-\frac{1}{3} e$. Quarks also carry colour quantum numbers r, g, b and interact with the strong interaction. Each fermion possesses a corresponding anti-particle with identical mass but electric charge of opposite sign. As a consequence of the maximum parity violation observed in the charged weak interaction, only left-handed particle states are placed in the weak isospin doublets, right-handed particles are in weak isospin singlet states.

Bosons are particles with integer spin values. Bosons with spin 1 are called *gauge bosons* and serve as mediator of the three interactions described by the Standard Model. The massless and electrically neutral photon γ is the mediator of the electromagnetic interaction which couples to particles carrying electric charge. The eight massless gluons g are the mediators of the strong interaction, which couple to particles carrying color charges. Gluons carry color charge themselves, which allows gluon-gluon self-interactions. The charged W^+ , W^- bosons

and the neutral Z bosons are the mediators of the weak interaction. While the photons and gluons are massless, the W^+ , W^- and Z bosons are massive. The W Bosons only interact with left-handed particles and right-handed anti-particles, while the Z boson interacts with left- and right-handed particles. Bosons with spin 0 and CP quantum number $+1$ are called *scalar bosons*. The only known fundamental scalar boson is the *Higgs* boson H , which is a consequence of the Higgs mechanism for the generation of W^{pm} , Z boson and fermion masses in the SM, which otherwise would be massless.

The interactions in the Standard Model can be described by vector gauge fields corresponding to the local gauge symmetries of the interactions, with B_μ the field of the $U(1)$ group, the three fields $W_\mu^{1,2,3}$ the fields of the weak isospin $SU(2)$ group and $G_\mu^{1,\dots,8}$ the fields of the color $SU(3)$ group of the strong interaction. With the field-strengths

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu \quad (2.2)$$

$$W_{\mu\nu}^i = \partial_\mu W_\nu^i - \partial_\nu W_\mu^i - g_2 \epsilon^{ijk} W_\mu^j W_\nu^k \quad (2.3)$$

$$G_{\mu\nu}^i = \partial_\mu G_\nu^i - \partial_\nu G_\mu^i - g_3 f^{ijk} G_\mu^j G_\nu^k \quad (2.4)$$

with the structure constants ϵ^{ijk} and f^{ijk} of $SU(2)$ and $SU(3)$, and the gauge coupling constants g_1 , g_2 and g_3 of the weak hypercharge, the weak isospin and the strong interaction, kinematic terms of the gauge boson fields to the Lagrangian density of the Standard Model are:

$$L_{kin,Gauge} = -\frac{1}{4} B^{\mu\nu} B_{\mu\nu} - \frac{1}{4} W_i^{\mu\nu} W_{\mu\nu}^i - \frac{1}{4} G_a^{\mu\nu} G_{\mu\nu}^a. \quad (2.5)$$

By introducing the gauge covariant derivative of spinor fields ψ describing the fermions,

$$D_\mu^L \psi_L = (\partial_\mu + ig_1 Y_L B_\mu + ig_2 \tau_I W_\mu^I + ig_3 \lambda_a G_\mu^a) \psi_L \quad (2.6)$$

$$D_\mu^R \psi_R = (\partial_\mu + ig_1 Y_R B_\mu + ig_3 \lambda_a G_\mu^a) \psi_R \quad (2.7)$$

with the hypercharge Y , the Pauli matrices τ_I and the Gell-Mann matrices λ_a as generators of the $U(1)$, $SU(2)$ and $SU(3)$ symmetry groups, kinematic and interaction terms of the fermions in the Lagrangian density are given by:

$$L_{kin,int.,Spinor} = \bar{\psi}_L i D_\mu^L \gamma^\mu \psi_L + \bar{\psi}_R i D_\mu^R \gamma^\mu \psi_R \quad (2.8)$$

with the left-handed spinor doublets ψ_L and the right-handed spinor-singlets ψ_R , and the Dirac matrices γ^μ .

2.2. The Higgs Boson

The Higgs Boson H is a scalar boson that is neutral with regards to electric charge and colour charge [14] [15]. It transforms evenly under charge (C), parity (P) and charge-parity (CP) conjugations.

The Higgs Boson was discovered in 2012 at the *Large Hadron Collider* (LHC) by the ATLAS and CMS experiments[9] [10]. Both experiments used data at center of mass energies of $\sqrt{s} = 7$ GeV and $\sqrt{s} = 8$ GeV with integrated luminosity of 4.8 fb^{-1} and 5.8 fb^{-1} respectively in the case of ATLAS and and of 5.1 fb^{-1} and 5.3 fb^{-1} respectively in the case of CMS. Clear signatures of a new particle resonance at a mass around ≈ 125 GeV were observed above the expected background in the Higgs to four lepton ($H \rightarrow 4l$) and Higgs to di-photon ($H \rightarrow \gamma\gamma$) decay channel as shown in Figure 2.2 and Figure 2.3. The current combined result of the CMS and ATLAS experiments for the mass of the Higgs Boson is $125.11 \pm 0.09 \text{ (stat)} \pm 0.06 \text{ (sys)}$ GeV [16].

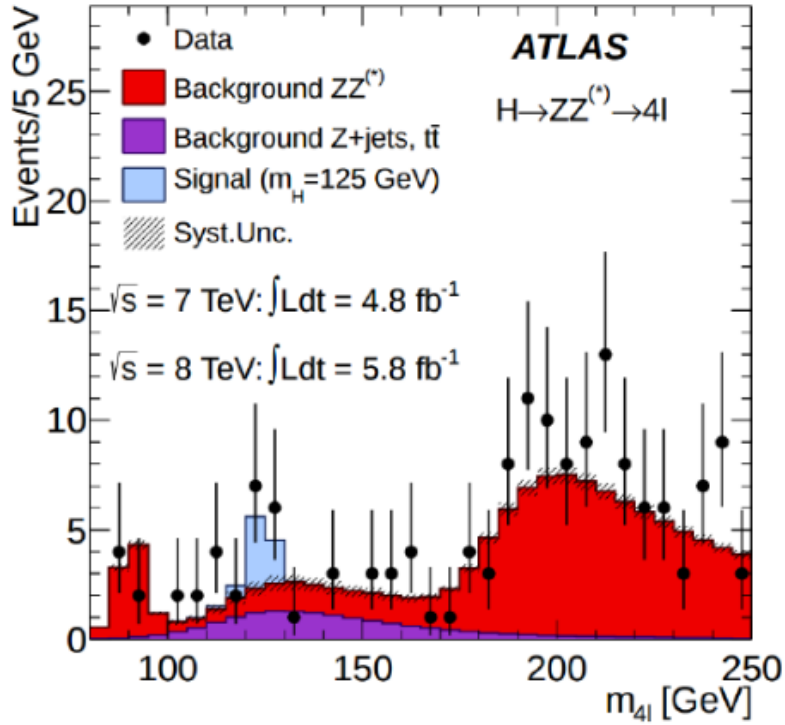


Figure 2.2.: Four lepton invariant mass distributions from the ATLAS and CMS samples for the Higgs boson decay to four leptons at 7 TeV and 8 TeV which contributed to the discovery of the Higgs boson. The fits of the signal and background are shown [9].

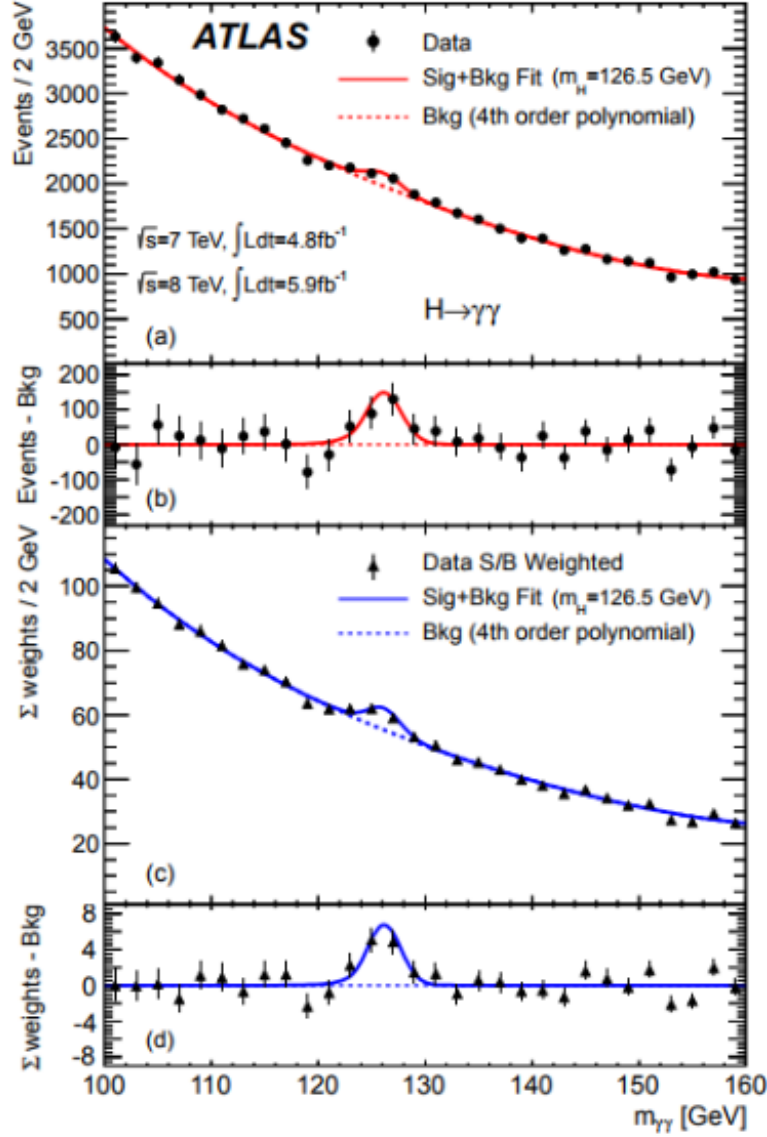


Figure 2.3.: Di-photon invariant mass distributions from the ATLAS and CMS samples for the Higgs boson decay to two photons at 7 TeV and 8 TeV which contributed to the discovery of the Higgs boson. The fits of the signal and background are shown [9].

2.2.1. The Higgs Mechanism

It can be shown that introduced masses of fermions and weak gauge bosons break the local gauge invariance of the Standard Model Lagrangian and make it non-renormalizable [14] [15].

To preserve the gauge invariance, spontaneous electroweak gauge symmetry breaking is introduced via the *Higgs mechanism*.

The symmetry breaking is realized by introducing a complex scalar weak isospin doublet field

$$\boldsymbol{\phi} = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}, \quad (2.9)$$

with an electrically charged component ϕ^+ and a neutral component ϕ^0 and a *Higgs potential*, the term of the scalar field containing no derivative operators in the Lagrangian, of the form

$$V(\boldsymbol{\phi}) = -\mu^2 \boldsymbol{\phi}^\dagger \boldsymbol{\phi} + \lambda (\boldsymbol{\phi}^\dagger \boldsymbol{\phi})^2 \quad (2.10)$$

with a mass coefficient μ and a positive self-coupling coefficient λ . For $\mu^2 < 0$, the Higgs potential has a local maximum for $\boldsymbol{\phi} = 0$ and an infinite number of minima (groundstates) for $\boldsymbol{\phi}$ satisfying

$$\boldsymbol{\phi}^\dagger \boldsymbol{\phi} = \frac{v^2}{2}, \quad (2.11)$$

with *Vacuum Expectation Value* (VEV) $\langle 0 | \boldsymbol{\phi} | 0 \rangle \equiv v \equiv \sqrt{-\frac{\mu}{\lambda}} \neq 0$ as illustrated in Figure 2.4. For reasons of charge conservation, only the neutral part of the Higgs field doublet yields a finite VEV. Choosing a particular groundstate,

$$\boldsymbol{\phi} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ \phi^0 \end{pmatrix}, \quad (2.12)$$

leads to spontaneous breaking of the electroweak gauge symmetry $\times SU(2)_L \times U(1)_Y$ in the groundstate while preserving the local gauge invariance of the Lagrangian.

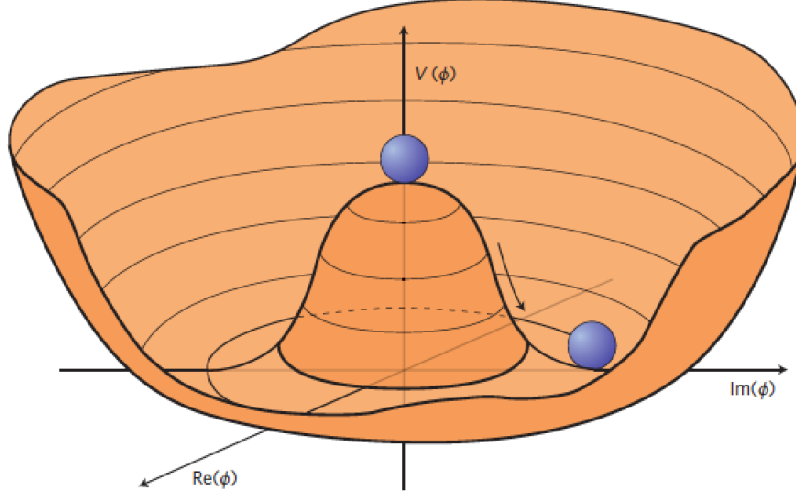


Figure 2.4.: Illustration of the Higgs potential in the case of a complex $U(1)_Y$ Higgs field ψ [17].

In the Standard Model the ground state is three-dimensional in the four-dimensional space of complex Higgs fields. Radial excitations from this groundstate correspond to the physical Higgs field $H_{(x)}$:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + H(x) \end{pmatrix}. \quad (2.13)$$

Defining

$$W_\mu^\pm = \frac{1}{\sqrt{2}} (W_\mu^1 \mp iW_\mu^2) \quad (2.14)$$

$$Z_\mu = \frac{g_2 W_\mu^3 - g_1 B_\mu}{\sqrt{g_2^2 + g_1^2}} \quad (2.15)$$

leads to the Higgs field contribution to the Lagrangian density of the Standard Model:

$$\begin{aligned} L_{Higgs} = & \frac{1}{2} \partial_\mu H \partial^\mu H + \mu - \lambda v^2 H^2 - \lambda v H^3 - \frac{1}{4} \lambda H^4 \\ & + \frac{g_2^2}{4} (2vH + H^2) (W_\mu^+ W^{-\mu} + \frac{1}{2 \cos^2(\theta_W)} Z_\mu Z^\mu). \\ & + \frac{g_2^2 v^2}{8} (W_\mu^+ W^{+\mu} + W_\mu^- W^{-\mu} + \frac{2}{\cos^2(\theta_W)} Z_\mu Z^\mu) \end{aligned} \quad (2.16)$$

The first line of Equation 2.16 contains the kinematic, mass and self-interaction terms of the Higgs field. The second line describes the interactions of the Higgs boson with $W^\pm$ and Z bosons. The third line gives the mass terms for the $W^\pm$ and Z bosons.

The fermion masses are generated by the so called *Yukawa couplings* to the Higgs field to the fermions, which leads to Yukawa coupling terms in the Standard Model Lagrangian:

$$L_{Yukawa} = -g_{Yukawa}^f \frac{v + H(x)}{\sqrt{2}} (\bar{\psi}_L \psi_R + \bar{\psi}_R \psi_L), \quad (2.17)$$

with specific Yukawa coupling constant g_{Yukawa}^f for each fermion, which are proportional to the fermion masses:

$$m_f = \frac{g_Y v}{\sqrt{2}} \quad (2.18)$$

An overview of the relations between Higgs coupling strength and particle mass for different types of particles is shown in Table 2.1.

Table 2.1.: Relationship between Higgs coupling strengths and the particle masses

Higgs Coupling type	Coupling Strength wrt. Mass
Higgs Self-Coupling (H^3)	$g^{H^3} \propto \frac{3m_H^2}{v}$
Higgs Self-Coupling (H^4)	$g^{H^4} \propto \frac{3m_H^2}{v^2}$
Coupling to Gauge Bosons (HVV)	$g^{HVV} \propto \frac{2m_V^2}{v}$
Coupling to Gauge Bosons ($HHVV$)	$g^{HHVV} \propto \frac{2m_V^2}{v^2}$
Coupling to Fermions (Hff)	$g^{Hff} \propto \frac{m_f}{v}$

2.2.2. Higgs Production Modes at the LHC

At the LHC, Higgs bosons are produced in proton-proton collisions with a centre-of-mass energy of $\sqrt{s} = 13.6$ TeV in the current Run 3. As can be seen in Table 2.1, the Higgs boson strength increases with the mass of the particles. An overview of the cross-section of different Higgs production modes at the LHC is shown as a function of the centre-of-mass energies in Figure 2.5. The most important production modes are discussed in more detail in the following[18]

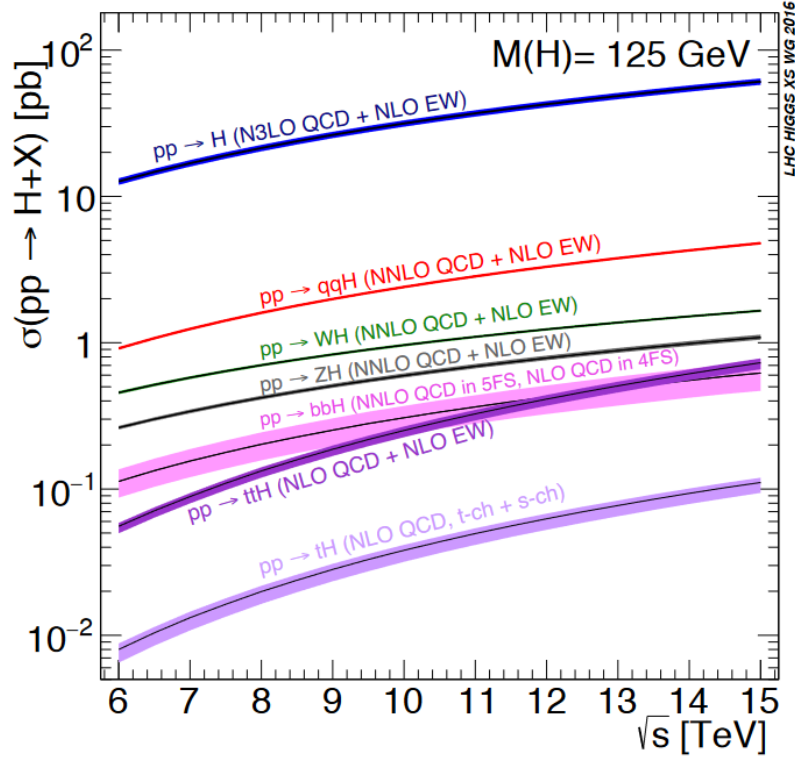


Figure 2.5.: Cross Sections of different Higgs boson production modes at the LHC as a function of the centre-of-mass energy [18]. The production modes shown are, from top to bottom: ggF , VBF , WH , ZH , bbH , ttH , tH .

Gluon-Gluon-Fusion (ggF) is the dominant production mode at the LHC with a cross section of 52.2 pb at $\sqrt{s} = 13.6$ TeV. The Higgs boson does not directly couple to gluons, but only via a quark-loop dominated by top quarks (see Figure 2.6a). The gluons are emitted by the incoming protons.

Vector-Boson Fusion (VBF) has the second largest contribution with a cross section of 4.08 pb at $\sqrt{s} = 13.6$ TeV. In vector-boson-fusion, two incoming quarks from opposite proton beams scatter by emitting weak gauge bosons which combine to produce a Higgs boson (see Figure 2.6b). This channel shows a distinct signature of two jets from the scattered quarks with accompany the Higgs boson and which are preferably boosted in forward direction with regards to the proton beams.

Associated Vector Boson Production (VH), also often referred to as *Higgsstrahlung*, has a cross section of 2.41 pb at $\sqrt{s} = 13.6$ TeV. In the VH production mode, a weak gauge boson is created for example via quark-anti-quark annihilation or gluon-gluon-fusion. The

resulting weak gauge boson then scatters by emitting a Higgs boson (see Figure 2.6c). Due to the decay of the associated weak gauge boson, this channel also has a typical two-jet signature.

Associated Heavy Quark Pair Production (ttH, bbH) is the smallest contributions to the total cross section of Higgs production, with cross sections of 0.57 pb and 0.52 pb for top-anti-top (ttH) and bottom-anti-bottom (bbH) associated production at $\sqrt{s} = 13.6$ TeV. In the ttH and bbH production modes the Higgs boson is produced from t or b quarks from $t\bar{t}$ or $b\bar{b}$ pairs emitted by gluons from the proton beams (see Figure 2.6d).

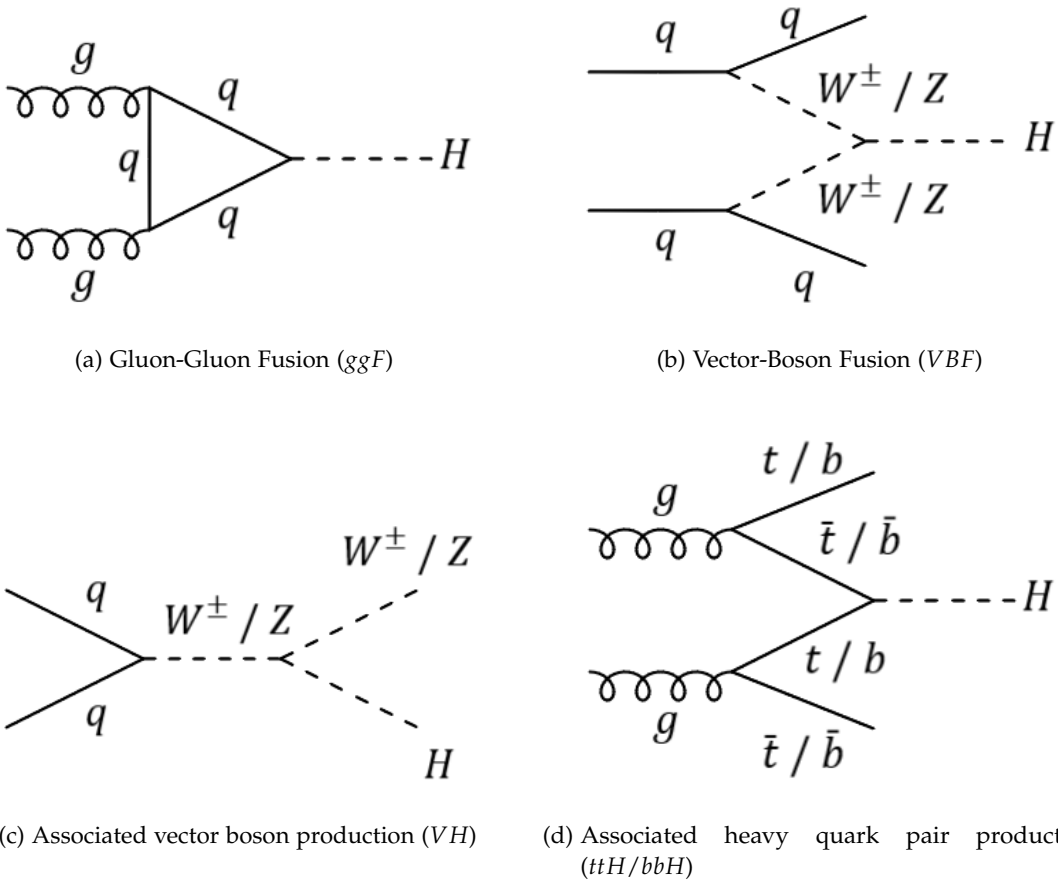


Figure 2.6.: Tree-level Feynman diagrams of the dominant Higgs production modes at the LHC.

2.2.3. Higgs Decay

As the Higgs boson couples strongest to heaviest particles, decays into the heaviest particles show the highest branching ratios, as can be seen in Figure 2.7 [19]. The dominant decay

mode of the Higgs boson consequentially is the decay into a $b\bar{b}$ pair as the Higgs boson is not heavy enough to decay into a top quark pair. Due to the hadronic $b\bar{b}$ final state, separating this decay channel from the QCD background, which is naturally very high in hadron collider experiments, is very difficult and this decay channel suffers from poor signal-to-background separation in analysis. The decay modes with the next highest branching ratios are the decay into W^+W^- pairs, followed by the decays into charged τ lepton pairs. As the W bosons and τ leptons often decay hadronically these decay modes are also rather difficult to detect.

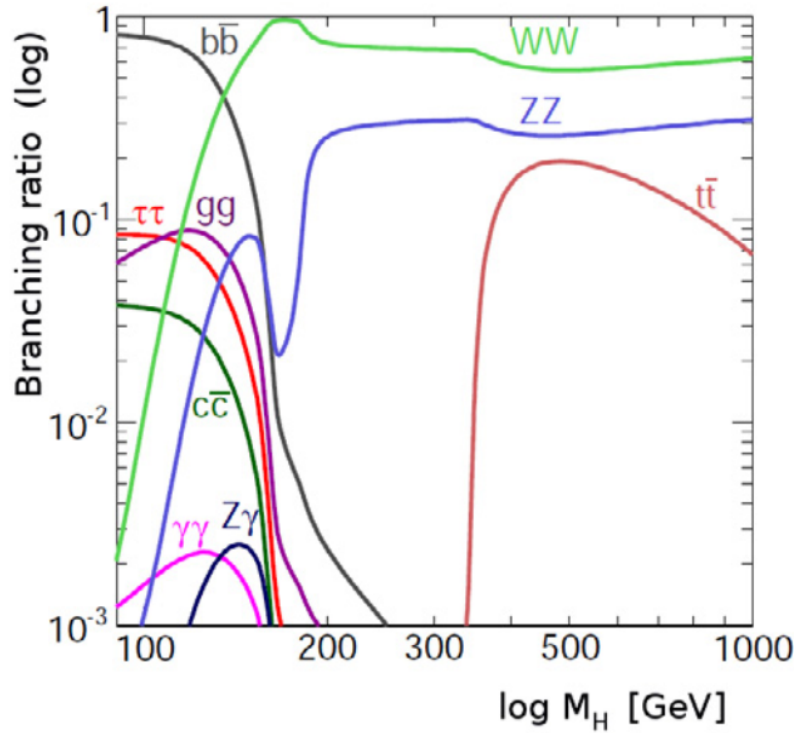


Figure 2.7.: Higgs boson decay branching ratios as a function of the Higgs mass [19]. The measured Higgs boson mass is 125.11 GeV.

The two decay channels that offer the best sensitivity and were vital in the discovery of the Higgs boson are the $H \rightarrow \gamma\gamma$ and $H \rightarrow ZZ \rightarrow 4l$ ($l = e, \mu$) channels as high energy photons and leptons are relatively easily identified and separated from background processes at a hadron collider. As the neutral Higgs boson does not directly couple to photons, they have to be created via a fermion loop, dominated by top quarks, or W^+W^- loops (see Figure 2.9).

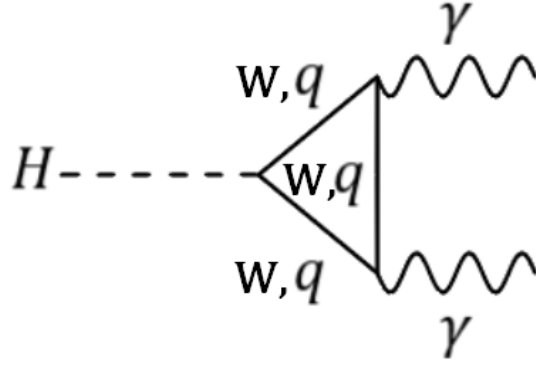


Figure 2.8.: Tree level Feynman diagram of the $H \rightarrow \gamma\gamma$ decay.

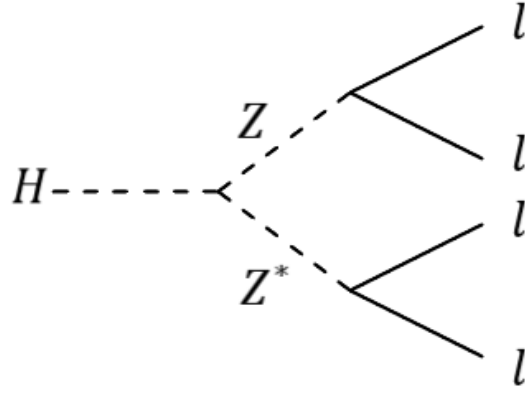


Figure 2.9.: Tree level Feynman diagram of the $H \rightarrow ZZ \rightarrow 4l$ decay.

3. The ATLAS Experiment at the LHC

The Large Hadron Collider is the world's largest and most energetic particle collider with a circumference of about 27 km. During the current Run 3 of the LHC, which started in July 2022, proton-proton collisions occur at a centre-of-mass energy of $\sqrt{s} = 13.6$ TeV. The LHC features four major experiments: ATLAS (A Toroidal LHC Apparatus), CMS (Compact Muon Solenoid), ALICE (A Large Ion Collider Experiment) and LHCb (Large Hadron Collider beauty). The ATLAS and CMS detectors are general-purpose experiments that cover a wide range of physics topics, from Standard Model precision measurements to searches for physics beyond the Standard Model. The ALICE experiment focuses on heavy-ion collisions and investigates properties of the strong-interaction, especially of the quark-gluon-plasma. The LHCb experiment focuses on the study of hadrons containing b -quarks to investigate the CP -violation in the Standard Model and rare B meson decays. As this thesis is performed using ATLAS data, the ATLAS sub-detectors are discussed in the following.

The ATLAS detector has a cylindrical shape with a length of about 46 m and a height of about 25 m[20], see Figure 3.1. It has an onion-like structure with several layers serving specialized functions to detect and identify different kinds of particles. The different layers are discussed from the inside out.

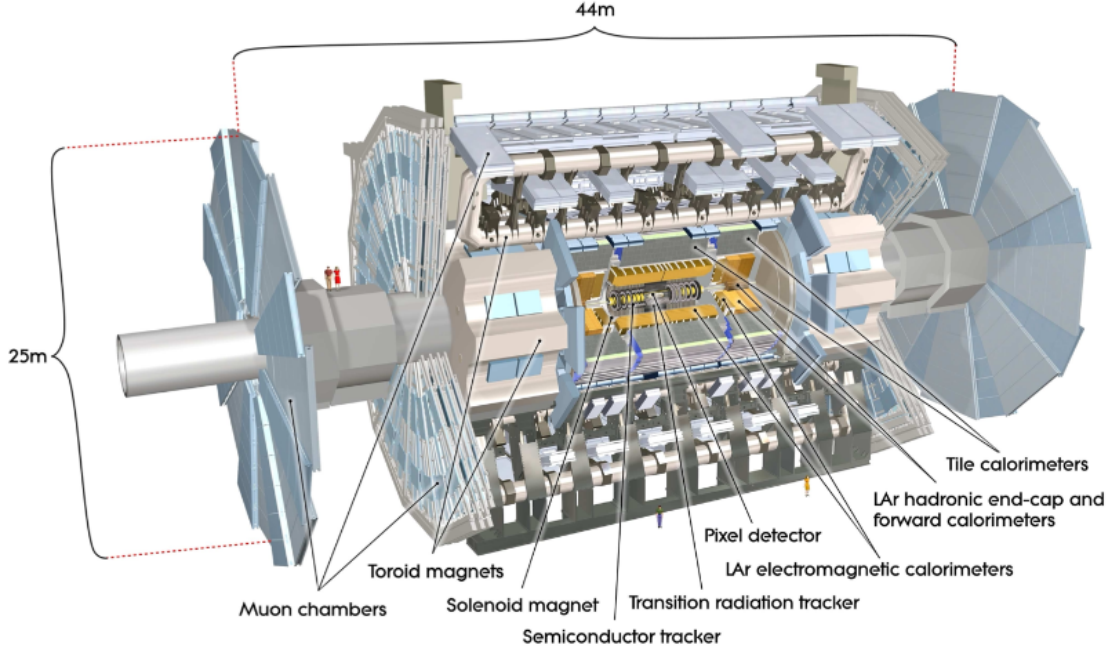


Figure 3.1.: Schematic cut-away view of the ATLAS detector [20].

3.1. Inner Detector

The innermost part of the ATLAS detector located most closely to the interaction point is the *Inner Detector* (ID). The Inner Detector measures the tracks and momenta of charged particles with $p_T > 0.5$ GeV and $|\eta| < 2.5$ inside a 2 T magnetic field aligned with the beamline, and allows for reconstructing the primary production and secondary decay vertexes. The ID consists of three independent sub-detectors, the Silicon Pixel Detector, the Semiconductor Tracker (SCT) and the Transition Radiation Tracker (TRT). An overview of the Inner Detector is shown in Figure 3.2 [20].

The innermost part of the Inner Detector is the Silicon Pixel Detector. The Pixel Detector is designed to provide excellent spatial resolution and granularity for the measurement of the primary and secondary vertexes. The Silicon Pixel Detector features three cylindrical layers around the beamline at radii of 50.5 mm, 88.5 mm and 122.5 mm (barrel part), containing sensors with pixel size of $50 \times 400 \mu\text{m}^2$, as well as three end-cap discs on each side. A fourth layer closest to the beamline at $r = 33.25$ mm, called the Insertible B-Layer (IBL) with $50 \times 250 \mu\text{m}^2$ pixels, was added during the first long shutdown phase of the LHC after Run 1 to further improve the primary and secondary vertex resolution [20].

Following the Silicon Pixel Detector comes the Semiconductor Tracker (SCT), which consists

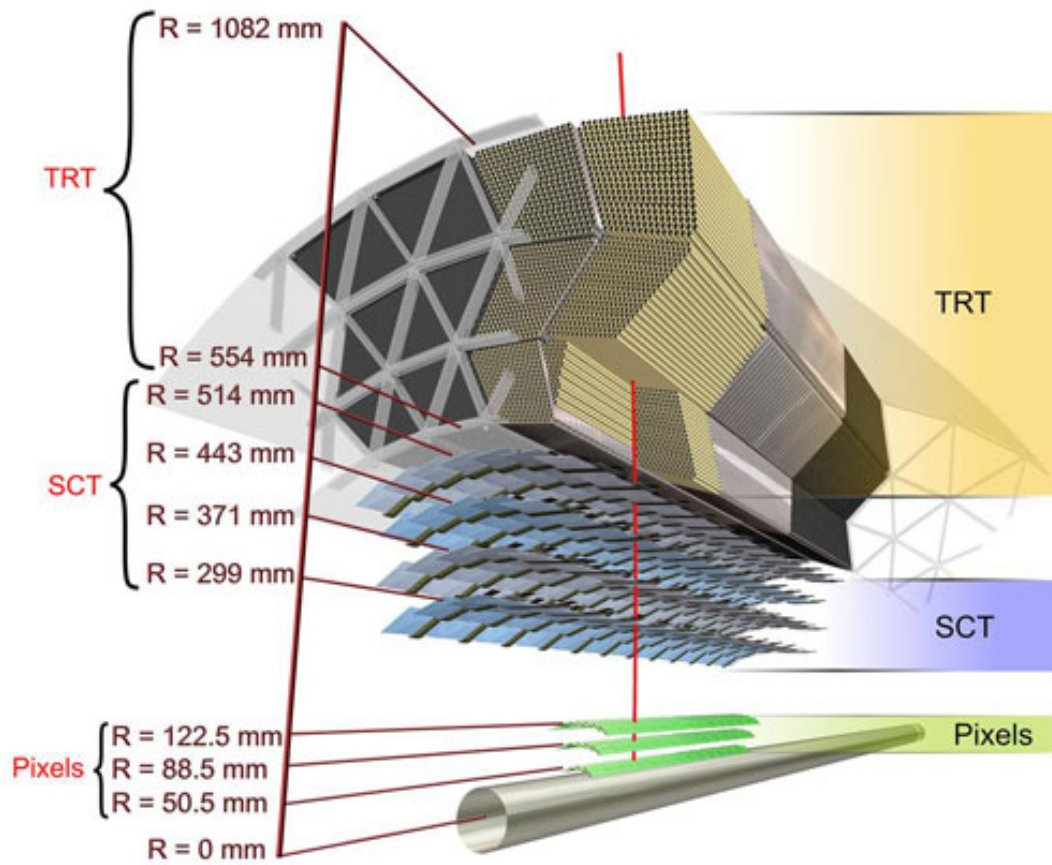


Figure 3.2.: Layout of the Inner Detector of the ATLAS experiment [21].

of silicon micro-strip sensors with a strip pitch of 80 μm that are arranged in four layers in the detector barrel part and nine end-cap discs on both ends of the detector. The SCT detector modules consist of p^+ -on- n silicon strip sensors with a thickness of 285 μm [20].

The outermost part of the Inner Detector is the Transition Radiation Tracker (TRT). The TRT is a straw drift tube tracker with a gas mixture of $\text{Xe}/\text{CO}_2/\text{O}_2$ (70%/27%/3%). 370.000 drift tubes with a diameter of 4 mm are installed in the TRT. Charges induced in the gas by traversing ionizing particles are collected by gold-plated tungsten wires in the centre of the drift tubes [20].

3.2. Calorimeters

Following the Inner Detector, the ATLAS detector contains two calorimeters to measure the energy and direction of particles. They cover pseudorapidity up to $|\eta| = 4.9$ [20]. Incoming particles deposit energy in the calorimeter material, producing showers of secondary particles until the energies of the shower particles are low enough to be absorbed by the calorimeter material.

The first calorimeter is the electromagnetic calorimeter which primarily captures electromagnetically interacting particles, such as photons, electrons and other charged leptons and hadrons. It consists of alternating layers of liquid Argon (LAr) as active detector material and lead absorbers in an accordion-like structure. The electromagnetic calorimeter consists of a barrel region ranging up to $|\eta| < 1.475$ and two endcap wheels on each side covering $1.375 < |\eta| < 3.2$. The electromagnetic calorimeter features an energy resolution of $\frac{\sigma_E}{E} = \frac{10\%}{\sqrt{E}} \oplus 7\%$.

Outside the electromagnetic calorimeter lies the hadronic calorimeter which captures the particles interacting via the strong interaction, jets of mesons and baryons. The hadronic calorimeter consists of three regions: The barrel region with $|\eta| < 1.7$ features plastic scintillator tiles as active material and steel as absorber medium, the endcap regions with $1.5 < |\eta| < 3.2$, which use liquid Argon as active detector material and copper as absorber, and a forward calorimeter with $3.1 < |\eta| < 4.9$ that also uses liquid Argon as active material and an absorber matrix of copper and tungsten. The hadronic calorimeter has an energy resolution of $\frac{\sigma_E}{E} = \frac{50\%}{\sqrt{E}} \oplus 3\%$ in the barrel region and of $\frac{\sigma_E}{E} = \frac{100\%}{\sqrt{E}} \oplus 10\%$ in the endcap regions.

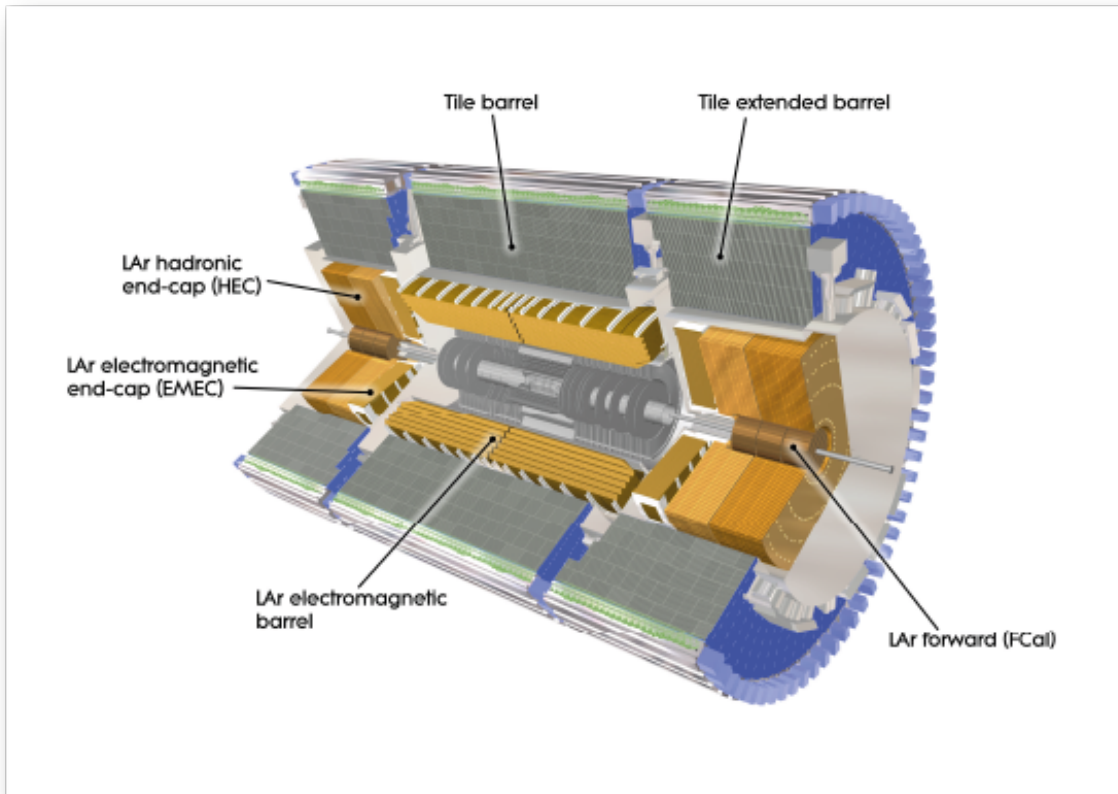


Figure 3.3.: Cut-out view of the ATLAS calorimeter system [22].

3.3. Muon Spectrometer

As muons deposit very little energy in the detector, they frequently escape the ATLAS calorimeters. To measure muon momenta and trajectories with high efficiency and provide muon triggers, a muon spectrometer is used as outermost layer of the ATLAS detector. The magnetic field for the muon tracker is provided by a large air-core toriodal magnet in the barrel region ($|\eta| < 1.4$) and two smaller air-core toriodal magnets in the endcap regions ($1.6 < |\eta| < 2.7$) with field strengths between 0.5 T and 1 T. For precision muon tracking, Muon Drift Chambers (MDT) are used, while Resistive Plate Chambers (RPC) and Thin Gap Chambers (TGC) are used as trigger chambers. For Run 3, the Small Wheels constituting the inner layer of the endcaps were replaced with New Small Wheels (NSW) which feature small strip-thin Gap Chambers (sTGC) and Micromegas (MM) detectors for tracking and triggering at high background rates counting rates [20].

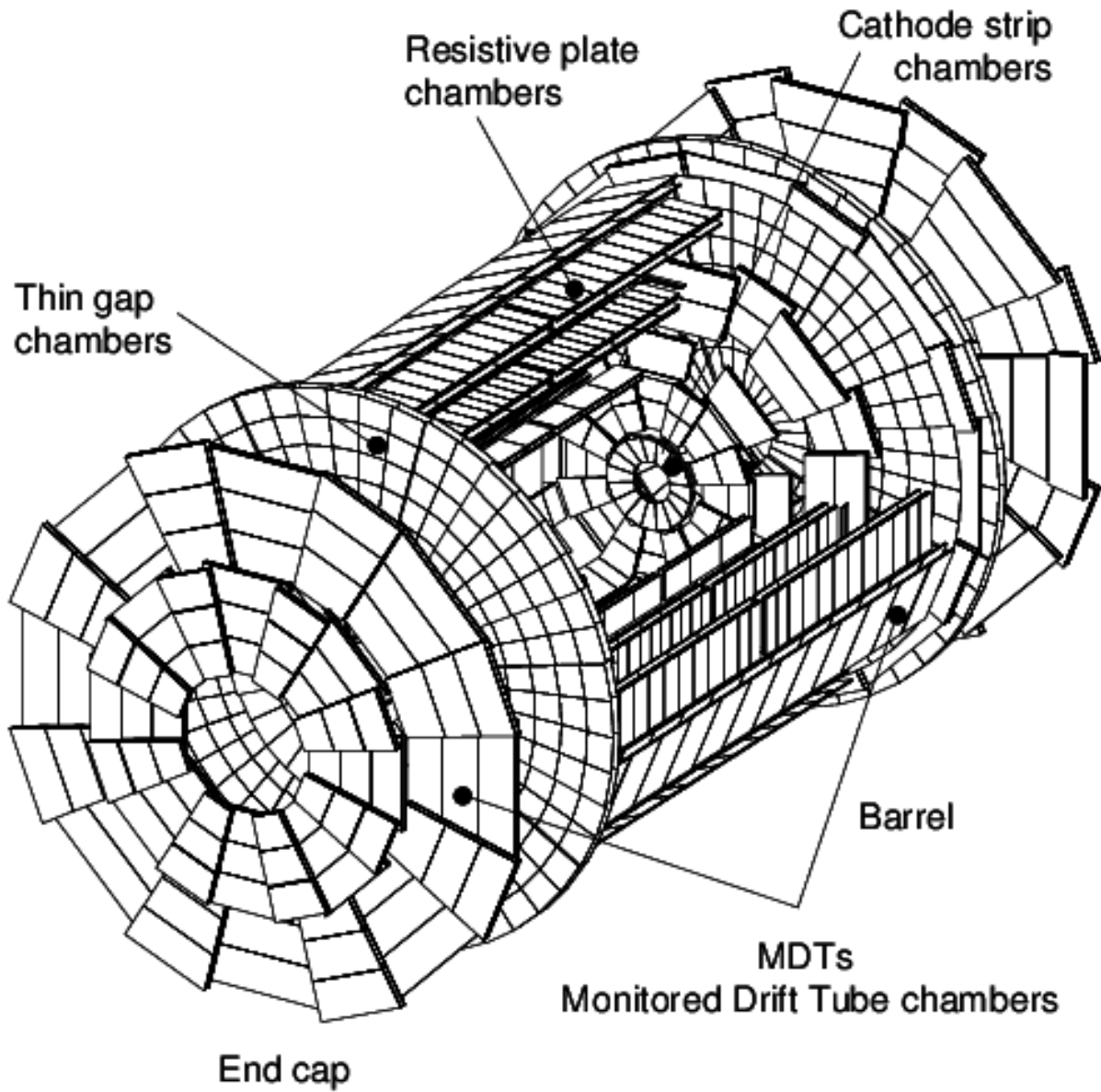


Figure 3.4.: Layout of the ATLAS muon spectrometer [23].

3.4. Coordinate System

The kinematic properties of the particles are fully characterized by their four-momenta,

$$p^\mu = \begin{bmatrix} \frac{E}{c} \\ p_x \\ p_y \\ p_z \end{bmatrix}. \quad (3.1)$$

Due to the cylindrical nature of the ATLAS detector, a polar coordinate system is usually chosen. Cylindrical coordinates are used with the radius $r = \sqrt{x^2 + y^2}$ perpendicular to the z-axis, pointing parallel to the beam-line, the azimuthal angle ϕ and the polar angle θ . Key variables to describe particle kinematics at the LHC are:

- the transverse momentum $p_T = \sqrt{p_x^2 + p_y^2}$ with respect to the beamline
- the *pseudorapidity* η defined as:

$$\eta = -\ln\left(\frac{\tan \theta}{2}\right). \quad (3.2)$$

with the polar angle θ . In the relativistic limit, where $E \gg m$, η approaches the rapidity y , which is defined as:

$$y = \frac{1}{2} \ln\left(\frac{E + p_z}{E - p_z}\right) \quad (3.3)$$

which is invariant under Lorentz boosts along the z-axis. The angular distance between two particles in $\eta - \phi$ coordinates is given by:

$$\Delta R = \sqrt{\eta^2 + \phi^2}. \quad (3.4)$$

4. The $H \rightarrow ZZ^* \rightarrow 4l$ Decay Channel

The focus of this analysis is the $H \rightarrow ZZ^* \rightarrow 4l$ decay channel. In the $H \rightarrow ZZ^* \rightarrow 4l$ channel, one of the Z bosons is produced virtually, or *off-shell*, as the restmass of the Higgs ($m_H \approx 125$ GeV) is not enough to allow the production of two on-shell Z bosons ($m_Z \approx 91$ GeV). Due to the fast decay of the tau leptons, only electron and muon final states are considered in this analysis, which results in four leptonic final states considered, depending on whether the leptons were produced by the on- or off-shell Z boson: $4e$, 4μ , $2e2\mu$ and $2\mu2e$.

The $H \rightarrow ZZ^* \rightarrow 4l$ decay channel is often referred to as the *Golden Channel* due to its clear experimental signatures and despite its comparatively low branching ratio of $\approx 3\%$. Not only does it feature a high signal-to-background-ratio, it also has a very clear trigger signature due to the high p_T signal leptons in the final state [9].

4.1. Event Samples

A large amount of events are necessary to effectively train a complex Neural Network. The amount of $H \rightarrow ZZ^* \rightarrow 4l$ events measured at the LHC is far too small to train an effective Neural Network classifier. Instead, large $H \rightarrow ZZ^* \rightarrow 4l$ event data samples are generated using *Monte Carlo* simulation [24]. Different Monte Carlo generators are used for the different production channels. For Run 2 and Run 3, the ggF , VBF and VH Higgs production modes were simulated using the POWHEG-Box v2. Monte Carlo generator [25] [26] [27] with the parton shower being simulated with PYTHIA8 [28]. The VH and VBF production modes were produced next-to-leading-order NLO QCD, while the ggF mode were produced at next-to-next-to-leading-order (NNLO QCD). Parton Shower matching was performed with the NNLOPS [29] and MINLO [30] algorithms for the ggF and VH channels, respectively. For the ZZ^* background, SHERPA2.2.2 was used as Monte Carlo Generator for Run 2, while SHERPA2.2.12 was used in Run 3 [31]. The QCD tree-level of the background event data samples depends on the phase space region and is discussed in more detail in Reference [32].

4.2. Background

The background contributions to the $H \rightarrow ZZ^* \rightarrow 4l$ channel can be divided into an irreducible category, with four leptons originating from the main collision, referred to as *prompt*

leptons and a reducible background containing non-prompt or light-flavor jets which are misidentified as leptons ("fake" leptons).

The main contributions to the background in the $H \rightarrow ZZ^* \rightarrow 4l$ decay channel are irreducible from non-resonant ZZ^* production. The largest contribution to this process is quark-anti-quark annihilation, followed by gluon-gluon fusion. As the Z boson is not a colour-charge carrying particle, gluon-gluon production necessitates an additional loop. Smaller contributions stem from processes involving tri-gauge boson (VVV) production, ZZZ , WWZ , WZZ and processes with a Z boson produced with a top-anti-top pair ($t\bar{t}Z$). The Feynman diagrams for the processes are shown in Figure 4.1.

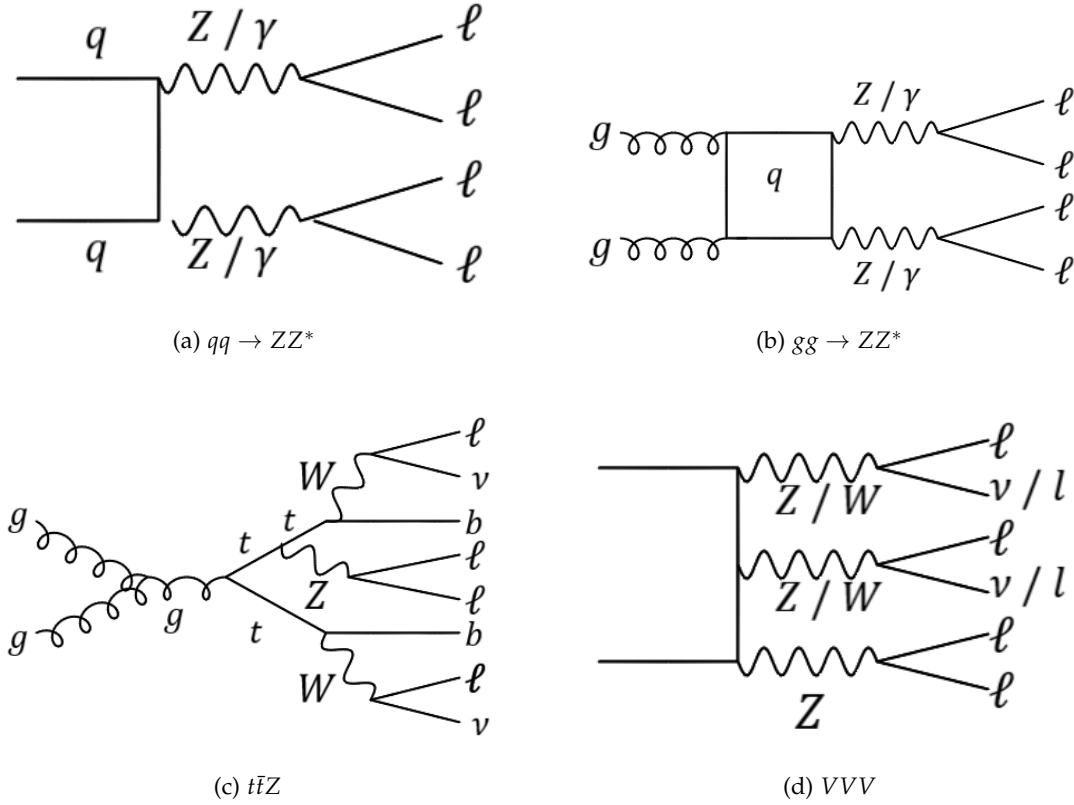


Figure 4.1.: Tree-level Feynman diagrams of the most important irreducible background processes to the $H \rightarrow ZZ^* \rightarrow 4l$ decay channel.

Estimations of the irreducible ZZ^* background have previously been obtained using Monte Carlo simulations, which were validated in the side-band regions $105 < m_{4l} < 115$ GeV and $130 < m_{4l} < 160$ GeV. Due to increasing amounts of data collected at the LHC, it has become possible to use a data-driven approach instead, especially for the dominant $qq \rightarrow ZZ^*$

background [32]. In the data driven approach, the normalization of the ZZ^* background is obtained by fitting the aforementioned side-band regions and the resonance mass region $115 < m_{4l} < 130$ GeV depending on the number n_{jets} of jets with bins for zero, one and more than one jets. The advantage of the data-driven approach is that it avoids theoretical systematic uncertainties and uncertainties in the luminosity. A data-driven approach also lends itself to the irreducible $t\bar{t}$ background contribution, while the tri-gauge boson still requires the Monte Carlo simulation approach.

The reducible background is smaller than the irreducible background. The main contributions to the reducible background stem from processes involving the production a Z boson with additional jets ($Z + jets$), as well as $t\bar{t}$ pair production with semileptonic decays of both top quarks and WZ production. The Feynman diagrams for these processes are shown in Figure 4.2. Reducible background contributions are distinguished based on the flavor of the sub-leading lepton pair, $ll + ee$ and $ll + \mu\mu$. In the $ll + \mu\mu$ case, the non-prompt muons mostly stem from semi-leptonic decays of heavy-flavor hadrons in the $Z + jets$ and $t\bar{t}$ channels. The $ll + ee$ case mostly receives contributions from misidentified light-flavored hadron jets and electrons produced by photon conversion, as well as small contributions from the decay of heav-flavored hadrons.

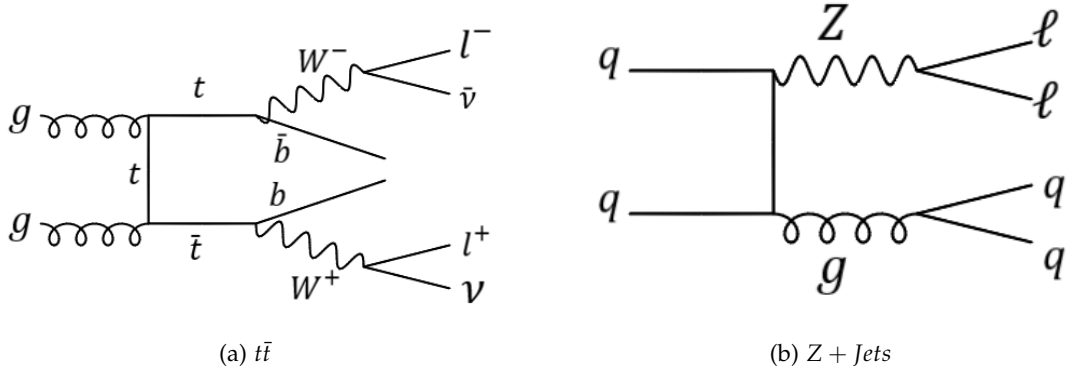


Figure 4.2.: Tree-level Feynman diagrams of the most important reducible background processes to the $H \rightarrow ZZ^* \rightarrow 4l$ decay channel.

4.3. Event Selection

To suppress the background contributions, event selection cuts are applied. Higgs candidate events are selected using particle identification, as well as general event quality criteria. The event selection criteria used in this analysis are discussed in the following.

4.3.1. Object Reconstruction and Identification

To reconstruct the final state, the tracks and productional decay vertices of the produced particles need to be first reconstructed in the Inner Detector. The tracks are reconstructed using a χ^2 fit using information from the Inner Detector sub-components, the Pixel Detector, the Semiconductor Tracker and the Transition Radiation Tracker. Using the track information, vertices are also reconstructed using χ^2 fits, where the primary vertex is defined as the vertex with the largest sum of squared transverse momenta of contributing tracks $\sum_i^N (p_T^i)^2$. The distance between the tracks and vertices is an important parameter for the event selection, which is measured by the transverse and longitudinal impact factors d_0 and z_0 .

Electrons are reconstructed using Inner Detector track information and energy deposition clusters in the Electromagnetic Calorimeter with a clustering algorithm [33]. They are matched to Inner Detector tracks after passing cluster shape requirements. A likelihood-based multivariate discriminant is used to reject background electrons with an electron efficiency above 95% at a *Loose* working point. It is further required that the electrons satisfy pseudorapidity $|\eta| < 2.47$ and transverse energy $E_T > 7$ GeV, as well as longitudinal impact parameter $|z_0 \sin \theta| < 0.5$.

Due to their very low energy deposition in the calorimeters, muons are primarily reconstructed using track information from the Inner Detector and the Muon Spectrometer. As the routing read out cables of the Inner Detector necessitate a gap in the muon spectrometer instrumentation for $|\eta| < 0.1$, tracks are identified as muons in this region if their energy deposit in the calorimeter are compatible with a minimum-ionising particle. Four types of muon reconstruction are used, depending on whether the muon track is reconstructed in both the Inner Detector and the Muon Spectrometer (*Combined Muon Reconstruction*), only reconstructed in the Inner Detector (*Segment-Tagged Muon Reconstruction*), only reconstructed by the Muon Spectrometer without a match to an Inner Detector track (*Stand-Alone Muon Reconstruction*) or identified by its energy deposit in the calorimeters where there is no Muon Spectrometer coverage (*Calorimeter-Tagged Muon Reconstruction*) [34]. Similarly to the electron identification, a *Loose* working point is used in this analysis. While all muons must satisfy transverse momentum $p_T > 5$ GeV, further selection criteria are used depending on the reconstruction type. In the *Segment-Tagged* case, muon tracks in the Inner Detector must be associated with at least one hit in the Muon Detector. The *combined* case is the main type of muon reconstruction, where muons are reconstructed with a combined fit of Muon Spectrometer and Inner Detector hits. The *Stand-Alone* case requires the muon track to traverse at least two layers in the Muon Spectrometer and is mainly used to extend acceptance in the forward region with $2.5 < |\eta| < 2.7$, as this area is not covered by the Inner Detector. The *Calorimeter-Tagged* case is mainly optimized for the Muon Spectrometer gap region $|\eta| < 0.1$ and a low momentum range with $15 < p_T < 100$ GeV.

The hadronisation of gluons and quarks causes narrow, hadronic, particle showers which are

called *jets*. Jets are reconstructed through the energy deposition in the hadronic calorimeter using the *anti* - k clustering algorithm [35]. As many proton-proton collisions happen at the same time during LHC collision events, interferences from secondary collisions need to be taken into account, which are generally referred to as *pile-up*. To reduce the impact of Pile-Up contributions, a Jet-Vertex-Tagger method is applied which uses Inner Detector information to associate jets to the different primary collision vertices and to reject pile-up jets. All jets are required to satisfy $p_T > 30$ GeV and $|\eta| < 4.5$.

As it is possible that different objects are associated to the same detector information, overlap corrections need to be applied. If an electron and a muon are reconstructed from the same Inner Detector track, the electron will be removed, unless the muon stems from Calorimeter-Tagged reconstruction, in which case the muon will be removed. Electrons and muons that appear within a cone of $\Delta R = 0.2$ are also removed.

4.3.2. Event Selection Criteria

A summary of the criteria used for the selection of $H \rightarrow 4l$ events in Run 3 data is given in the following [33].

Quadruplet Selection To select $H \rightarrow 4l$ candidate events, they are required to contain at least four reconstructed leptons, which can be combined in two pairs with identical flavor and opposite sign. The three leptons with highest p_T are required to satisfy $p_T > 20$ GeV, $p_T > 15$ GeV and $p_T > 10$ GeV respectively. The lepton pair system with a mass m_{12} closest to the rest mass of the Z boson is referred to as the *leading* pair, which must satisfy $50 < m_{12} < 106$ GeV, while the remaining lepton pair is interpreted to result from the off-shell Z candidate, required to satisfy $m_{thres} < m_{34} < 134$ GeV. m_{thres} depends on the invariant mass m_{4l} of the four lepton system, starting at 12 GeV for $m_{4l} < 140$ GeV and linearly rising with m_{4l} to 50 GeV for $m_{4l} > 140$. Events containing sub-leading lepton pairs with $m_{34} < 5$ GeV are rejected to reduce background contributions from J/ψ decays. The leptons of each pair are also required to have a minimum angular separation of $\Delta R_{(l,l')} > 0.1$.

Isolation As the signal leptons are generally produced well isolated from other particles in the $H \rightarrow 4l$ event, isolation criteria based on track- and calorimeter information are used for event selection. Track-based isolation uses the sum of the transverse momenta of object within a cone of $\Delta R = 0.3$ around muons and of $\Delta R = 0.2$ around electrons. Calorimeter-based isolation uses the sum of the transverse energies E_T of particles surrounding the lepton with a cone of $\Delta R = 0.3$. For both the track based and the calorimeter based isolation additional kinematic criteria are applied on the objects within the cones, which are detailed in Ref. [33]. An isolation working point is defined by combining the track- and calorimeter based isolation variables in the requirement $I_{track} + 0.4I_{cal} < 0.16$.

Impact Parameter Significance As the Higgs decays are very fast, the signal leptons are expected to originate from the primary vertex unlike leptons originating from heavy flavor decays, which tend to originate from the secondary vertices. To reject the heavy flavor background, requirements on the transverse impact parameter significance are placed, $\frac{d_0}{\sigma_{d_0}} < 5$ for electrons and $\frac{d_0}{\sigma_{d_0}} < 3$ for muons, where σ_{d_0} is the uncertainty on d_0 .

Vertex Selection The four signal leptons are required to originate from the same vertex. The tracks of the corresponding leptons are fitted to the same vertex, requiring that the number of degrees of freedom $\frac{\chi^2}{N_{dof}} < 5$ for 4μ final state and $\frac{\chi^2}{N_{dof}} < 9$ for all other final state flavor combinations.

Best Quadruple It is possible that several lepton quadruple combinations resulting from the same collision event pass all previous criteria. The best quadruplet is selected by choosing the quadruplet with leading and subleading lepton pairs closest to the Z boson mass, separately for each of the final states ($4e, 4\mu, 2e2\mu, 2\mu2e$). If there are such Higgs boson candidates in more than one of the final states, then the final state with the highest Higgs boson signal selection efficiency is chosen. The selection efficiencies of the different final states are 31%, 21%, 17% and 16% for $4\mu, 2e2\mu, 2\mu2e$ and $4e$ respectively.

5. $H \rightarrow 4l$ Event Categorization

Accurate $H \rightarrow ZZ^* \rightarrow 4l$ cross-section measurements are vital for the investigation of Higgs boson properties. To enhance the sensitivity of these measurements for potential new physics beyond the Standard Model, the Simplified Template Cross Sections (STXS) framework was developed. In this framework, the production cross sections are measured in several exclusive production phase space regions, based on kinematic properties such as Higgs transverse momentum, jet multiplicity in association with the Higgs decay, and di-jet invariant mass [32].

The production bins of the STXS framework are motivated by several considerations. The production phase space can be chosen such as to reduce the theory uncertainties related to the recovery of the particle-level Higgs boson from the reconstructed selected events. The more granular measurement provides higher sensitivity to new physics at large energy scales. In addition, the common definition of STXS bins allows for the combination of measurements in different Higgs decay channels. The kinematical regions are to minimize the statistical uncertainties [32].

The first and least granular step of the STXS binning scheme is the "Production Mode Scheme" (see Figure 5.1), where the different Higgs production modes (ggF , VBF , VH , $t\bar{t}H$, tH and $b\bar{b}H$) are discriminated in different STXS bins. Due to the limited sensitivity with the current amount of data, the tH and $b\bar{b}H$ bins are merged with the $t\bar{t}H$ and ggF bins, respectively [32].

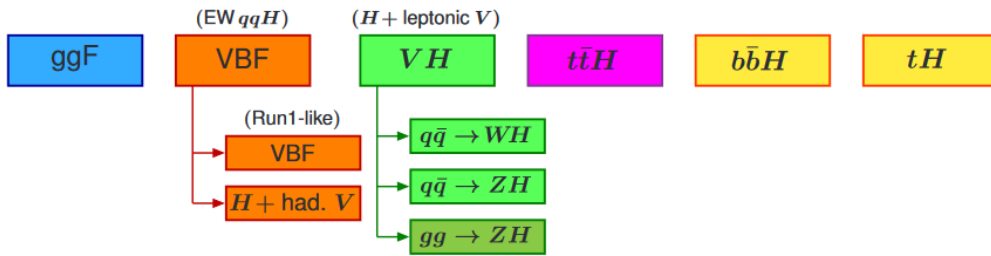


Figure 5.1.: STXS production mode scheme [32].

A more granular version that will serve as baseline for this analysis is the Stage 1.1 Reduced STXS binning scheme, which is optimized for the $H \rightarrow 4l$ decay channel. Here, bins of the Production Mode scheme are further divided according to the number of jets, n_{jets} , produced

in Higgs production, the Higgs transverse momentum, p_T^H , and the invariant mass of the two leading jets, m_{jj} . The binning in p_T^H provides additional sensitivity to BSM physics which is expected to appear predominantly in higher p_T^H Higgs boson production. The n_{jets} and m_{jj} selection criteria, while providing additional BSM sensitivity, provide also kinematic signatures that allow for the separation of the ggF , VH and VBF production modes. The kinematic cuts are chosen to result in bins that allow for good separation between the production modes. A detailed overview of this binning scheme can be found in Ref. [32]. A schematic overview is shown in Figure 5.2.

5. $H \rightarrow 4l$ Event Categorization

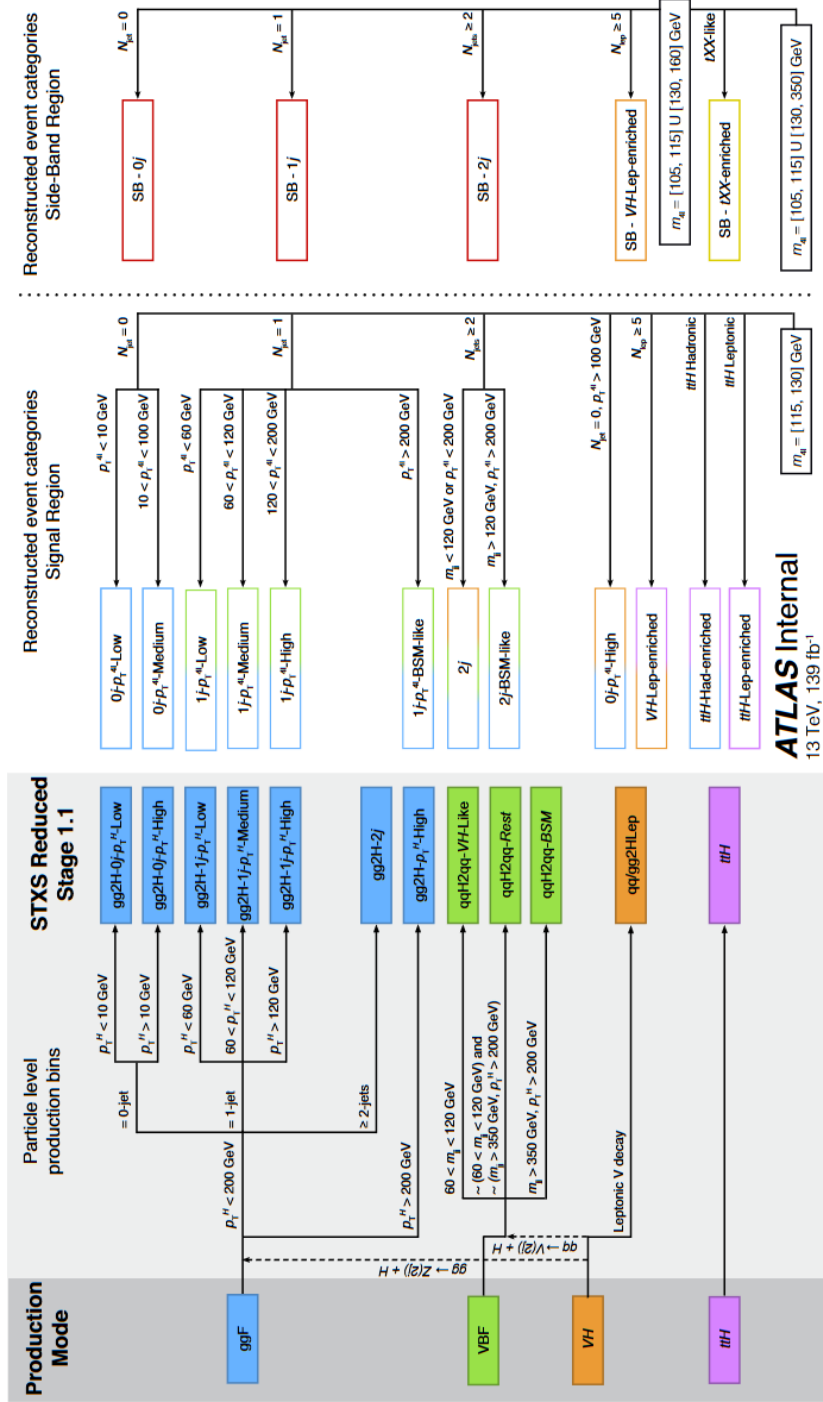


Figure 5.2.: Overview of the STXS binning schemes, showing, from left to right: the Production Mode scheme; the Reduced Stage 1.1 scheme; and the corresponding categories of reconstructed events in the signal and control-sideband regions of collected data. [32].

The STXS production bins introduced here are defined via information that can not be accessed by the detector, such as the production mode of an event, but rather assumes full knowledge of the particle generation, which is generally referred to as *particle level*, as opposed to the information available from the detector, which is referred to as *detector-level*. Particle-level objects need to be recovered from the information from the reconstructed detector signal (here referred to as *reco-level* information). Reco-level selection criteria are chosen to be as close as possible to the particle-level binning. The mapping of the measured event yields at reco-level to the cross-section at particle-level is then performed using Monte Carlo simulation of the full detector response. The reco-level event categories for the Reduced Stage 1.1 STXS scheme are also summarised in Figure 5.2. The event categorization for the reconstructed Higgs boson candidates is performed in the main window with $115 \text{ GeV} < m_{4l} < 130 \text{ GeV}$, based on the number of jets, the invariant mass of the two leading reconstructed jets and the transverse momentum of the Higgs boson candidate.

As the particle-level and reco-level production categorization criteria are not identical, each reco-level category will be contaminated with events from other production modes and the ZZ background. Therefore, an as accurate as possible assignment of each reconstructed event to the correct Higgs production mode is important for the STXS measurement. In a previous STXS measurement in the $H \rightarrow 4l$ channel for ATLAS Run 2 data at $\sqrt{s} = 13 \text{ TeV}$ using 140 fb^{-1} of data [32], a machine learning based approach using Neural Networks was used to distinguish between the different Higgs production modes at reco-level. Neural Networks are trained on Monte Carlo simulated signal and background samples. While an approach based on *Recurrent Neural Networks* (RNNs) [36] was chosen for the previous Run 2 analysis, an alternative, Neural Network approach implementing invariance of the signal final state under permutation of the Higgs decay leptons is investigated in this analysis as a potential improvement for measurements with Run 3 data. This new approach is further discussed in Section 6.3.3.

A different event categorization scheme than the one chosen during Run 2 is currently used for the training of the RNNs on Run 3 data. To ensure comparability with the results of the RNNs, both event categorization schemes are considered in this study, referred to as Run 2 and Run 3 categorization schemes. Six kinematic regions are used in the Run 2 categorization scheme and four in the Run 3 categorization scheme, based on the number of jets n_{jets} , the transverse momentum of the four-lepton system, p_T^{4l} , and the invariant mass of the two leading jets, m_{jj} . An overview of the event categories and the contributing channels for the Run 2 and Run 3 schemes is given in in Table 5.1.

Table 5.1.: Kinematic regions and contributing production modes in the reconstructed event categories used in the training of the $H \rightarrow 4l$ Classification Neural Networks. The targeted Higgs production mode in each category is marked in red. $H \rightarrow 4l$ categorization scheme used for the Neural Network training in Run 2 [37] and a less granular scheme for application on Run 3 data are investigated. In the scheme for Run 3, the $0 - Jet$ $2e2\mu$ and $4l$ event categories, as well as the $1 - Jet$ low- p_T^{4l} and med.- p_T^{4l} event categories are merged into inclusive $0 - Jet$ and $1 - Jet$ categories respectively. Also, the ggF and VBF processes are treated as a combined background in the $2 - Jet$ low- p_T^{4l} category instead of being treated separately

Run 2 Event Categorization Scheme		
Event Categories	Event Selection Criteria	Dominant Processes
$0 - Jet \mid 4l$	$n_{jet} = 0$ p_T^{4l} -inclusive $eeee/\mu\mu\mu\mu$ final state	ggF ZZ
$0 - Jet \mid 2e2\mu$	$n_{jet} = 0$ p_T^{4l} -inclusive $ee\mu\mu/\mu\mu ee$ final state	ggF ZZ
$1 - Jet \mid \text{low-}p_T^{4l}$	$n_{jet} = 1$ $p_T^{4l} < 60 \text{ GeV}$	VBF ggF ZZ
$1 - Jet \mid \text{med.-}p_T^{4l}$	$n_{jet} = 1$ $60 \text{ GeV} < p_T^{4l}$	VBF ggF ZZ
$2 - Jet \mid \text{low-}p_T^{4l}$	$n_{jet} = 2 \text{ or } n_{jet} = 3$ $p_T^{4l} < 200 \text{ GeV}$ or $m_{jj} < 120 \text{ GeV}$	VH VBF ggF
$2 - Jet \mid \text{high-}p_T^{4l}$	$n_{jet} = 2 \text{ or } n_{jet} = 3$ $p_T^{4l} > 200 \text{ GeV}$ and $m_{jj} > 120 \text{ GeV}$	VBF ggF
Run 3 Event Categorization Scheme		
Event Categories	Event Selection Criteria	Dominant Processes
$0 - Jet$	$n_{jet} = 0$ p_T^{4l} -inclusive	ggF ZZ
$1 - Jet$	$n_{jet} = 1$ p_T^{4l} -inclusive	VBF ggF ZZ
$2 - Jet \mid \text{low-}p_T^{4l}$	$n_{jet} = 2 \text{ or } n_{jet} = 3$ $p_T^{4l} < 200 \text{ GeV}$ or $m_{jj} < 120 \text{ GeV}$	VH $ggF + VBF$
$2 - Jet \mid \text{high-}p_T^{4l}$	$n_{jet} = 2 \text{ or } n_{jet} = 3$ $p_T^{4l} > 200 \text{ GeV}$ and $m_{jj} > 120 \text{ GeV}$	VBF ggF

6. Principles of Neural Network Algorithms

As discussed in Section 5, it is important for the study of Higgs couplings to accurately separate events produced by a targeted Higgs mode from events produced by other, contaminating Higgs modes and the ZZ background. Events produced by differing $H \rightarrow 4l$ modes (or the ZZ background) can result in identical reconstructed final state topologies in the detector. Due to this, it is not possible to determine with absolute certainty from which mode an $H \rightarrow 4l$ candidate was produced in the detector. A probabilistic model is trained on simulated data to predict the probabilities of an $H \rightarrow 4l$ candidate having been produced by each respective Higgs production mode or ZZ background. A common approach in modern machine learning to classification problems such as this are *Neural Networks*.

Neural Networks [38] [39] [40] [41] are computational models that can be applied to a variety of tasks, like predictive modeling, generative modeling or clustering. Predictive Neural Networks, such as in classification cases, aim to approximate the true relationship $f(x)$ between a set of input variables x and desired output values y , generally referred to as *targets*. In a classification task, the targets y are the classes that are to be separated from each other. In the case of the production mode classification of $H \rightarrow 4l$ events presented here, the input variables are kinematic properties of the reconstructed final state, like the transverse momentum p_T^{4l} of the four lepton system, the di-jet mass m_{jj} or the pseudorapidity η^{lep} of the leptons from the Z boson decay, while the classes y are the dominant Higgs production modes (or ZZ^* background) in each event category.

The structure of Neural Networks is inspired by the human brain, in which neurons process and then pass information to other, connected neurons in, complex networks. In Neural Networks, the function of human neurons is mimicked in a mathematical model, generally referred to as *nodes*. A Neural Network consists of multiple layers of interconnected nodes. Nodes receive and process information from connected nodes in previous layers of the network and then pass an output signal to connected nodes in the proceeding layer, as showcased in Figure (6.1). Neural Networks in which all nodes of neighboring layers are connected to each other are referred to as *Dense Neural Networks*. The input variables x are initially passed to the first layer of the Neural Network, the *input layer*. The last layer, generally referred to as the *output layer*, yields an output $\hat{y}$ based on the given input variables, which is desired to be as close as possible to the true targets y . The output layer can consist of a single or multiple nodes, depending on the underlying task of the Neural Network. Additional layers of nodes between input and output layer are known as *hidden layers*, which allow the Neural Network

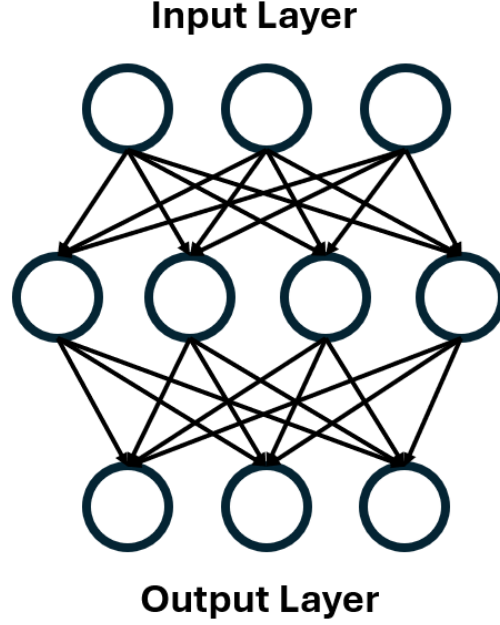


Figure 6.1.: Schematic example of the interconnected structure of Neural Networks. The connections between the circular nodes are represented by arrows.

to model more complex relationships between input and output. Neural Networks with at least one hidden layer are referred to as *Deep Neural Networks*.

The strength of the connections between two respective nodes is represented by weights. As such, the nodes processes incoming inputs as weighted sum. Specifically, the output a_j^l of the j -th node in a layer with inputs x_i is,

$$a_j = \sum_{i=0} w_{ij} x_i + b_j, \quad (6.1)$$

where w_{ij} is the weight assigned to the i -th input for the j -th node and b_j is an offset, generally referred to as *bias*, which allows the Neural Network to model more complex functions. To not limit the Neural Network to linear relationships between input and output, the weighted sums a_j of nodes are passed through non-linear functions, referred to as *activation functions*.

In a classification task, the target y of an event is represented by a vector of length n , where n is the number of classes. Each index of the vector represents a class, or in this case, more specifically, a Higgs production mode (or ZZ^* background). The target vector assigned to a Higgs candidate event is set to 1 at the index corresponding to its true production mode and 0 otherwise, which is generally referred to as *one-hot encoding*. In event categories with only

two classes, the target vector y can be simplified to a single integer value which is set to 1 or 0, where 1 represents the targeted Higgs production mode and 0 represents other Higgs production modes (or ZZ^* background).

Given a dataset X of input variables x from a distribution of input variables $p(x)$, a Neural Network produces a corresponding set of outputs $\hat{Y}$ from an output posterior distribution $p(\hat{y}|x)$, which is a step generally referred to as *inference*. To achieve an output posterior distribution $p(\hat{y}|x)$ as close as possible to the unknown, true, posterior distribution $p(y|x)$ of the Higgs production mode classes, Neural Networks are trained by iteratively fitting their weights and biases of all nodes, referred to as the Neural Network parameters θ , to known input data and target data (training data). The training procedure itself will be discussed in detail in Chapter (7).

In the case of $H \rightarrow 4l$ production mode classification, the output vector of the Neural Network $\hat{y}$ is desired to represent the probabilities, with which the Neural Network predicts, that a Higgs candidate event was produced by each of the Higgs production modes or the ZZ^* background, where each index relates to a given Higgs production mode. To achieve this, the format of the output layer must be chosen accordingly. Additionally, an activation function must be chosen with a range of $[0, 1]$. For event categorizations with exactly two contributing Higgs/ background processes, the *Sigmoid* function $\sigma(x)$ fulfills these requirements (see Figure 6.2):

$$\sigma(a_{out}) = \frac{1}{1 + \exp(-a_{out})}, \quad (6.2)$$

where a_{out} is the outgoing weighted sum for each of the final layer nodes. In event categorizations where more than two Higgs and/or background processes must be distinguished from another, it must further be required that the Neural Network outputs for each of the Higgs modes (or ZZ^* background) sum up to one to represent a proper probability distribution. This is realized with the *Softmax* function, which gives an output for the c -th class of:

$$\text{softmax}(a)_c = \frac{\exp(a_c)}{\sum_{j=1}^{N_{\text{Classes}}} \exp(a_j)}, \quad (6.3)$$

with the outgoing weighted sum a_c of output node at the index of the c -th class.

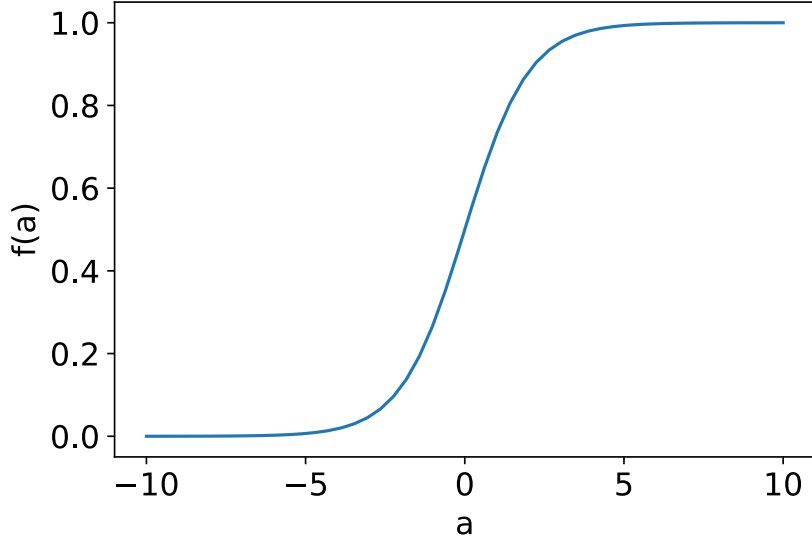


Figure 6.2.: The curve of the sigmoid function depending on the outgoing weighted sum a of the output node.

6.1. Analysis Requirements on the Neural Network Architecture

In order to employ a Neural Network for the classification of Higgs production modes in $H \rightarrow ZZ^* \rightarrow 4l$ decays using reconstructed data, a suitable Neural Network architecture must be chosen. The model should fulfill two major requirements.

Firstly, kinematic properties of the reconstructed events originating from different Higgs production modes in the $H \rightarrow 4l$ decay channel provide key information for the discrimination between the Higgs production modes and the main ZZ background. Since the jets originate from the Higgs production and leptons from the decay, no correlation is expected between these two types of reconstructed objects, motivating the introduction of two separate networks - the lepton and the jet network. Secondly, since the models are trained with different event categories (see Section 5), the Neural Network architecture should be able to handle input sets of varying size of leptons and jets. Two different architectures which fulfill these requirements are compared in this analysis: a *Recurrent Neural Network* (RNN) approach [42] and a *Deep Sets* approach [zaheer_deep_2018]. Lastly, as leptons and jets are very specific physical objects, that are produced separately from each other, the Neural Network architecture should reflect that fact. Because of this, the input variables used for the training are split into three groups: a lepton variable group with transverse momentum p_T^{lep} and pseudorapidity η^{lep} of the leptons, a jet variable group with transverse momentum p_T^{jet} and pseudorapidity η^{jet} of the jets, and a group for the remaining, "global" input variables, such as the dijet mass m_{jj} or the transverse momentum p_T^{4l} of the four lepton system. Three different Neural Networks are trained

separately for each of these input groups, a lepton-, jet- and global-Neural Network, the outputs of which are then further processed by another, final Neural Network, which then produces final output, as shown in Figure (6.3).

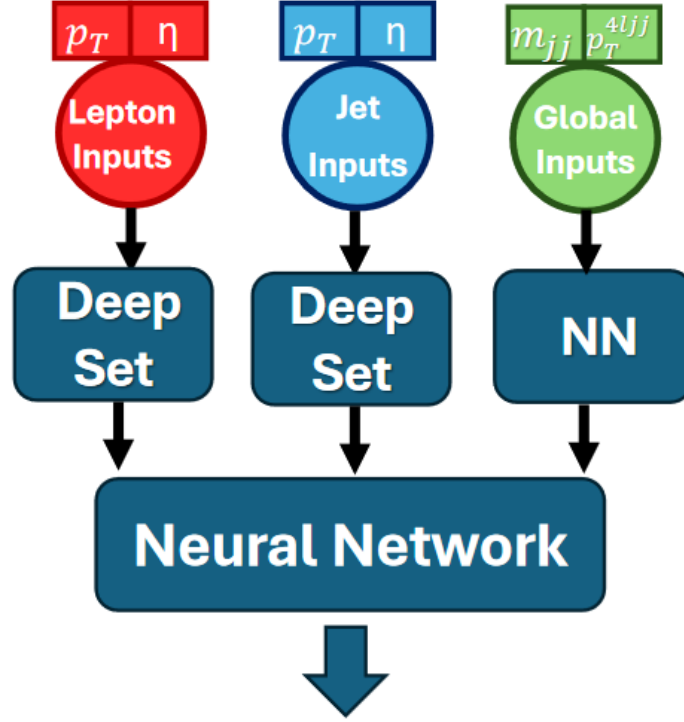


Figure 6.3.: Schematic of Neural Network architecture for the classification of $H \rightarrow ZZ^* \rightarrow 4l$ candidate events.

In the following, two Neural Network architectures that are suitable for the lepton- and jet-networks are introduced.

6.2. RNN-based Analysis

6.2.1. Recurrent Neural Networks

The architecture of Recurrent Neural Networks (RNNs) was developed specifically for the analysis of sequential data [36] [42]. In case of the $H \rightarrow ZZ^* \rightarrow 4l$ analysis, the lepton and jets can be ordered in a sequence using criteria such as their transverse momenta. Since RNNs allow for arbitrary sequence lengths during training and inference, they can be applied to event categories with varying jet multiplicity. An RNN architecture has been employed previously in Run 2 for the classification of $H \rightarrow 4l$ decay candidates according to the production mode. For the current Run 3 $H \rightarrow 4l$ data, a new RNN-based framework is currently being

developed [32].

The characteristic feature of Recurrent Neural Networks is that they take a sequence of inputs one after the other. The model then "remembers" information from previous inputs when processing later inputs of the sequence. This is achieved by introducing the so called hidden state, a vector of fixed length which stores the relevant information. This "memory vector" serves as an additional input to the network and is updated for every input element of the input sequence. For the inference and during each input step, the RNN generates an output that is a function of the current hidden state. This hidden state is calculated by a feed-forward Neural Network. The updated hidden state is then forwarded into another Neural Network for the output prediction. An overview of this process is shown in Figure 6.4. A simple three-layer implementation of an RNN with an input layer, a hidden state and an output layer, is thus described by:

$$h_t = \sigma_h(x_t W_h + h_{t-1} H_h + b_h), \quad (6.4)$$

$$\hat{y}_t = \sigma_{\hat{y}}(h_t H_{\hat{y}} + b_{\hat{y}}) \quad (6.5)$$

where the first term describes the calculation of a new hidden state h_t in the t -th step, with the t -th set of inputs X_t , the weights for the input data W_h , the hidden state of the previous step h_{t-1} , the weights for the hidden state H_h and the bias of the hidden state b_h . The second line describes the calculation of the RNN output $\hat{y}_t$ of the t -th step, with the weights for the hidden state $H_{\hat{y}}$ and a bias for the RNN output $b_{\hat{y}}$. σ_h and $\sigma_{\hat{y}}$ are task-specific activation functions [36].

Since the previous hidden state is used in the calculation of the updated hidden state, information from earlier input elements can be passed on to future inference steps. The predicted output of the Recurrent Neural Network for the last element is therefore based on the data of the entire input sequence of a Higgs candidate event.

The final hidden state thus serves as a low-dimensional representation of the whole sequence. In the case of the $H \rightarrow 4l$ analysis, the sequence of lepton inputs is fed lepton-by-lepton into the lepton-RNN, where each reconstructed lepton in the final state represents one step in the input sequence. Similarly, the jet information is fed jet-by-jet into the jet-RNN with varying input sequence length depending on the jet multiplicity. A single input from a final state lepton or jet consists of their transverse momentum and pseudorapidity.

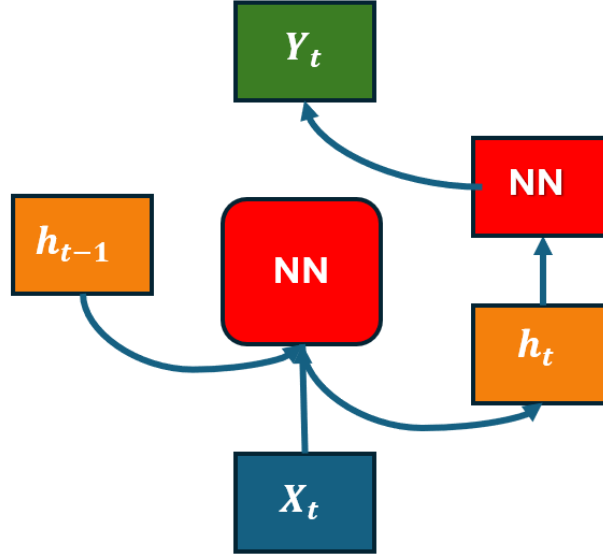


Figure 6.4.: Schematic of a simple Recurrent Neural Network. h_t and h_{t-1} are the hidden states of the current and previous input step, respectively. X_t is the t -th set of inputs and $\hat{y}_t$ the calculated RNN output.

6.2.2. Gated Recurrent Units

The previously described basic RNN architecture has been shown to be difficult to train such that it grasp more complex relationships between different elements of the sequential data [43]. Since early input elements within the input sequence are passed through the hidden layer many times there is much higher risk that the key information of these elements get lost and contribute less to the output compared to more recent elements in the sequence. This can cause the gradients in the neural network parameter space to become increasingly smaller during the training, which generally uses gradient-based algorithms, inhibiting the further training of the model. This is generally referred to as the vanishing gradients problem.

In the context of the $H \rightarrow 4l$ event classification, it is important that the features of the entire lepton and jet input sequences, is accounted for in the training. One possibility to account for the long-range dependencies within the sequence is to control of the information flow via trainable gates within the Neural Network. The gates are installed to decide what information is important to be passed to the next step.

One such extension of the RNN architecture is the so called Gated Recurrent Unit (GRU) [43], which uses two gates for the information flow control (see 6.5): the reset gate r_t and the update gate z_t , defined by

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r), \quad (6.6)$$

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z). \quad (6.7)$$

For the t -th set of input data x_t and the previous hidden state h_{t-1} , the first line defines the output of the reset gate r_t , while the second line defines the output of the update gate in the t -th in the t -th step, where W_r , U_r , b_r and W_z , U_z , b_z refer to the weights for the input data, the weights for the previous hidden state and the bias for the reset and update gate, respectively and σ is the Sigmoid function. Instead of directly calculating a new hidden state h_t , a candidate hidden state $\tilde{h}_t$ is first constructed in a GRU with

$$\tilde{h}_t = \tanh(W_{\tilde{h}} x_t + U_{\tilde{h}}(r_t \odot h_{t-1}) + b_{\tilde{h}}), \quad (6.8)$$

with the weights $W_{\tilde{h}}$ for the input dataset, weights $U_{\tilde{h}}$ of the element-wise multiplication between the reset gate and the previous hidden state. Here, the reset gate controls the impact of the previous hidden state vector h_{t-1} on the new candidate hidden state $\tilde{h}_t$. Subsequently, the previous hidden state h_{t-1} is updated with information from the candidate hidden state $\tilde{h}_t$ to form the new hidden state h_t . The update gate z_t determines, how much information from $\tilde{h}_t$ is allowed to be transmitted to h_t [43]:

$$h_t = (1 - z_t) \odot \tilde{h}_t + z_t \odot h_{t-1}. \quad (6.9)$$

Important context information can thus be protected from being overwritten in an update step while still accounting for the key information from new inputs.

In, GRUs have been shown to outperform RNNs in many applications in terms of performance and trainability and also perform well in comparison to other RNN-extensions, such as e.g. Long-Short Term Memory (LSTM) Neural Networks [43]. For that reason, GRUs are used in the RNN-based implementation instead of basic RNNs. In the following, the term "RNN" refers to the GRU implementation of a Recurrent Neural Network.

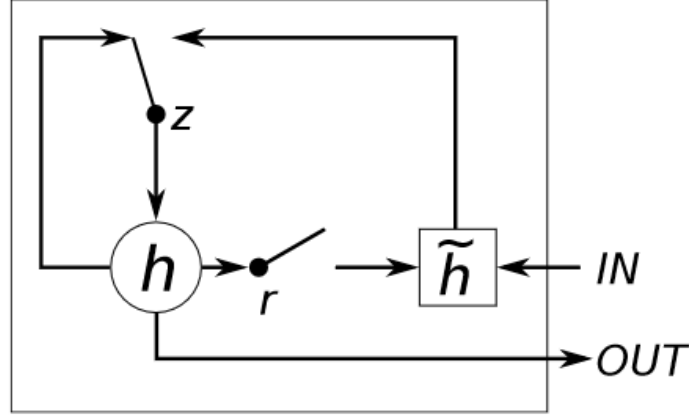


Figure 6.5.: Schematics of a Gated Recurrent Unit (GRU). h and $\tilde{h}$ are the hidden state and the candidate hidden state, respectively. r and z are the reset- and update functions, respectively [43]

6.3. Deep Set Based Analysis

6.3.1. RNNs and causality

The RNN based architecture is a natural candidate for the $H \rightarrow 4l$ classification task because of its ability to handle varying jet input sequence lengths. However, it also introduces assumptions which present a bias during the training of the Neural Network. The RNN treats the lepton and jet information input sequences with a causal direction as illustrated in Figure 6.6. Earlier input elements in the sequence, in this case reconstructed leptons, inform subsequent elements in the sequence, but not the opposite. This reflects, however, not necessarily a property of the trained physical system.

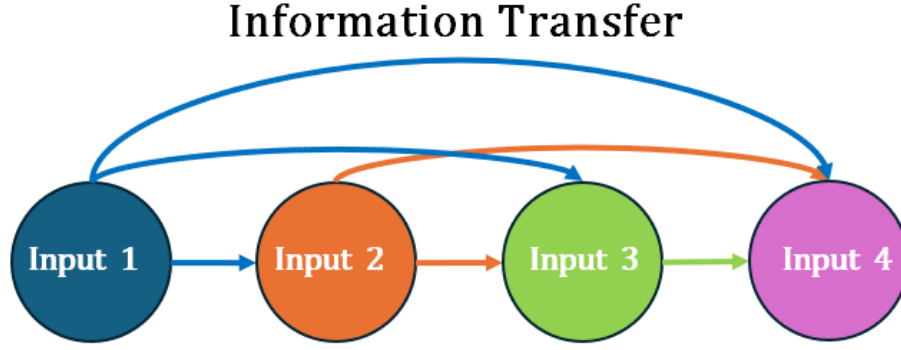


Figure 6.6.: Schematic of the information flow via causal connections between elements of an input sequence as processed by an RNN.

There is no causal direction among the leptons and the jets created in an $H \rightarrow 4l$ event. For example, the two leptons produced from the on-shell Z boson decay are created simultaneously. Consequently, there is no obvious preference which of the two leptons should inform the other in the architecture. The choice of order in the input sequence is arbitrary. An alternative Neural Network architecture without such a sequential causality may thus be a better option.

As on the other hand the sequential ordering of the data may help to account for inherent physical properties of the data, these properties may also be taken into account by introducing them explicitly as so-called *inductive bias* explicitly to the architecture, which opens the possibility to account for symmetries in the underlying data, as described in the following.

6.3.2. Symmetry in Neural Networks

It is often desirable to embed the known inherent symmetries in the data into the Neural Network architecture to ensure better convergence of the Neural Network towards well performing solutions [44]. The two most common cases of symmetry in Neural Networks are *invariance* and *equivariance*. Invariance is the property that a certain transformation $g(X)$ on the input sequence X does not change the model output $f(x)$:

$$f(g(x)) = f(x) | x \in X. \quad (6.10)$$

Equivariance is the property that a certain transformation g yields the same final result when applied to the input as compared to applying it to the output:

$$f(g(x)) = g(f(x)). \quad (6.11)$$

In the particular case of the $H \rightarrow 4l$ classification, the freedom in the choice of lepton and jet ordering can be considered as invariance under transformations that change the order of the inputs, referred to as *permutation invariance*. Permutation invariance plays an important role in many Neural Network tasks, which led to the development of architectures that are designed to account for this symmetry. Graph Neural Networks (GNNs) or Point Nets are examples of such architectures designed for graph-like and point cloud-like data. Here the more generic Deep Sets approach is considered, which is applicable to any data that can be represented by unordered sets [44].

6.3.3. Deep Sets Neural Networks

As a proof of concept of the permutation invariant Neural Network approach to the $H \rightarrow 4l$ classification, a simple NN architecture adopted for permutation-invariant datasets has been tested as an alternative to the RNN approach, a Deep Sets architecture [12].

The Deep Sets architecture consists of three parts. First, each element of the input sequence is processed by a preliminary dense Neural Network to obtain a more informed representation $\phi(x_l)$ of the l -th input set in the sequence that is already aware about key correlations of the data. The same preliminary Neural Network is shared for the processing of all elements of the input sequence, to ensure that each element is processed in the same way independent of its position in the sequence [44].

Subsequently, permutation invariance is introduced to the model by a permutation invariant function that combines the information of all representations $\phi(x)$. Typical choices for the permutation invariant function, also called aggregation function, are summation, the average value (average-pooling) or the maximum value (max-pooling) [44]. The model used in this analysis employs summation as aggregation function. For the example of a sequence of four lepton input sets, the pooling function would yield:

$$f_{aggr}(\phi(x_1), \phi(x_2), \phi(x_3), \phi(x_4)) = \phi(x_1) + \phi(x_2) + \phi(x_3) + \phi(x_4) \quad (6.12)$$

where $\phi(x_1), \phi(x_{t2}), etc.$ is the representation of first, second, etc. set of lepton input data (p_T^{lep} and η^{lep}) in the sequence. In the final step, the combined representation of all sequence members given by the aggregation function is passed through an additional dense Neural Network to produce the output.

The implementation of the Deep Sets-based Neural Network for $H \rightarrow 4l$ classification is designed analogously to the previously described RNN-based implementation: the separate treatment of lepton and jet information, a dense Neural Network for global input variables and a final dense Neural Network for the generation of the output remain the same. The lepton and jet RNNs are replaced by Deep Set models.

7. Deep Sets Optimization of the $H \rightarrow 4l$ Event Categorization

Neural Networks are trained by fitting the Neural Network parameters θ (weights and biases) to a set of known input and output data (training data) by minimizing the so called *loss* [38] [39] [40] [41]. The loss is a metric of the accuracy of the Neural Network that signifies how large the difference between the Neural Network output and the true targets of the dataset is, as described by the *loss function*. As such, a smaller loss is indicative of a better performing Neural Network. The choice of loss function depends on the type of task the Neural Network is trained for. A common loss function choice used in classification tasks with more than one class in the event categorization is the *Categorical Cross Entropy Loss*, $L_{(X,\theta)}$, as defined by:

$$L_{(X,\theta)} = -\frac{1}{N} \sum_{i=0}^N \sum_{c=0}^C y_{i,c} \log \hat{y}_{i,c}, \quad (7.1)$$

where, in the case of $H \rightarrow 4l$ production mode classification, N is the number of events in a training dataset, C is the number of production modes (or ZZ^* background) in the event categorization, $y_{i,c}$ is an index that is 1 if c matches the true Higgs mode of an event i and 0 otherwise and $\hat{y}_{i,c}$ is the Neural Network output for the Higgs mode c for an event i . For an event categorization with two classes, also known as *binary classification*, $y_{i,c}$ and $\hat{y}_{i,c}$ can be simplified to y_i and $\hat{y}_i$, where $y_i = 1$ represents events from the targeted Higgs mode and $y_i = 0$ represents events from the background if the event. This yields the *Binary Cross Entropy Loss*,

$$L_{(X,\theta)} = -\frac{1}{N} \sum_{i=0}^N (y_i \log \hat{y}_i + (1 - y_i) \log (1 - \hat{y}_i)). \quad (7.2)$$

Loss minimization is typically performed with methods based on the *Gradient Descent* [39] approach, in which the parameters θ of the Neural Network model are updated according to the gradient $\nabla L_{(X,\theta_i)}$ of the loss function,

$$\theta_{i+1} = \theta_i - \tau \nabla_{\theta_i} L_{(X,\theta_i)} \quad (7.3)$$

where θ_i are the parameters in the i -th update step of the Neural Network parameters, or *iteration*, respectively, $L_{(X,\theta)}$ is the loss function for a dataset X and Neural Network parameters

θ_i , τ is a unit-less scaling factor of for the gradient of the loss function, which defines the size of the update steps, generally referred to as *learning rate*, and θ_{i+1} are the updated Neural Network parameters.

As the datasets used in the training of Neural Networks tend to be very large, calculating the gradient of the loss function with regards to the Neural Network parameters over the entire dataset is very demanding in terms of computational resources. To reduce the necessary computational resources, the gradient update shown in Equation 7.3 is not performed over the entire training dataset, but on a smaller subset sampled from the training dataset, referred to as *mini – batch*, which is resampled after every iteration. As such, Gradient Descent when using mini-batches is also referred to as *Mini Batch Gradient Descent*. Events in a mini-batch sampled from the training data can not be sampled again, until the entire training dataset is exhausted. The number of training iterations necessary to exhaust the training dataset is generally known as *epoch*.

Since the mini-batch can be much smaller than the training set, the uncertainty in the estimate of the loss gradient increases, leading to larger deviations from the "true" gradient value. This is, however, not necessarily a disadvantage. These random deviations can serve as an explorative mechanism in the optimization process helping the model to escape regions in which the true gradient approaches zero, such as local minima or saddle-points. The batch size has a similar effect as the learning rate, as it determines whether the optimizer acts more exploratively or conservatively during a training iteration. This makes the batch size an important hyperparameter to consider during the training.

Multiple batch size configurations have been tested for the training of the $H \rightarrow 4l$ Deep Set Neural Networks. In general, choosing a batch size of 16 events for the first training iteration and then continuing the training with batch sizes of 32 and 64 events over 200 epochs for each for "fine-tuning" demonstrated very good performance and training speed results in the case of the Deep Set optimization.

Gradient Descent does not generally ensure convergence towards global minima of the loss function in the space of the Neural Network parameters, but only convergence towards local minima [39]. Therefore, great care has to be taken during the training of the Neural Network to achieve convergence towards robust solutions with low loss. The training method and choice of so-called hyperparameters, which refers to variables in the model and training configuration, that are not automatically updated via the optimization algorithm in the training procedure, but manually chosen, impact the learning efficiency.

These choices will be discussed in detail in this chapter, focusing on the $H \rightarrow 4l$ classification task.

To compare the performance of the Deep Set based classification and the previously intro-

duced RNN architecture employed for Run 2 data and to evaluate the differences between using Run 2 and Run 3 data, Deep Set Neural Networks are trained separately for all Run 2 and Run 3 reconstructed event categories defined in Section 5.

All optimization studies are performed in the open-source Python-based machine-learning framework PyTorch [45]. PyTorch offers support for a large variety of building blocks for deep learning that are used in the development of the Deep Set models. Furthermore, the library PyTorch Geometric, specifically developed for data with irregular structures, is used motivated by the set-like structure of the lepton and jet data in the $H \rightarrow 4l$ events [46]. In addition to the PyTorch format, the optimized Deep Set models were exported to the Open Neural Network Exchange (ONNX) format [47]. This format allows the Neural Networks to be transferred between different frameworks simplifying the implementation into different software environments, such as the common $H \rightarrow 4l$ event selection software framework .

7.1. $H \rightarrow 4l$ Input Data

7.1.1. Input variables for the $H \rightarrow 4l$ event selection

Since the kinematic properties of a $H \rightarrow 4l$ system can vary significantly depending on the reconstructed event category, different dedicated sets of input variables are chosen separately for each event category. A summary of these input variables is given in Table 7.1 and Table 7.2 for the Run 2 and Run 3 event categories. The complete set of distributions of these input variables is shown in Appendix A. It should be noted that while the event categories alter from Run 2 to Run 3 data, the lepton and jet input variables defining the respective event categories are kept the same. The main difference is in the set of global event variables, which depend on the targeted Higgs production mode of a given event category.

The inputs used in the training of the Deep Set Neural Networks are almost identical to the ones used for the training of the RNN models [32]. The only difference is an additional input variable, $\mathbb{1}_{Z_{off}}^{lep}$, for the Deep Set training on the lepton data associated to each lepton in order to distinguish between leptons from on- and off-shell Z boson decay:

$$\mathbb{1}_{Z_{off}}^{lep} = \begin{cases} 0, & \text{if lepton from on-shell } Z, \text{ and} \\ 1, & \text{if lepton from off-shell } Z. \end{cases} \quad (7.4)$$

The addition of this input variable is further discussed in Section 9.2.1.

The input variables are chosen to ensure high learning efficiency during the training of the Neural Network, providing high discrimination power between different Higgs production

modes. A brief discussion of these variables is given in the following. A selection of important kinematic variables is shown in Figure 7.1 and Figure 7.2, while the full sets of kinematic distributions for all event categories are shown in Appendix A.

The kinematic properties of the four leptons from the Higgs boson decay do not significantly differ between the Higgs production modes. Their transverse momenta p_T^{lep} and pseudorapidities η^{lep} provide discrimination power between the Higgs signal and the ZZ background. The lepton p_T in ggF events is boosted to higher values compared to ZZ events (see Figure 7.2c). The differences between signal and background are most pronounced for the lowest lepton p_T in an event, as these leptons are most likely to originate from the off-shell Z boson decay, as opposed to ZZ background, which is dominated by two on-shell Z bosons.

Besides the leptons, jet kinematic properties strongly depend on the Higgs production mode. Unlike in ggF production, jets mostly originate from the initial proton hard-scattering process in the VBF production mode and from the vector boson decay in the VH production mode. Thus, the jet kinematic properties provide distinct signatures for the different production modes. In particular, jet pseudorapidity provided discrimination between all three of the mentioned Higgs production modes (see Figure 7.1a). While the jets are produced much more centrally in the VH process compared to the ggF and VBF production modes, the VBF process tends to result in forward directed jets, which leads to a depleted central region as opposed to the ggF and VH production modes, as can be seen in the distribution of the angular distance between the four-lepton system and the leading jet, R_{ZZj} (see Figure 7.2a). In addition to the lepton and jet kinematics, there are several global input variables related to the full event topology. One such variable is the invariant mass of the two leading- p_T jets, m_{jj} (see Figure 7.1b). In event categories with two or more jets, this variable discriminates between the three Higgs production modes: VBF yields an m_{jj} distribution shifted towards higher mass values, while it peaks near weak vector boson masses for VH production. Jets in the VBF production mode also have a larger angular separation R_{ZZj} from the $ZZ^* \rightarrow 4l$ system compared to the ggF process (see Figure 7.2a). The transverse moment of the combined lepton- and jet-system p_T^{4lj} tends to be lower for ggF events compared to VBF and VH events (see Figure 7.1c). ggF and ZZ background events tend to contain jets with lower transverse momenta p_T^{jet} compared to VBF and VH . This is especially the case for the lowest- p_T jet p_T^{jet3} in an event (see Figure 7.1d). The most important signature to separate ZZ background from the Higgs production, especially the dominant ggF mode in the $0 - Jet$ and $1 - Jet$ event categories, is the kinematic discriminant D_{ZZ} , which is defined as

$$D_{ZZ} = \log\left(\frac{\mathbb{M}_{HZZ}}{\mathbb{M}_{ZZ}}\right), \quad (7.5)$$

where $\mathbb{M}_{HZZ}$ and $\mathbb{M}_{ZZ}$ are the matrix elements for the $H \rightarrow ZZ^*$ and the $ZZ \rightarrow 4l$, respectively for the transverse momenta of the final state leptons of the given $H \rightarrow 4l$ candidate event. D_{ZZ} is a measure of the relative likelihoods of observing an $H \rightarrow ZZ$ or a ZZ background event. The D_{ZZ} distribution for ZZ background events is shifted to lower values

compared to the distribution for the Higgs signal (see Figure 7.2b).

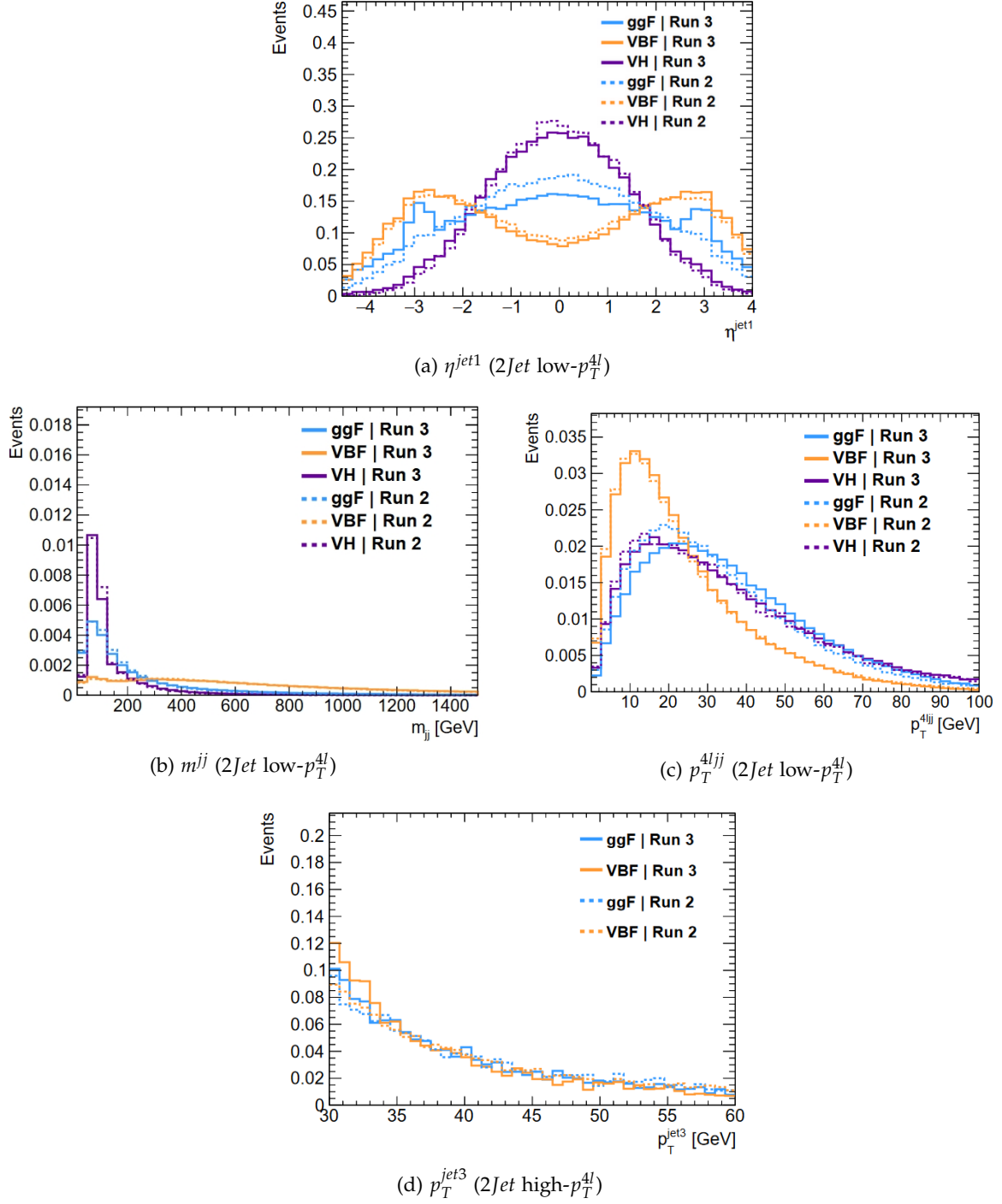


Figure 7.1.: Distributions of kinematic input variables with highest discrimination power between signal and background in the 2 – Jet event categories. The variables are explained in the text.

7. Deep Sets Optimization of the $H \rightarrow 4l$ Event Categorization

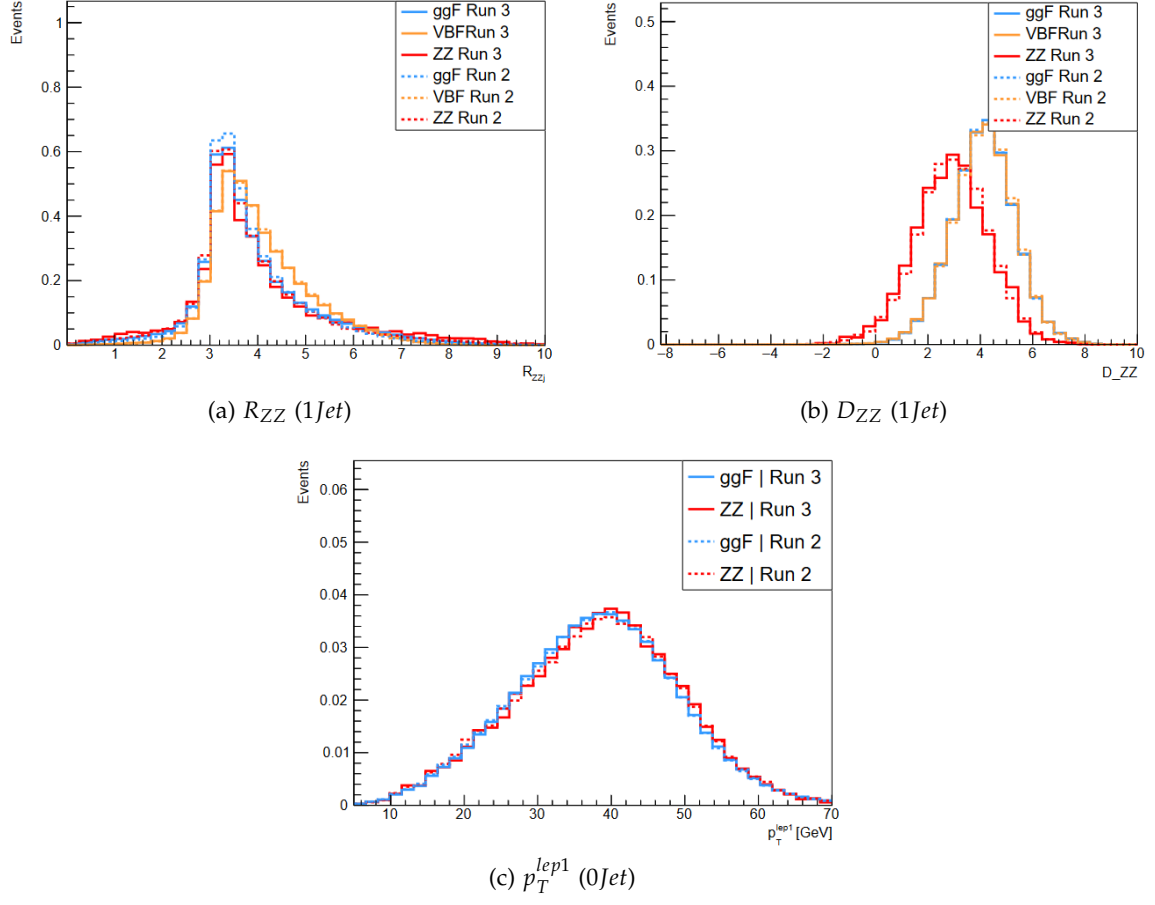


Figure 7.2.: Distributions of kinematic input variables with highest discrimination power between signal and background in the 1 – Jet and 0 – Jet event categories. The variables are explained in the text.

Table 7.1.: Summary of the input variables used in the training of the Deep Set Neural Networks for the Run 3 $H \rightarrow 4l$ analysis. The targeted Higgs production mode treated as signal is shown in red.

Reconstructed Event Category	Target Processes	Global Event Inputs	Lepton properties	Jet properties
0 – Jet 4l	ggF ZZ	$p_T^{4l}, m_{1,2}, m_{3,4}, D_{ZZ}, \cos \theta_1, \cos \theta^*, \phi_{ZZ}$	$p_T^{lep,i}, \eta^{lep,i} (i = 1, \dots, 4)$	
0 – Jet 2l2l	ggF ZZ	$p_T^{4l}, m_{1,2}, m_{3,4}, D_{ZZ}, \cos \theta_1, \cos \theta^*, \phi_{ZZ}$	$p_T^{lep,i}, \eta^{lep,i}$	-
1 – Jet low- p_T^{4l}	VBF ggF ZZ	$p_T^{4l}, \Delta R_{ZZj}, D_{ZZ}, p_T^{jet}, \eta^{jet}$	$p_T^{lep,i}, \eta^{lep,i}$	- -
1 – Jet medium- p_T^{4l}	VBF ggF ZZ	$p_T^{4l}, \eta^{4l}, \Delta R_{ZZj}, D_{ZZ}, E_T^{Miss}, p_T^{jet}, \eta^{jet}$	$p_T^{lep,i}, \eta^{lep,i}$	-
2 – Jet low- p_T^{4l}	VH VBF ggF	p_T^{4ljj}, m_{jj}	$p_T^{lep,i}, \eta^{lep,i}$	$p_T^{jet,i}, \eta^{jet,i} (i \geq 2)$
2 – Jet high- p_T^{4l}	VBF ggF	$p_T^{4ljj}, \eta_{ZZ}^{Zep}$	$p_T^{lep,i}, \eta^{lep,i}$	p_T^{jet}, η^{jet}

 Table 7.2.: Summary of the input variables used in the training of the Deep Set Neural Networks for the Run 2 $H \rightarrow 4l$ analysis. The targeted Higgs production mode treated as signal is shown in red.

Reconstructed event category	Target processes	Global event inputs	Lepton properties	Jet properties
0 – Jet	ggF ZZ	$p_T^{4l}, m_{1,2}, m_{3,4}, D_{ZZ}, \cos \theta_1, \cos \theta^*, \phi_{ZZ}$	$p_T^{lep,i}, \eta^{lep,i} (i = 1, \dots, 4)$	
1 – Jet p_T^{4l} -inclusive	VBF ggF ZZ	$p_T^{4l}, \eta^{4l}, \Delta R_{ZZj}, D_{ZZ}, E_T^{Miss}, p_T^{jet}, \eta^{jet}$	$p_T^{lep,i}, \eta^{lep,i}$	-
2 – Jet low- p_T^{4l}	VH VBF/ggF	p_T^{4ljj}, m_{jj}	$p_T^{lep,i}, \eta^{lep,i}$	$p_T^{jet,i}, \eta^{jet,i} (i \geq 2)$
2 – Jet high- p_T^{4l}	VBF ggF	$p_T^{4ljj}, \eta_{ZZ}^{Zep}$	$p_T^{lep,i}, \eta^{lep,i}$	$p_T^{jet,i}, \eta^{jet,i}$

7.1.2. Pre-Processing of Input Data

Before the input data is used in the training of a Neural Network, it is transformed into a format more suitable for the model. This step is crucial for improving the performance and accuracy of the Neural Network.

One of the common pre-processing techniques is a simple so called Z-score normalization [48], which rescales the variables according to

$$x_{i,j}^{norm.} = \frac{x_{i,j} - \mu_i}{\sigma_i}, \quad (7.6)$$

where $x_{i,j}$ is the j -th data point of the i -th input variable with a mean value μ_i and variance σ_i , such that the normalized values have a mean value equal to zero and a variance equal to one.

This normalization is applied to the input variables of the lepton- and jet Deep Set networks, as well as the network of the global variables. It leads to noticeably better convergence during the training of the network, as shown in Figure 7.3 for the Run 3 data in the $0 - Jet$ event category.

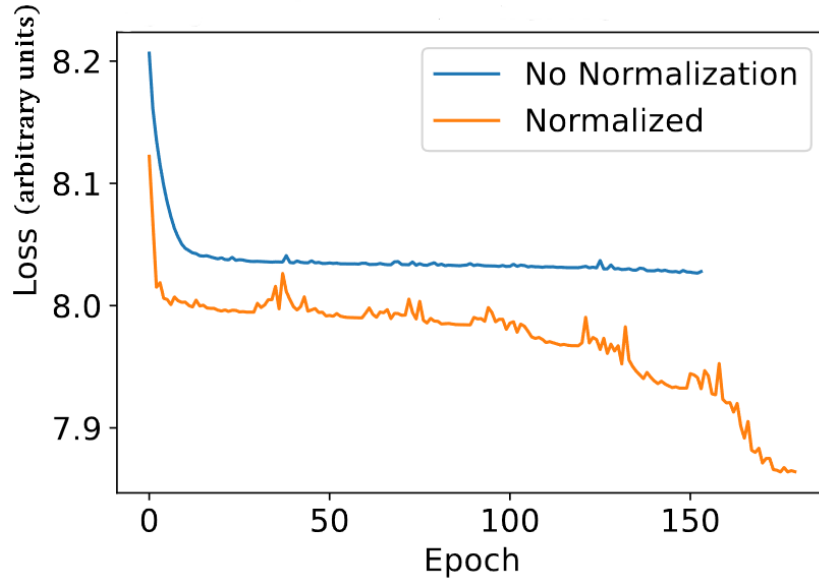


Figure 7.3.: Evolution of the loss function value over epoch during the training of the Higgs classification in the $0 - Jet$ event category for Run 3 data. The blue (orange) curve represents training on input variables with (without) a prior application of Z-score normalization. The training was performed with a learning rate τ of 10^{-5} and a batch size equal to 16 events. The value of the loss function is minimized by processing the entire training data in several passes (epochs), during which the weights of the Neural Network are refined. If a Z-score normalization is applied on the input variables, a smaller, i.e. better performing, loss value can be achieved compared to the loss without Z-score normalization.

7.2. Optimization of the Deep Set Training Setup

7.2.1. Optimizer Algorithm

As discussed previously, Neural Network parameters (weights and biases) are generally optimized in the training process by minimizing the loss by means of an optimizer algorithm, such as, the a Gradient Descent method (see Equation 7.3). The goal of the optimization is to descend to the minimum value of the loss function in the network parameter space. Beyond the basic Gradient Descent, several other optimizer algorithms have been developed to improve the speed and stability of the training [49]. These are discussed briefly in the following:

Adaptive Moment Estimation (ADAM) combines the moment and adaptive learning rate extensions of the basic Gradient Descent algorithm [50]:

Moment: If successive optimizer steps point in a similar direction, the optimization proceeds in increasingly larger steps. This can be seen as an acceleration of the optimizer as it continues to move in a certain direction. This is achieved by also taking gradients of previous training iterations into account during the Neural Network parameter update, which modifies Equation 7.3 to:

$$\theta_{i+1} = \theta_i - \tau m_i \quad (7.7)$$

$$, m_i = \beta m_{i-1} + (1 - \beta) \nabla L_{(x, \theta)}, \quad (7.8)$$

where, i -th training iteration, the Neural Network parameters θ_i are updated by the so called moment m_i , which is a combination of the moment of the previous training iteration m_{i-1} and the current gradient of the loss function $\nabla L_{(x, \theta)}$, where the hyperparameter $\beta < 1$ determines the relative contributions of the two terms in the moment update step. As such, the impact of previous gradients on future θ updates decreases exponentially with regards to β with increasing training iteration number.

This extension to the θ update step does not only speed up the descent towards minima of the loss function in the network parameter space, but also helps the model to escape from unfavorable local minima [50].

Adaptive Learning Rate: In the standard Gradient Descent approach, a fixed, common learning rate τ is used for all network parameters. This is not necessarily ideal in terms of convergence speed. The loss might tend to decrease rather steeply along some directions in the parameter space, which may call for lower learning rates compared to other directions to avoid an overshoot of a potential minimum. In contrast, for updates in directions where the loss is only slowly changing, the learning rate could be increased to speed up the optimization. The adaptive learning rate approach addresses this by scaling the learning rate for each network parameter separately based on the magnitudes of preceding gradients for that parameter[50].

Compared to the basic Gradient Descent approach for the training of the Deep Set Neural Networks, the ADAM optimizer approach results in faster convergence towards well performing network parameters with regards to minimization of the loss. This is shown in Figure 7.4 on an example for the $2 - Jet$ low- p_T event category, comparing Mini Batch Gradient Descent and ADAM optimizers for the training of the Deep Set Neural Network.

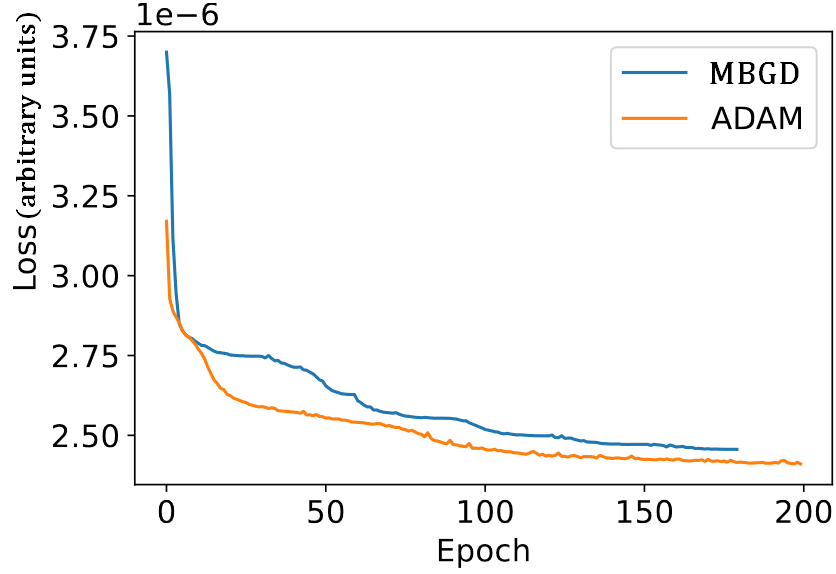


Figure 7.4.: Evolution of the loss value over epochs during the training of the Higgs event classification in the 2 – Jet low- p_T category for Run 3 data with Mini Batch Gradient Descent (MBGD)(blue) and ADAM (orange) as optimizer. The training was performed with a learning rate τ of 10^{-5} and a batch size equal to 16 events.

7.2.2. Learning Rate Scheduling

The choice of the learning rate hyperparameter value is vital for the success of Neural Network training. Learning rates that are too large can lead to the oscillation or even divergence of loss values during training, while learning rates that are too small will slow down the convergence or even stall it completely if the optimization is unable to escape a local [51].

Even with the ADAM optimizer that utilizes adaptive learning rates, which often improve the convergence towards well performing parameter ensembles, of the Neural Network, the training of the $H \rightarrow 4l$ classification with Deep Set Neural Networks converged very slowly in some of the event categories such as in the case of the 0 – Jet case for Run 3 data.

Another approach to address the problems of static learning rates is to cyclically vary the size of the learning rate over time by means of a so called learning rate scheduler. While this may seem as a redundant measure at first to use in combination with an adaptive learning rate optimizer, it has been shown that in certain scenarios, cyclical learning rates significantly speed up the training with optimizers like ADAM [52].

For the optimization of the $H \rightarrow 4l$ classification, the *Cosine Annealing Warm Restarts* (CAWR) scheduler [53] has been employed as a simple cyclical scheduler with CAWR, the learning

rate τ_t decreases over a certain number of epochs from a maximum τ_{max} to a minimum τ_{min} after a fixed number of epochs T_{rest} is reached, the learning rate τ_t is restored to τ_{max} and the cycle is restarted. In detail, the learning rate is scheduled according to:

$$\tau_t = \tau_{min} + \frac{1}{2}(\tau_{max} + \tau_{min})(1 + \cos \frac{T_{cur}}{T_{rest}}\pi), \quad (7.9)$$

where τ_t is the current learning rate and T_{cur} is the current number of epochs since the last restart. During each period T_{rest} , the usual evolution of the learning rate takes place. Compared to the case without scheduler, a visible improvement has been observed using CAWR for the Deep Set Neural Networks, for the $2 - Jet$ low- p_T event category for Run 3 data, as shown in Figure 7.6.

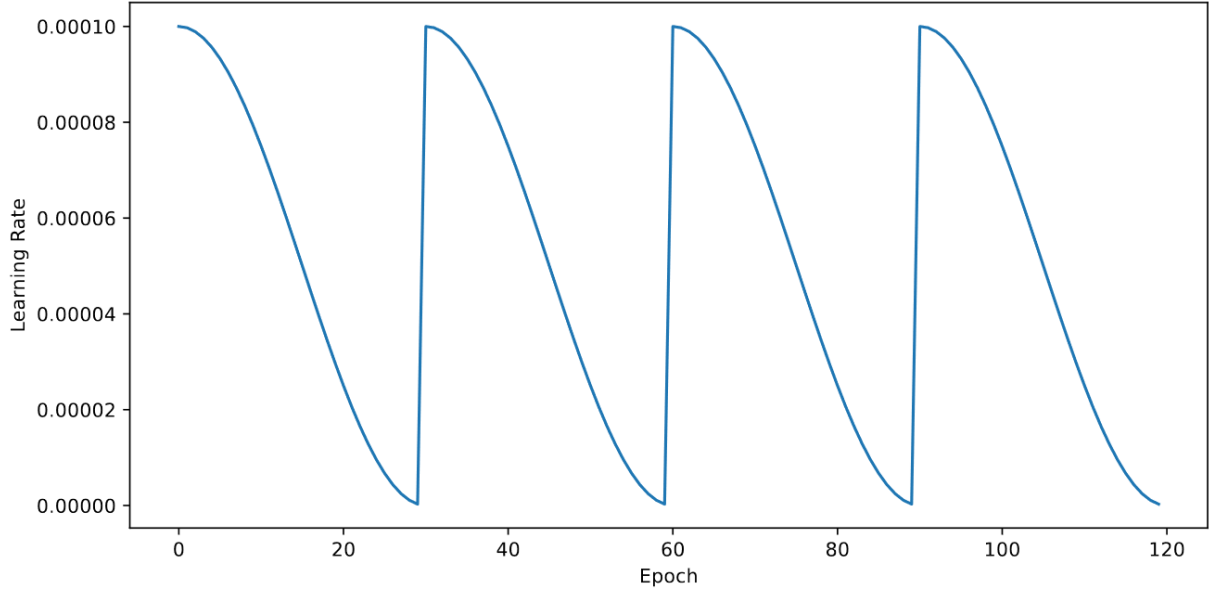


Figure 7.5.: Evolution of the learning rate over epochs obtained when using a Cosine Annealing Warm Restarts scheduler with $\tau_{max} = 10^{-4}$, $\tau_{min} = 10^{-7}$ and $T_{rest} = 30$.

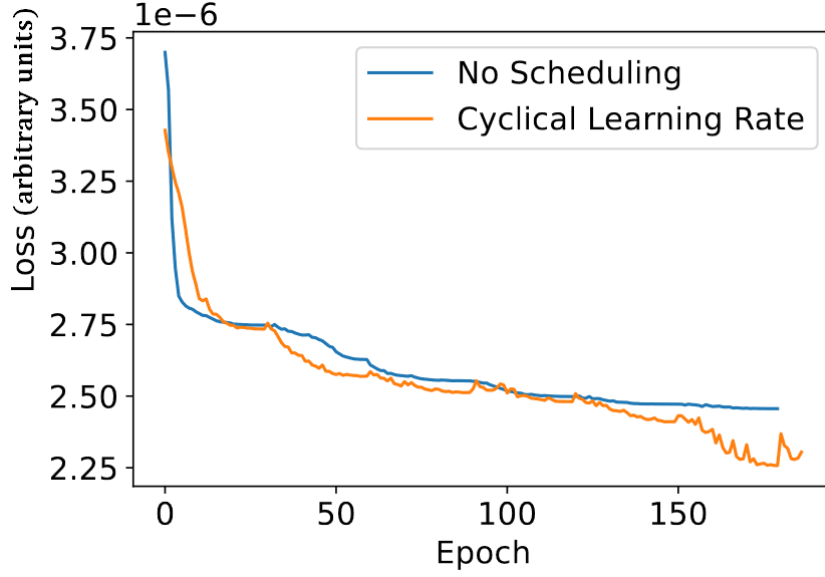


Figure 7.6.: Evolution of the loss over epochs for the training of the 2-Jet low- p_T event category for Run 3 data with (orange) and without (blue) the use of Cosine Annealing Warm Restarts (CAWR) learning rate scheduling. A fixed learning rate of 10^{-5} was used without CAWR, while maximum and minimum learning rates of 10^{-4} and 10^{-8} and $T_{rest} = 30$ epochs were used with CAWR. The Neural Network was trained with a batch size of 16 events. The restarts significantly improved the loss almost each time.

7.2.3. Scaling of Data Samples

One additional caveat of Neural Network training is that some of the data features in the training dataset may be significantly over- or underrepresented compared to other features. This can lead to biased results, since the loss function will have too little or too much contribution from under- or overrepresented features, respectively. In particular, two sources of misrepresentation in the $H \rightarrow 4l$ candidate events have been addressed: Non-uniform covering of the kinematic phasespace with simulated Monte Carlo events and differences in the production rates for the different Higgs production modes.

Due to statistical limitations, the phase space coverage by the Monte Carlo generated events employed for the Neural Network training is not necessarily uniform. This means that some events will be under- or overrepresented. In order to correct for this effect, a dedicated weight is assigned to each Monte Carlo generated event. These weights are taken into account during the loss calculation. The effective size of the Monte Carlo sample entering the training this is given by the sum of weights for all events in the sample.

It also needs to be considered, that the rates for different Higgs boson production modes as well as for the ZZ background differ from each other by up to two orders of magnitude. Scaling the relative contributions of each process according to these production rates would therefore lead to underrepresentation of certain data features. To ensure that no production mode is underrepresented, the number events of all processes are rescaled to the same value in the training sample.

7.2.4. Over-Fitting and Regularization

One of the common potential problems in machine learning is the so called *over-fitting* scenario, in which a trained model has a very high selective performance on the training data, while performing very poorly on different data. This is caused when the training interprets the noise fluctuations as actual data features, memorizing this non-physical information instead of learning the relevant discriminating information. To monitor the risk of over-fitting, the available data is split into three subsets: A training set, a validation set employed to evaluate the model performance during training and a test set for the final evaluation of the model performance on independent data. In the current study, the Monte Carlo data is split into 60% for training, 20% for validation and 20% for the final result.

In order to minimize the risk of over-fitting, dedicated regularization techniques are used during training. A common regularization method is the implementation of so-called *Drop-Out* layers [54] in the the Neural Network architecture. During training, a Dropout layer will randomly disables preceding nodes during each iteration with a certain, fixed probability, the *Dropout rate*, by setting their output to zero. By dropping the nodes, the model becomes less complex and relies less on individual nodes, making it less prone to over-fitting. An illustration of the dropout-based regularization procedure is shown in Figure 7.7.

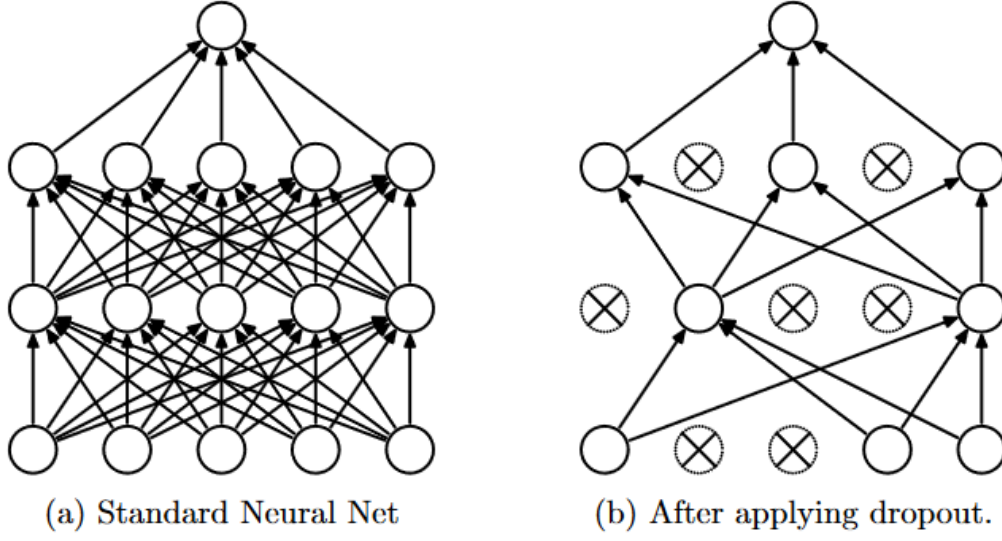


Figure 7.7.: Schematic of a Neural Network before a) and after b) the implementation of Dropout [54].

Another benefit of the Dropout regularization method is that it makes Neural Network more robust against slight changes in the input features. Neural Networks over-focusing on a certain input feature with high discrimination power can lead to large changes in the output even for small perturbations in the input features. Since Dropout layers will randomly block these features from impacting a training iteration, the training will become less sensitive to such perturbations.

To ensure robustness and regularization, Dropout layers have also been implemented in the training of the Deep Set Neural Networks for the $H \rightarrow 4l$ event classification in all categories. The improved robustness is observed to be especially beneficial for the $0 - Jet$ event category, as seen in a more evenly matched relative impacts on the Neural Network output from the input variables, as will be discussed in detail in Section 9.1.2.

7.3. Model Specification for the $H \rightarrow 4l$ Classification

A general overview of the Deep Set Neural Networks employed for the $H \rightarrow 4l$ event classification was given in Section 6.3.3. In this section the detailed specifications of these models in terms of the hyperparameter values and the choice in activation functions are discussed.

The complexity of a Neural Networks is predominantly determined by its number of layers and nodes per layer. The sub-networks of the Deep Set models are kept relatively light, some

only having two hidden layers with a small number of nodes. While testing variations of the depth and the amount of nodes, it was observed that additional complexity does not affect the performance of the models while significantly increasing the training time.

It was observed that the use of different hidden layer activation functions for the different sub-networks of the Deep Set model lead to the best performance. For the lepton and jet sub-networks, the Rectified Linear Unit (ReLU)

$$ReLU(x) = \begin{cases} 0, & \text{if } x \leq 0 \\ x, & \text{if } x > 0, \end{cases} \quad (7.10)$$

led to a good model performance after the training. Since the ReLU is also very undemanding activation in terms of computational resources, this choice positively affected the training time of the optimization runs [40].

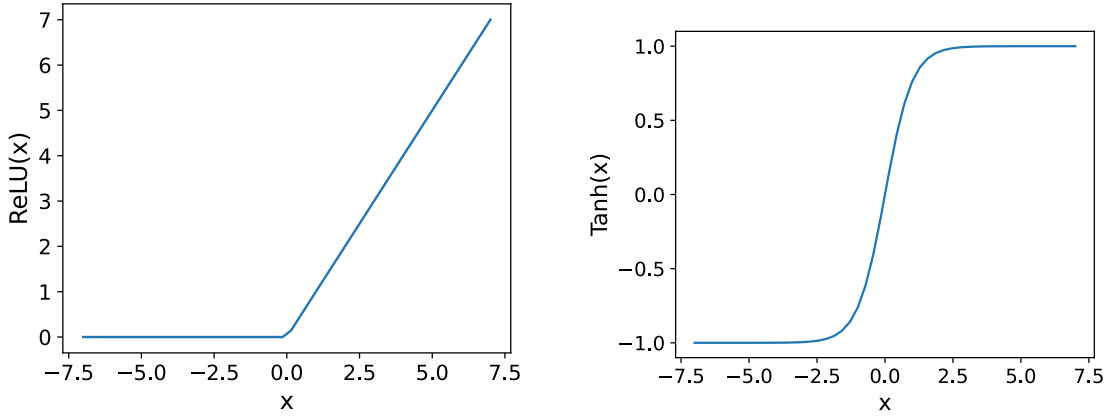


Figure 7.8.: Curves of the *ReLU* (left) and *Tanh* (right) Activation Functions

For the global and final Neural Networks, the tangens hyperbolicus $\tanh(x)$, another common activation function, was chosen instead. While the ReLU is less resource consuming during computation, their use in these sub-networks unfortunately led to performance degradation. This is because ReLUs can unintentionally "die" during training, outputting only zero. Since this leads to vanishing gradients, the gradient based optimization of the network stops. This is a well known problem in the Neural Network field and is generally referred to as *Dying ReLU* [55].

A full summary of the hyperparameters of the Deep Set Neural Networks and the training setups used in the optimization of the Deep Sets is given in Appendix B.

8. Cross Section Fitting Procedure

Measurements of the cross section of the Higgs production processes are an excellent window to the couplings of the Higgs boson. To obtain the Higgs production cross section in each event category (see Chapter 5), the separation of the targeted Higgs production mode from background processes by the Deep Set Neural Networks can be employed.

For the measurement of the product $\vec{\sigma} \cdot \mathcal{B}$ of the Higgs production cross section $\vec{\sigma}$ and the $H \rightarrow ZZ^*$ decay branching ratio $\mathcal{B}$ in each event categories, the Neural Network outputs NN of the production processes considered in each respective category (e.g. in the $1 - Jet$ category: NN_{ggF} , NN_{VBF} and NN_{ZZ}) can be employed as discriminant observables. $H \rightarrow 4l$ candidate events can then be separated into different discriminant bins, defined by the NN thresholds (e.g. bins for NN_{ggF} thresholds = $[0,0.2,0.4,0.6,0.8]$).

Resulting event counts for Monte Carlo generated data in the discriminant bins can then be fitted to the respective event counts for observed data to obtain the Higgs production cross section $\vec{\sigma} = \{\sigma_1, \sigma_2, \dots, \sigma_N\}$, where σ_p is the cross section of the p -th Higgs production mode [56]. A likelihood function $L(\vec{\sigma}, \vec{\Phi})$ depending on the Higgs production cross section and nuisance parameters $\vec{\Phi}$ is used, where nuisance parameters refer to fitting parameters not of primary interest in the likelihood model that need to be considered for the fit, which in this case account for the systematic uncertainties. The likelihood function used for the cross section fit is defined by:

$$L(\vec{\sigma}, \vec{\Phi}) = \prod_{i=1}^{N_{categories}} \prod_{j=1}^{N_{bins}} P(N_{i,j} | L \cdot \vec{\sigma} \cdot \mathcal{B} \cdot \vec{A}_{i,j}(\vec{\Phi}) + B_{i,j}(\vec{\Phi})) \times \prod_{m=1}^{N_{nuisance}} C_m(\vec{\Phi}) \quad (8.1)$$

with Poisson distributions P of the observed number of events $N_{i,j}$ in the i -th event category and j -th discriminant bin, with regards to the expectations for each background process $B_{i,j}(\vec{\Phi})$ and for the signal $S_{i,j} = L \cdot \vec{\sigma} \cdot \mathcal{B} \cdot \vec{A}_{i,j}(\vec{\Phi})$, where L is the integrated luminosity, and $\vec{A}_{i,j}(\vec{\Phi})$ are the signal acceptances for each Higgs production mode, defined as the fraction of events in an event category i that pass the event selection and reconstruction cuts presented in Section 4.3 and were generated in the event category j , relative to the total amount of events in the event category. Constraints on systematic uncertainties m are taken into account by functions $C_m(\vec{\Phi})$ of the nuisance parameters. The cross sections of the different production modes are fitted as independent parameters, which correlate across the different event cate-

gories i .

9. Results of the Deep Set based $H \rightarrow 4l$ classification

9.1. $H \rightarrow 4l$ Event Classification with Run 2 Data

9.1.1. Comparison of results using Deep Set and RNN architectures

In order to evaluate the performance of the Deep Set based-Neural Network architecture, it first needs to be compared to the performance of the reference classification using the RNN architecture. The RNN-based classification has only been fully optimized for Run 2 data so far. To allow for a fair comparison, the Deep Set architecture is also first trained on simulated Run 2 event samples. The performance of the two setups is evaluated using the same Run 2 event category definitions and for the same input variables - with the exception of the additional $\mathbb{1}_{Z_{off}}$ variable used for the Deep Set Neural Network (see Section 7.1.1). Following the comparison between the two Neural Network architectures with Run 2 data in the next two sections, an in-depth evaluation of the Deep Set Neural Network performance for Run 3 data will be given in Section 9.3.

The most important performance criterium for the Neural Networks discussed in this thesis is the signal-to-background separation, where here signal refers to the targeted Higgs production mode and background refers to the ZZ production introduced in Section 4.2 and other "contaminating" Higgs production modes. Due to the limited statistics of $H \rightarrow 4l$ events, it is vital for Higgs cross section measurements to preserve as much signal as possible while maintaining high background rejection. To distinguish between signal and background, a decision threshold is chosen. Events with a Neural Network output higher than the decision threshold are considered as signal, while those below are rejected as background.

The two key metrics to quantify the signal-to-background separation power are the signal and background efficiencies. The signal efficiency is defined as the ratio between the number of true signal events that are classified as signal and the total number of true signal events. The background efficiency is defined as the ratio between the number of background events classified as signal and the total number of background events. There is a trade-off between the desired large signal and low background efficiency. A low decision threshold will reject fewer signal events while letting more background events pass as signal, while a large decision threshold will filter background more reliably, but also falsely reject more

signal. To quantify the performance in the most general way, the relation between signal and background efficiency is evaluated for varying decision threshold, yielding the Receiver Operating Characteristic (ROC) curve. For the ROC curves, the range $[0,1]$ is divided into 30 equidistant decision thresholds. As the statistical uncertainties of the resulting signal and background efficiencies is observed to be of the order of 10^{-6} , the statistical uncertainties are not shown in the ROC curves as they are negligibly small.

To define a quantified comparison between the signal-to-background separation between the Deep Set and RNN architectures, the signal significance $\frac{N_{\text{signal}}}{\sqrt{N_{\text{background}}}}$ is chosen at a common working point of 60% signal efficiency. Similarly to the uncertainties of the signal and background efficiencies, the statistical uncertainties of the signal significance are several magnitudes smaller than the resulting values and thus considered negligible.

In addition to the ROC curves one can also directly compare the corresponding Neural Network output distributions for the signal and the background. The former are expected to peak near the optimal value of one, while the latter are expected to trend towards zero. To show that the Neural Networks also performs well on data different from the training set, the Neural Network outputs for the training dataset are compared to those for the test dataset. Four equally-sized sub-sets of the test set are chosen to evaluate the stability of the Deep Set predictions. The Neural Network outputs are divided into fifty equidistant bins.

In the following, the performance results for the Run 2-based event categories are discussed.

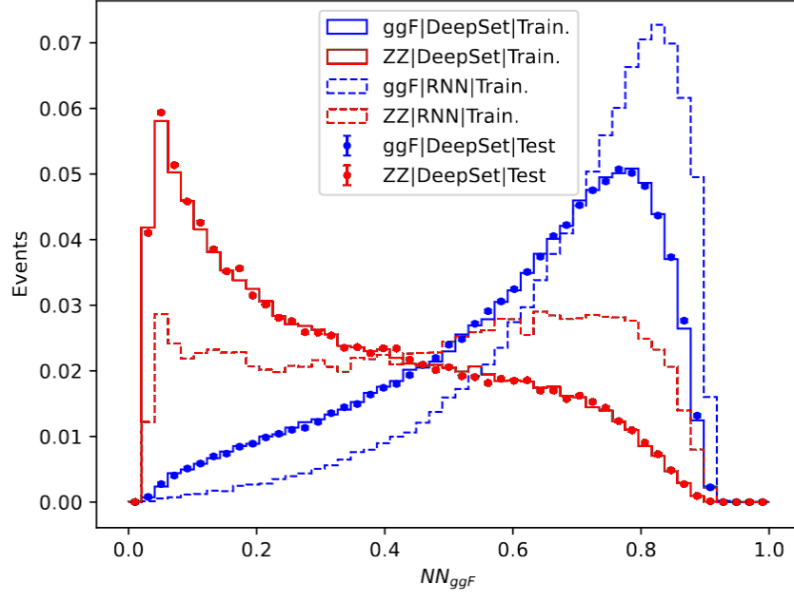
9.1.2. Classification in the Run 2 0-Jet categories

In the Run 2 event categorization, the $0 - \text{Jet}$ region is divided into the $2e2\mu$ and $4l$ regions. The former comprises events with mixed and the latter with the same charge lepton flavors (i.e. $4l = 4\mu$ or $4e$). This is done to enhance the discrimination power of the matrix element in the D_{ZZ} input variable introduced in Section 7.1.1 [32].

The $0 - \text{Jet}$ event category is dominated by ggF production, with the irreducible ZZ background as the dominant background process. Contributions from other Higgs production modes are negligible, since these contain additional jets at leading order. The resulting Neural Network outputs for the $2e2\mu$ and $4l$ cases are shown in Figure 9.1. The corresponding ROC curves with ggF production as signal and ZZ production as background are shown in Figure 9.2.

Compared to the RNN, the Deep Set Neural Network output NN_{ggF} for the ZZ background

peak significantly more towards lower values near zero. The RNN output has a much flatter distribution extending into the signal region of higher NN_{ggF} . The RNN output for the ggF signal shows a more pronounced peak towards higher values. Nevertheless, the Deep Set Neural Network allows for highly increased ZZ background rejection.



(a) Run 2 0 – Jet 4l

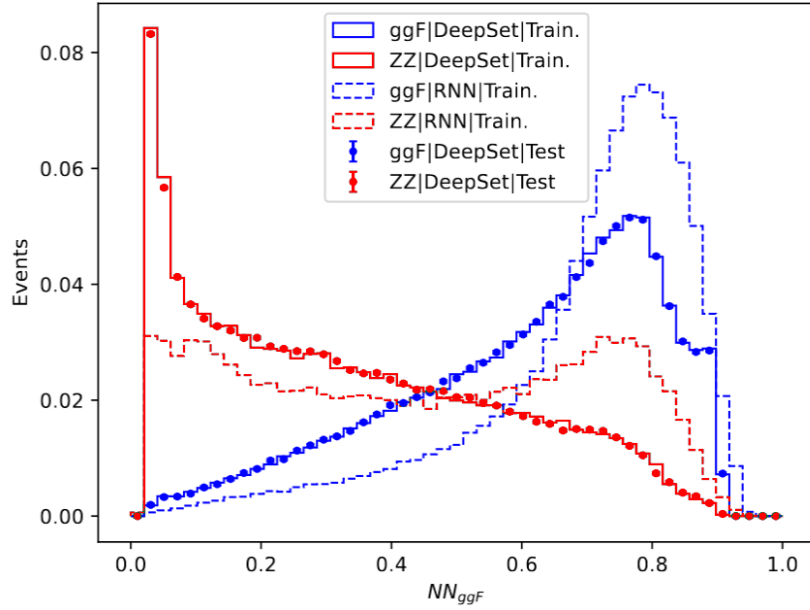

 (b) Run 2 0 – Jet 2e2 μ

Figure 9.1.: Comparison of RNN (dashed line) and Deep Set (solid line) Neural Network output distributions for the ggF signal (blue) ZZ background processes (green), for the $0 - Jet\ 2e2\mu$ (a) and $0 - Jet\ 4l$ (b) event categories. The Deep Set outputs for events from the training set (lines) and are compared to the outputs for events from the test set (full circles). Each distribution is normalized to unit area.

The ultimate RNN and Deep Set Neural Network performance comparison is provided by the ROC curves. In the region of low background efficiencies, the Deep Set Neural Network shows a clear advantage compared to the baseline RNN architecture. The improvement is slightly more pronounced for the $2e2\mu$ final state compared to the $4l$ case. This is due to the higher background rejection (see the Neural Network output distributions in Figure 9.1). Despite the weaker identification power for ggF signal, the Deep Set Neural Network yields significant improvements to the overall signal-to-background separation. The respective signal significancies for the Deep Set and RNN at a signal efficiency of 60% are 2.269 and 2.044 for the $0 - Jet\ 2e2\mu$ category and 2.240 and 2.060 for the $0 - Jet\ 4l$ category.

Since there is no jet sub-network in the case of the $0 - Jet$ event category, the main difference between the RNN and Deep Set Neural Network comes from lepton sub-network. As such, this category offers an important testbench to study the differences of the RNN and Deep Set lepton subnetworks.

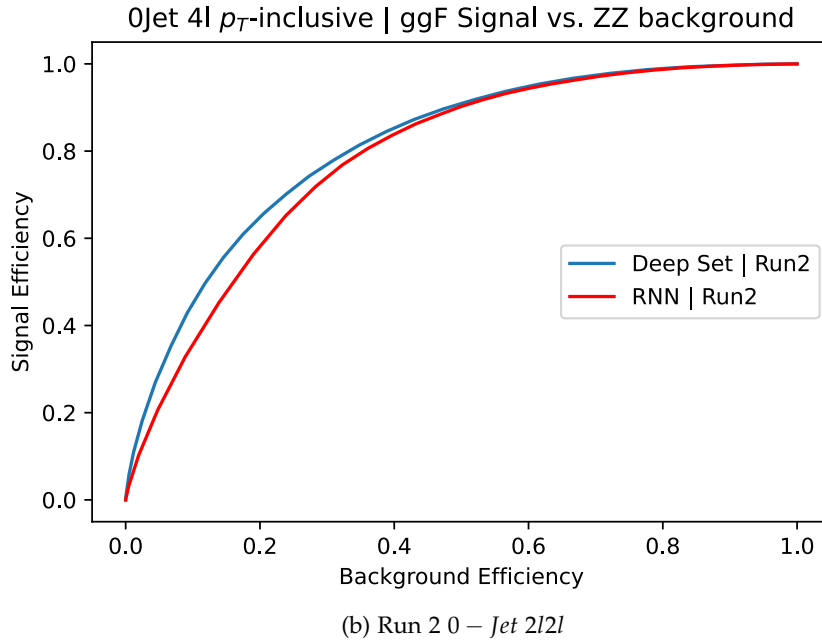
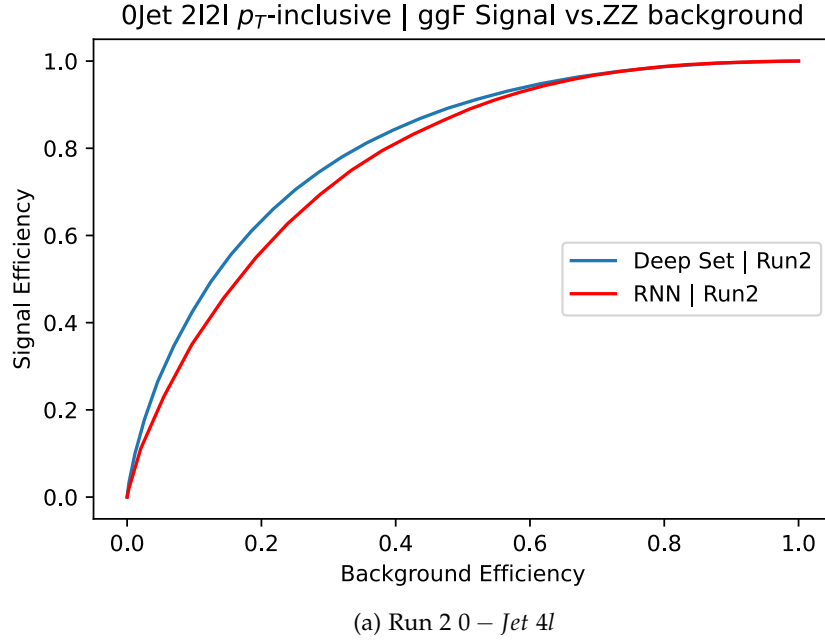


Figure 9.2.: The ROC curves obtained from the RNN (red) and Deep Set (blue) Neural Network training for the separation of the ggF Higgs production mode (signal) from the dominant ZZ background in the 0 – Jet event category with Run 2 data.

9.1.3. Classification in the Run 2 1-Jet event categories

The 1 – Jet event category is further divided into a low- p_T^{4l} region with $p_T^{4l} < 60$ GeV and medium- p_T^{4l} case with $p_T^{4l} > 60$ GeV in the Run 2 event categorization scheme. The high- p_T^{4l} region with $p_T^{4l} > 120$ GeV is merged with the medium- p_T^{4l} region due to the low number of expected events in the high- p_T^{4l} region.

The ggF and VBF Higgs production modes both have large contributions in the 1 – Jet event category with additional non-negligible contributions from ZZ background processes. While the VBF process is characterized by Higgs production with two forward facing jets, possible failure to detect one of the two jets due to limited detector acceptance leads to considerable VBF contributions in the 1 – Jet case. The ggF contributions, on the other hand, originate from substantial initial state radiation effects. Since both the ggF and VBF contribution are of interest for the $H \rightarrow 4l$ analysis, a classification between three classes (ggF , VBF and ZZ) is performed in the 1 – Jet event categories. This is done to ensure that both VBF and ggF can be treated as signal processes while enabling separation from the ZZ background. The resulting Neural Network output distributions are shown in Figure 9.3 and Figure 9.4, with the corresponding ROC curves for the low- p_T^{4l} and medium- p_T categories shown in Figure 9.5 and Figure 9.6.

Figure 9.3a shows the distributions of the Neural Network output for the ggF process NN_{ggF} in the 1 – Jet low- p_T^{4l} event category. Comparing RNN and Deep Set results, a more pronounced separation of ggF Higgs production from VBF and ZZ events is observed for the Deep Set architecture. The peak of the Neural Network output distributions for ggF events is broader for the RNN, reaching further towards higher Neural Network outputs for the ggF class of $NN_{ggF} > 0.6$. Compared to the RNN setup, the Deep Set setup yields a stronger shift of ZZ events towards lower Neural Network outputs for the VBF class NN_{VBF} . For the VBF and ggF events, both the RNN and Deep Set setups result in similar NN_{VBF} distributions.

The Neural Network output distributions for the ggF class NN_{ggF} in the 1 – Jet med- p_T^{4l} event category are shown in Figure 9.4a. Similar trends are observed for the VBF and ZZ processes as in the 1 – Jet low- p_T^{4l} event category. On the other hand, ggF events are less separated from the VBF and ZZ processes than in the 1 – Jet low- p_T^{4l} event category. This is due to the fact that the jets in ggF events are more energetic in the 1 – Jet medium- p_T^{4l} category and therefore harder to distinguish from the VBF and ZZ processes. In the case of NN_{VBF} output distributions in the 1 – Jet medium- p_T^{4l} category, the VBF and ZZ processes also show a similar behavior as in the 1 – Jet low- p_T^{4l} event category, while the ggF events tend towards lower NN_{VBF} values.

In general, substantial improvements in the rejection of the ZZ background can be observed in each cases, while the ggF signal identification weakens for the Deep Set compared to the

RNN. This is similar to the behavior observed in the 0 – *Jet* event categories, indicating a general feature of the difference between the Deep Set and RNN architectures. The *VBF* output distributions on the other hand are much less affected by the change from Deep Set to RNN.

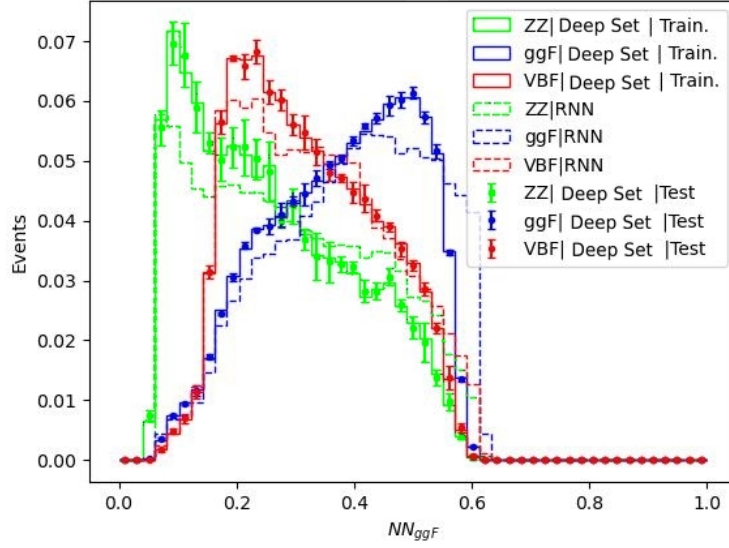
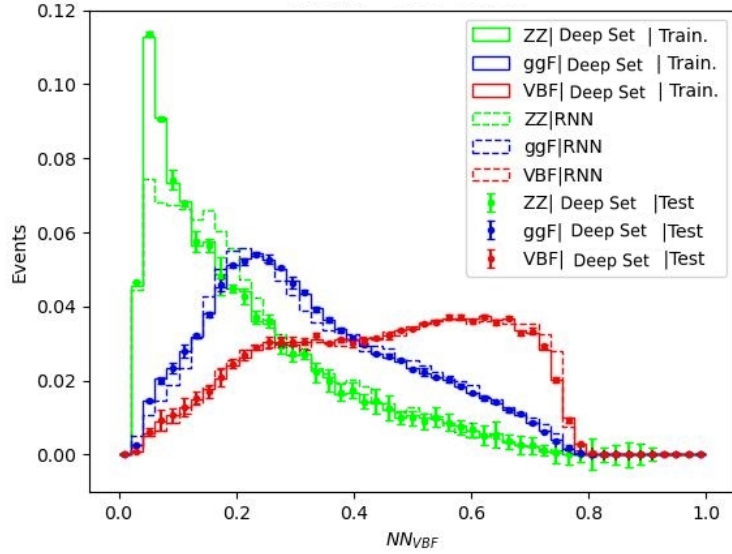

 (a) Run 2 low- p_T^{4l} Neural Network output distribution for the ggF class.

 (b) Run 2 low- p_T^{4l} Neural Network output distribution for the VBF class.

Figure 9.3.: Comparison of RNN (dashed line) and Deep Set (solid line) Neural Network output distributions for the ggF (blue), VBF (red) and ZZ (green) processes in the Run 2 1 – Jet low- p_T^{4l} event category. (a) NN output for the ggF class and (b) for the VBF class selection. The outputs for the test data set are shown as full circles compared to the results for the training set given as histograms. All distributions are normalized to unit area.

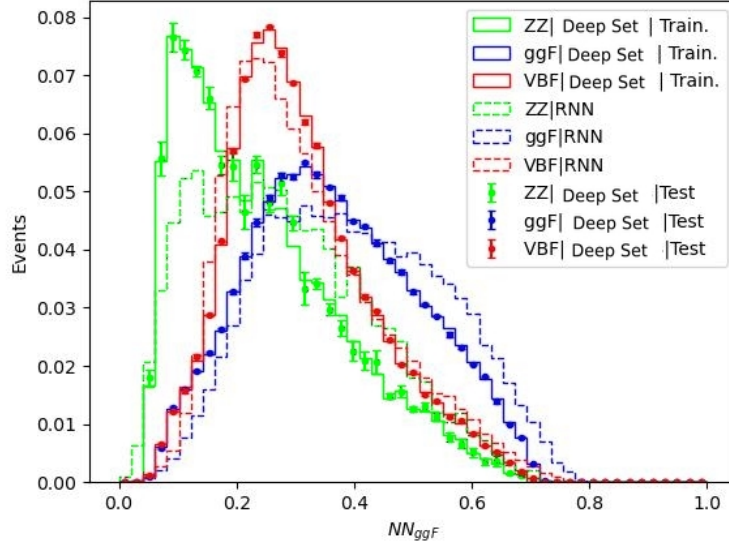
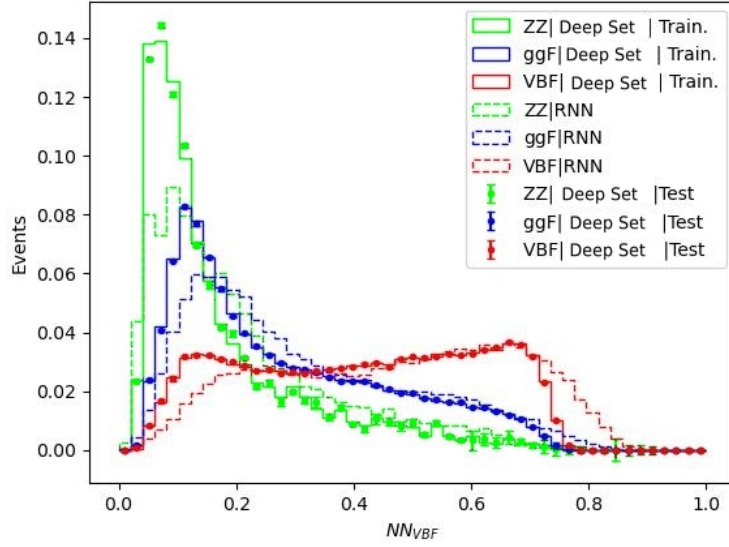
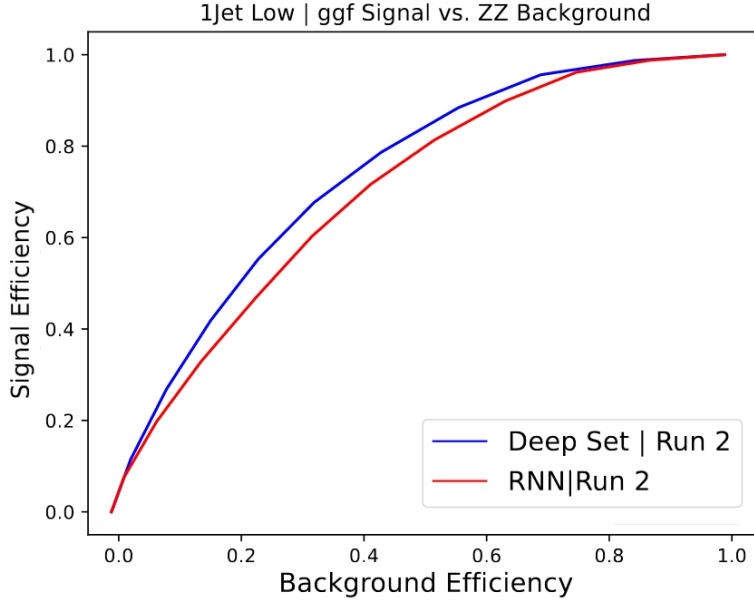
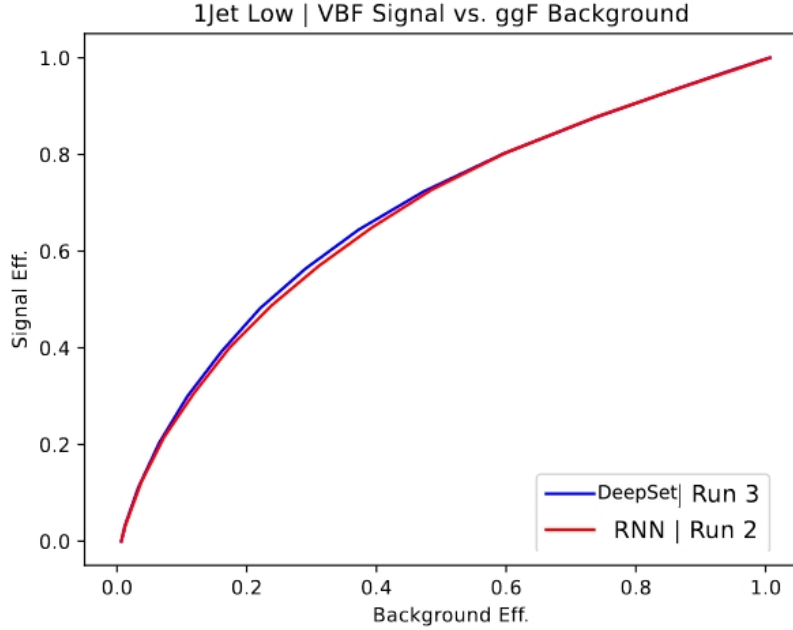

 (a) Run 2 medium- p_T^{4l} Neural Network output distribution for the ggF class.

 (b) Run 2 medium- p_T^{4l} Neural Network output distribution for the VBF class.

Figure 9.4.: Comparison of RNN (dashed line) and Deep Set (solid line) Neural Network output distributions for the ggF (blue), VBF (red) and ZZ (green) processes in the Run 2 $1 - Jet$ medium- p_T^{4l} event category. (a) NN output for the ggF class and (b) for the VBF class selection. The outputs for the test data set are shown as full circles compared to the results for the training set given as histograms. All distributions are normalized to unit area.

The ROC curves corresponding to the NN_{ggF} outputs for the separation of the ggF signal from the ZZ background in the $1 - Jet$ low- and medium- p_T^{4l} are shown in Figure 9.5a and Figure 9.6a. In both cases the Deep Set setup shows substantially steeper ROC curves compared to the RNN setup, with signal significancies (at a signal efficiency of 60%) for the Deep Set and RNN of 2.430 and 2.209 in the low- p_T^{4l} case and 2.640 and 2.296 in the medium- p_T^{4l} case, respectively. In the case of NN_{VBF} outputs for the separation of the VBF signal from the ggF process, as shown in Figure 9.5b and Figure 9.6b, the ROC curves of the Deep Set and RNN architectures are much more similar, with signal significancies (at a signal efficiency of 60%) for the Deep Set and RNN of 0.099 and 0.098 in the low- p_T^{4l} case and 0.092 and 0.091 in the medium- p_T^{4l} case, respectively. The improvements in the ZZ background rejection dominate compared to the decrease in ggF signal identification, while the signal-to-background separation is only slightly affected.

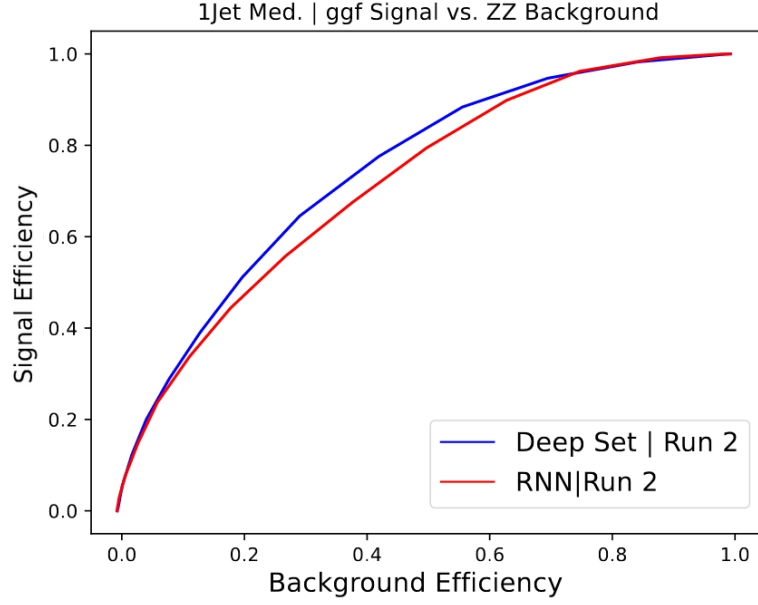


(a) Run 2 low- p_T^{4l} ROC curve for ggF signal discrimination against ZZ background.

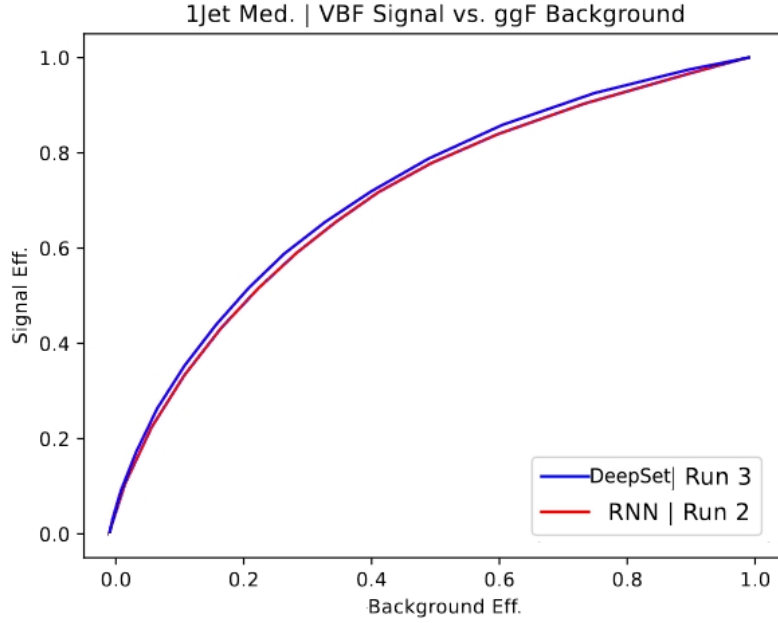


(b) Run 2 low- p_T^{4l} ROC curve for VBF signal discrimination against ggF background.

Figure 9.5.: The ROC curves obtained from the RNN (red) and Deep Set (blue) Neural Network training for the separation of a) the ggF Higgs production mode (signal) from the dominant ZZ background and b) the VBF Higgs production mode from the ggF process in the $1 - Jet$ low- p_T^{4l} event category with Run 2 data.



(a) Run 2 medium- p_T^{4l} ROC curve for ggF signal discrimination against ZZ background.



(b) Run 2 medium- p_T^{4l} ROC curve for VBF signal discrimination against ggF background.

Figure 9.6.: The ROC curves obtained from the RNN (red) and Deep Set (blue) Neural Network training for the separation of a) the ggF Higgs production mode (signal) from the dominant ZZ background and b) the VBF Higgs production mode from the ggF process in the $1 - Jet$ medium- p_T^{4l} event category with Run 2 data.

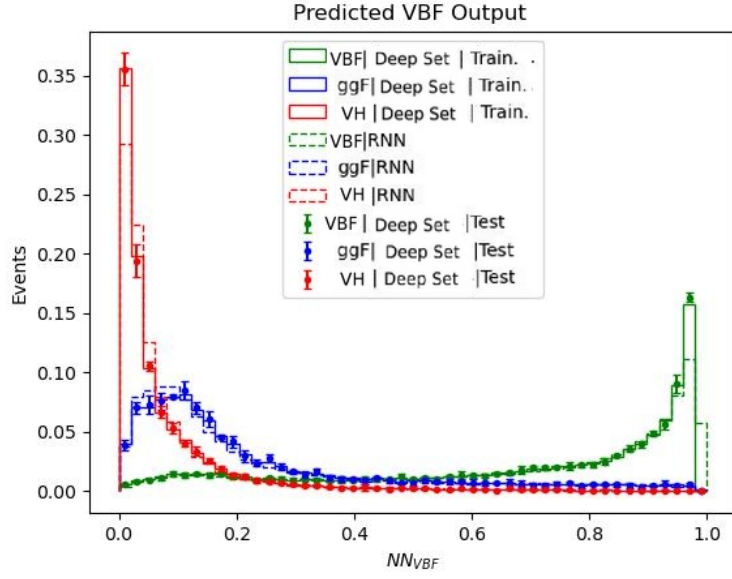
9.1.4. Classification in the Run 2 2-Jet $H \rightarrow 4l$ event categories

The 2-Jet region is divided into a VH -enriched low- p_T^{4l} region with either $p_T^{4l} < 200$ GeV, $m_{jj} < 120$ GeV or both and a VBF -enriched high- p_T^{4l} region with $p_T^{4l} > 200$ GeV and $m_{jj} > 120$ GeV. These event regions target the VH and VBF processes respectively, but are heavily contaminated by the ggF Higgs production, which are therefore considered as background. The contribution from ZZ events is negligible in comparison. Since the low- p_T^{4l} event category contains large contributions not only from the VH process, but also from the VBF process, a classification in three classes, VH , VBF and ggF , is chosen here. A binary classification (VBF and ggF) is chosen in the high- p_T^{4l} event category as the contributions from the VH process is very small in that region. The Neural Network output comparisons for the 2 – Jet low- p_T^{4l} and high- p_T^{4l} event categories are shown in Figures 9.7 and 9.10, while the resulting ROC curves are shown in Figures 9.8 and Figure 9.9 .

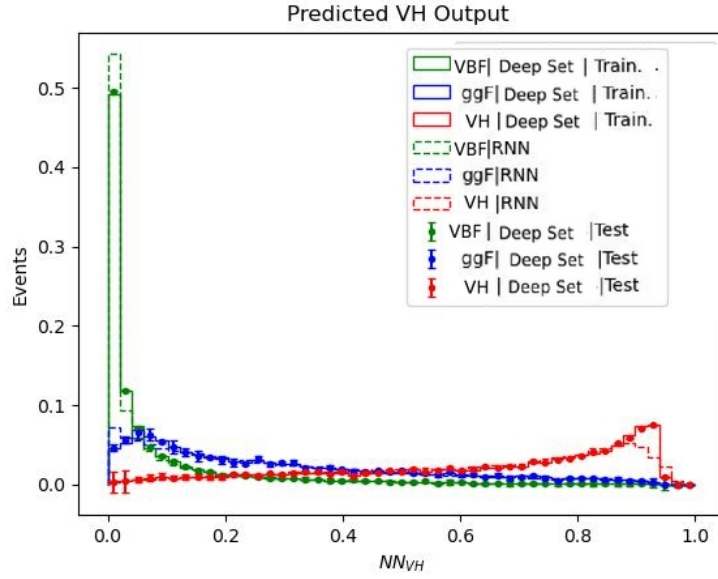
As seen in the Neural Network output distributions, the RNN and Deep Set architectures have a comparable in the 2 – Jet low- p_T^{4l} event category. The most notable difference can be seen Neural Network output distribution for the VH class NN_{VH} for the VH signal events. The Deep Set NN_{VH} distributions for these events peaks more sharply in the region with $NN_{VH} > 0.8$ compared to the RNN output. This results in slightly improved separation of the VH events from the ggF process, as seen in Figure 9.8a. For a signal efficiency of 60%, this yields signal significances of 0.263 and 0.245 for the Deep Set and RNN setups respectively. On the other hand, no improvements are observed for the separation of the VBF signal from the ggF background for the NN_{VBF} output class, as seen in Figure (9.8b), with corresponding signal significances of 0.850 and 0.846 for the Deep Set and RNN setups.

In the case of the 2 – Jet high- p_T^{4l} event category, there are less notable differences between the RNN and Deep Set Neural Network outputs as seen in Figure 9.10. Here the Deep Set architecture results in slightly improved rejection of ggF background while having lower efficiency for the VBF signal selection compared to the RNN. This yields signal significances of 1.357 and 1.339 for the Deep Set and RNN setups for a signal efficiency of 60%. This makes the 2 – Jet high- p_T^{4l} case the only event category in which the Deep Set outperforms the RNN architecture in terms of identification of ggF events.

Unlike in the case of the 0 – Jet and 1 – Jet event categories, both the RNN and Deep Set setups in the 2 – Jet categories include a jet subnetwork. As discussed in Section 7.1.1, the jet kinematic properties differ for different Higgs production modes, providing additional discrimination power. While the 0 – Jet event categories showcased the advantages of the Deep Set approach in the lepton subnetwork, the 2 – Jet categories offer an important insight into the effects jet subnetwork architecture. The observed improvements confirm that the permutation-invariant approach of the Deep Set setup is a viable alternative to the RNN approach for both the lepton and the jet subsets.

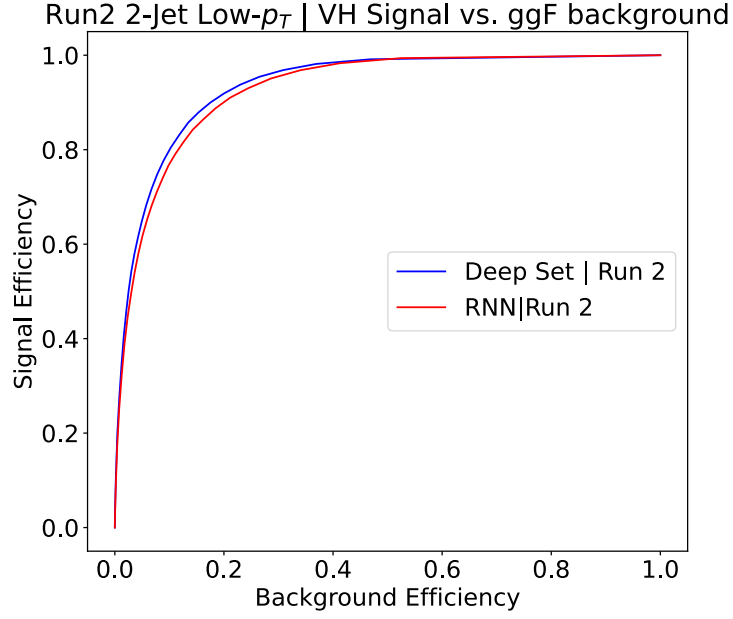


(a) VBF Class Probability Distribution

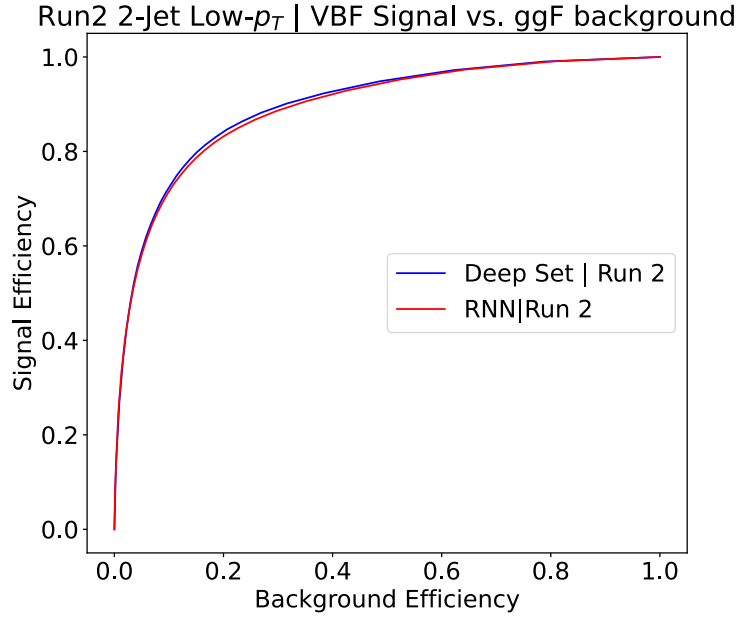


(b) VH Class Probability Distribution

Figure 9.7.: Comparison of RNN (dashed line) and Deep Set (solid line) Neural Network output distributions for the ggF (blue), VBF (yellow) and VH (purple) processes in the Run 2 $2 - Jet$ low- p_T^{4l} event category. (a) NN output for the VH class and (b) for the VBF class. The outputs for the test data set are shown as full circles compared to the results in the training set. Each distribution is normalized to unity.



(a) VH signal vs. ggF background



(b) VBF signal vs. ggF background

Figure 9.8.: The ROC curves obtained from the RNN (red) and Deep Set (blue) Neural Network training for the separation of a) the VH Higgs production mode and b) the VBF Higgs production mode from the ggF process in the $1 - Jet$ low- p_T^{4l} event category with Run 2 data.

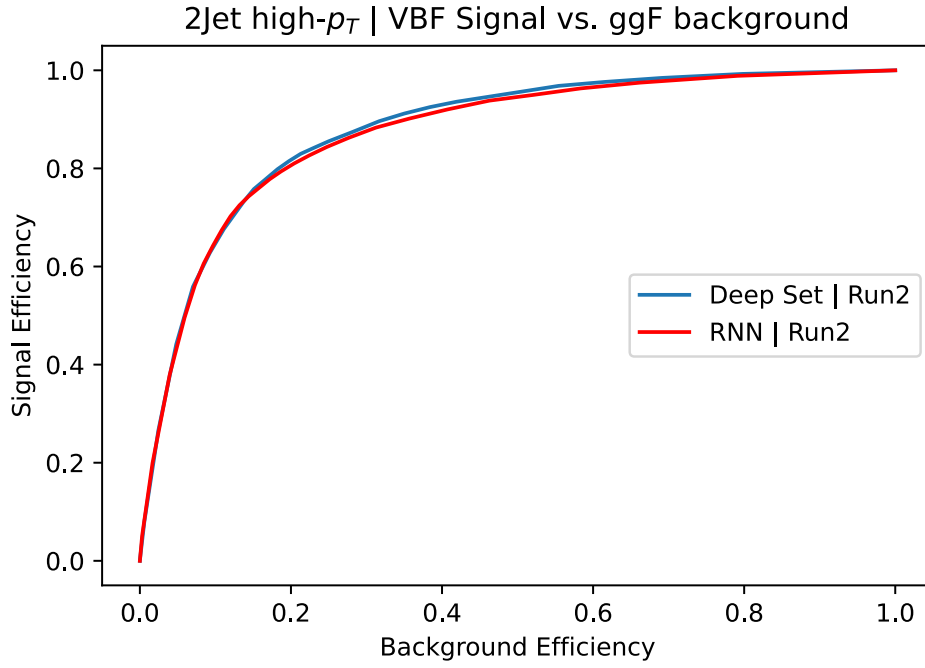


Figure 9.9.: The ROC curves obtained from the RNN (red) and Deep Set (blue) Neural Network training for the separation of the VBF Higgs production mode (signal) from the ggF background in the $2 - Jet$ high- p_T^{4l} event category with Run 2 data.

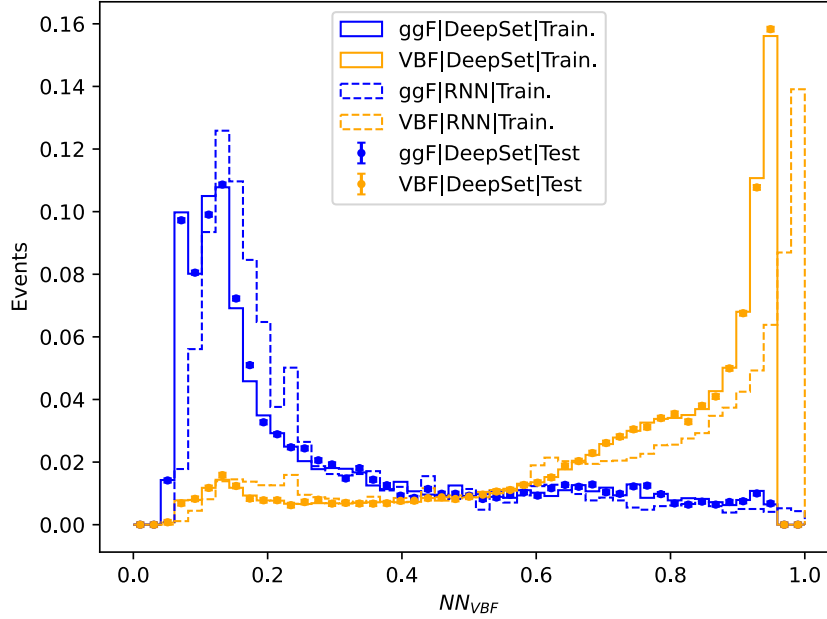


Figure 9.10.: Comparison of RNN (dashed line) and Deep Set (solid line) Neural Network output distributions for the VBF signal (yellow) and ggF background process (blue) in the Run 2 1 - Jet high- p_T^{4l} event category. The outputs for the test data set are shown as full circles compared to the results in the training set. Each distribution is normalized to unity.

9.2. Impact of Additional Input Information

In addition to the previously introduced set of lepton, jet and global input variables, additional lepton and jet information may be considered to be introduced as input variables for further signal-to-background separation. The impact of these additional input variables has been studied in the Deep Set architecture with Run 2 data, as detailed in the following section.

9.2.1. Association of the leptons to the on- and off-shell Z boson

In this section, the impact on the Neural Network performance is studied after introducing the $\mathbb{1}_{Z_{off}}$ lepton variable, which distinguishes the leptons originating from on- and off-shell Z bosons, as defined in Section 7.1.1.

The impact of the using $\mathbb{I}_{Z_{off}}$ as an additional input variable of the Deep Set has been studied with Run 2 Monte Carlo simulated samples in comparison to the original performance without this variable in otherwise identical training setups. For this explorative study, less sophisticated training setups were used than the ones ultimately used for the training of the Deep Set models discussed in Section 9.1, which causes slightly lower signal-to-background separations than previously shown. A noticeable improvement in the model performance has been observed after adding the $\mathbb{I}_{Z_{off}}$ variable, most notably in the $0Jet\ 4l$ event category, with a relative increase in signal efficiency of 17% (signal significancies of 1.721 and 2.013 for the Deep Set without and with $\mathbb{I}_{Z_{off}}$ as input variable, respectively), as shown in the Figure (9.11).

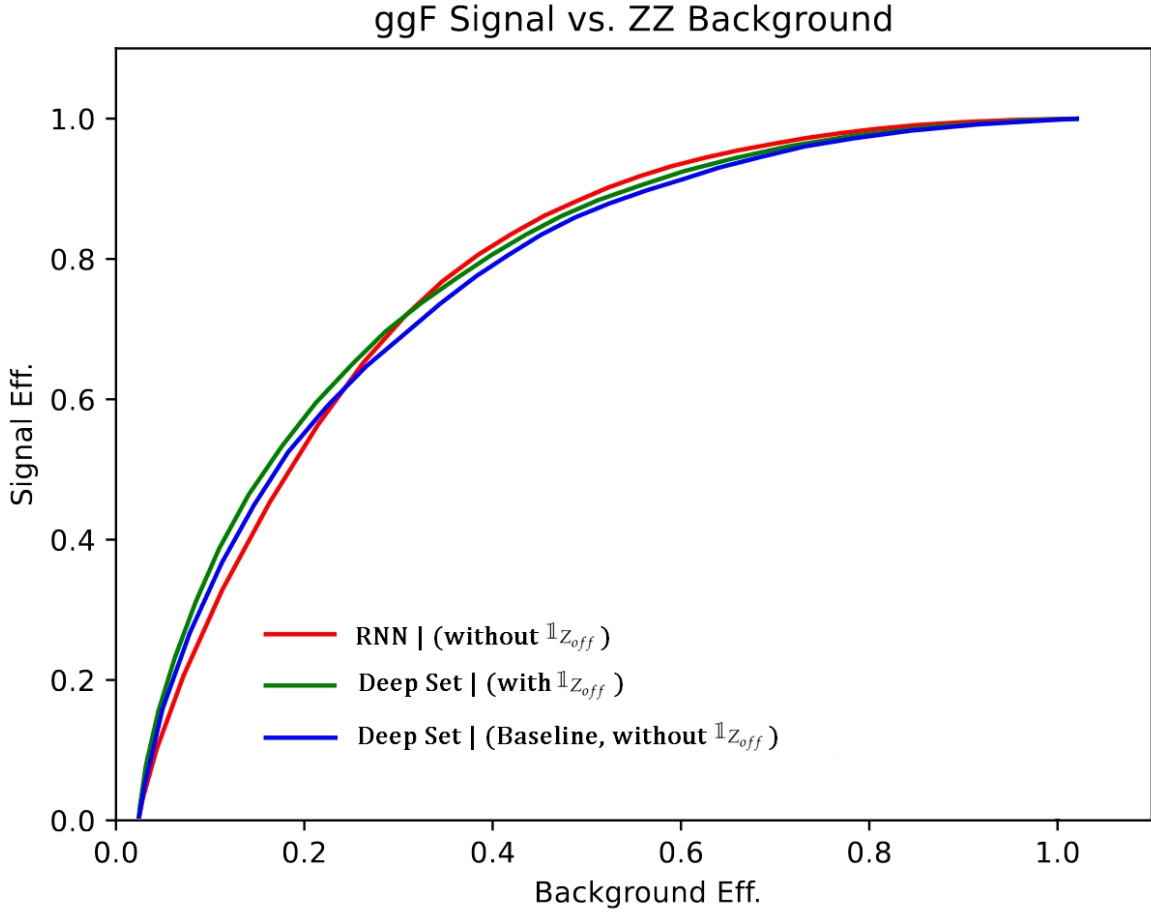


Figure 9.11.: The ROC curves for the separation of the ggF signal from ZZ background in the $0 - Jet\ 4l$ event category, as obtained for the Deep Set with and without the inclusion of the $\mathbb{I}_{Z_{off}}$ input variable. The corresponding RNN-based ROC curve without $\mathbb{I}_{Z_{off}}$ is shown for comparison.

This indicates that the Deep Set is less sensitive to the lepton origin when relying solely on the lepton p_T and η properties. The RNN setup, on the other hand, receives this information implicitly through the p_T based ordering of the lepton input sequence. Therefore, to ensure that the RNN and Deep Set architectures can be compared on an equal footing, the set of input variables for the Deep Set model is extended by adding the $\mathbb{1}_{Z_{off}}$ variable.

9.2.2. Lepton and Jet Four Momentum Variables

In the baseline training approach for the Deep Set models, only the transverse momenta and pseudorapidity properties of each lepton and jet have been taken into account instead of the full four-momentum information. By considering the full four-momentum vectors, an additional discrimination power may be provided via invariances of the four-lepton and multi-jet systems under Lorentz transformations. The invariant lepton or jet mass M and the azimuthal angle ϕ are thus considered as candidates for additional input variables for the training of the Deep Set models. The impact of these additional variables has been studied with Run 2 Monte Carlo samples separately in all event categories.

No noteworthy improvements have been observed after the addition of the full four-momentum information. On the contrary, in some event categories the performance is even shown to deteriorate. This is especially the case in the $2Jet, \text{low-}p_T^{4l}$ category, where a relative drop in signal significance of 8% (signal significances of 0.251 and 0.231 for the Deep Set with and without the additional four-momentum information, respectively) was observed after adding the additional lepton and jet properties. The employed Neural Network model cannot learn from the additional four-momentum information.

Additionally, the increased number of input variables considerably increases the consumption of computational resources. In order to further exploit the four-momentum information, more complex architecture approaches [57]. The optimization studies based on such architecture improvements are left for future work beyond the scope of this thesis.

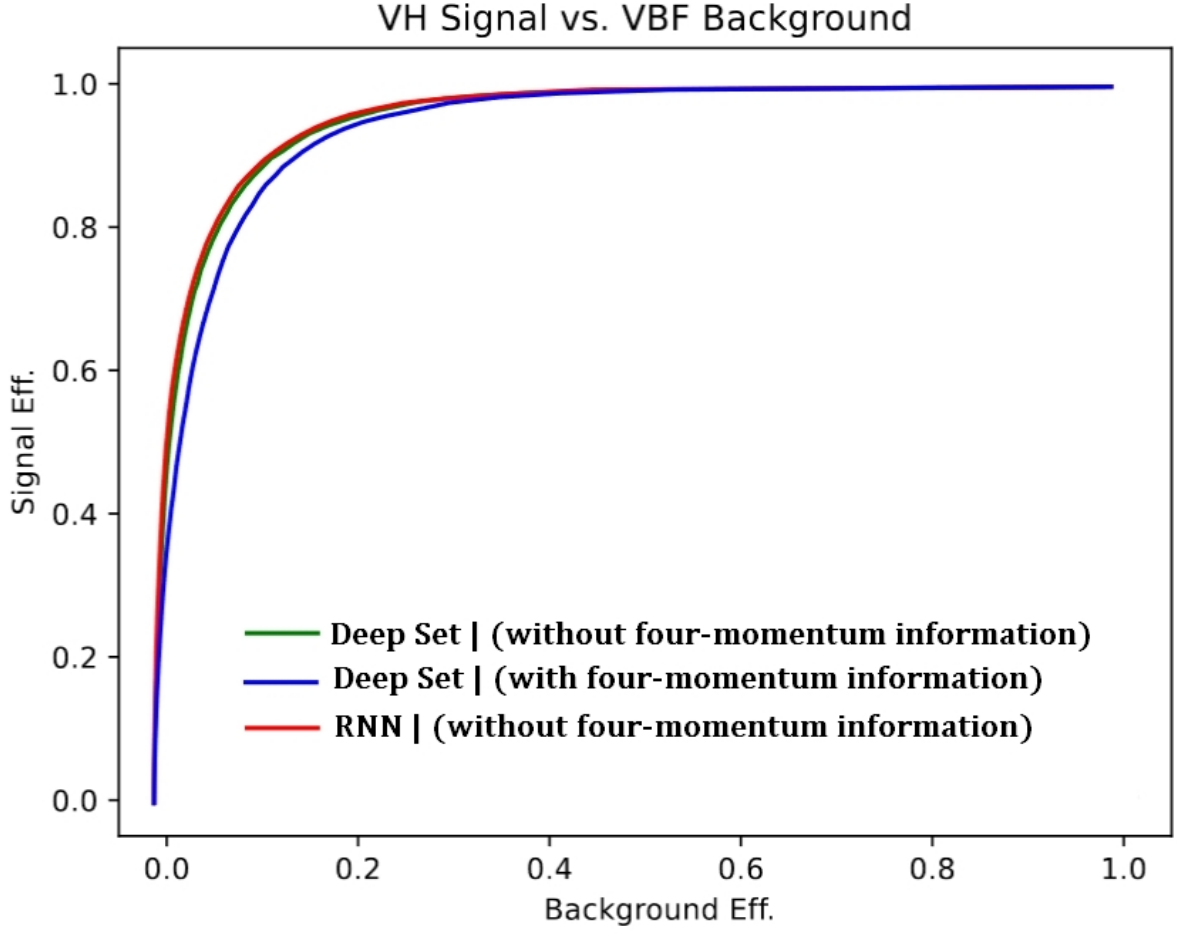


Figure 9.12.: The ROC curves for the separation of the ggF signal from ZZ background in the $2 - Jet$ low- p_T^{4l} event category, as obtained for the Deep Set with and without the inclusion of the additional lepton and jet four-momentum information. The corresponding RNN-based ROC curve without the four-momentum information is shown for comparison.

9.3. $H \rightarrow 4l$ Event Classification with Run 3 Data

The center-of-mass energy increased from 13 TeV in Run 2 to 13.6 TeV in Run 3 data taking at the LHC. This changes the kinematic properties of the Higgs boson decay into four leptons. For example, the Higgs boson produced in pp collisions at a higher center-of-mass energy is expected to have a higher transverse momentum and the individual lepton and jet four-momenta are expected to change accordingly. Therefore, the Neural Network models for the classification of $H \rightarrow 4l$ event candidates have to be retrained for the application on Run 3 data. In addition, at the early stages of the Run 3 data analysis, the available $H \rightarrow 4l$ data is expected to be about three times smaller than in the full set of Run 2 data. To compensate for the lower statistics, a less granular Run 3 event categorization (see Table 5.1) with four categories is used: the two $0 - Jet$ categories ($4l$ and $2e2\mu$) from the Run 2 analysis are merged into a single $0 - Jet$ category. Similarly, the $1 - Jet$ categories (low- p_T^{4l} and medium- p_T^{4l}) are merged into the inclusive- p_T^{4l} $1 - Jet$ category. While the events with two or more jets still remained separated into $2 - Jet$ low- p_T^{4l} and $2 - Jet$ high- p_T^{4l} categories,

While a comparison of the Deep Set architecture to the baseline RNN architecture is given in previous Sections based on the Run 2 data set, this section focuses on the comparison of Deep Set performances with Run 2 and Run 3 datasets. For this purpose, the Run 2 Deep Set models are retrained on the Run 2 Monte Carlo samples using the Run 3 event categorization scheme. At the same time, the same Deep Set architecture is also retrained with Run 3 data in the same event categorization scheme. As previously, the Neural Network output distributions and the corresponding ROC curves are used as performance metrics in each event category. The Neural Network output distributions obtained from the training data set are compared to those from the test data set as a cross-check that there was no overtraining of the model. In case of the ROC curves, the new Run 2 curves for the Run 3 event categorization scheme are compared to the previous results from the Run 2 categorization scheme. This helps to evaluate separately the impact of the coarser categorization scheme on the signal-to-background separation.

While an overview of the input variables used in each event category is given in Section 7.1.1, the impact of each of these variables on the final Neural Network output is quantified in detail in this section. To evaluate the impact of a given input variable, the so-called *Permutation Feature Importance* technique is used [58]. This technique randomly re-shuffles the values of the targeted input variable across all events within the training dataset. This breaks the relationship between the input variable and the "true" Higgs production modes of the simulated events. This is illustrated in Figure (9.13). The trained model is then applied to the dataset with the aforementioned shuffling applied to the targeted input variable. The importance of a given input variable is quantified by the relative increase in the resulting loss when comparing the output of the Neural Network with and without the shuffling applied, which will in the following sections be referred to as *relative performance impact*.

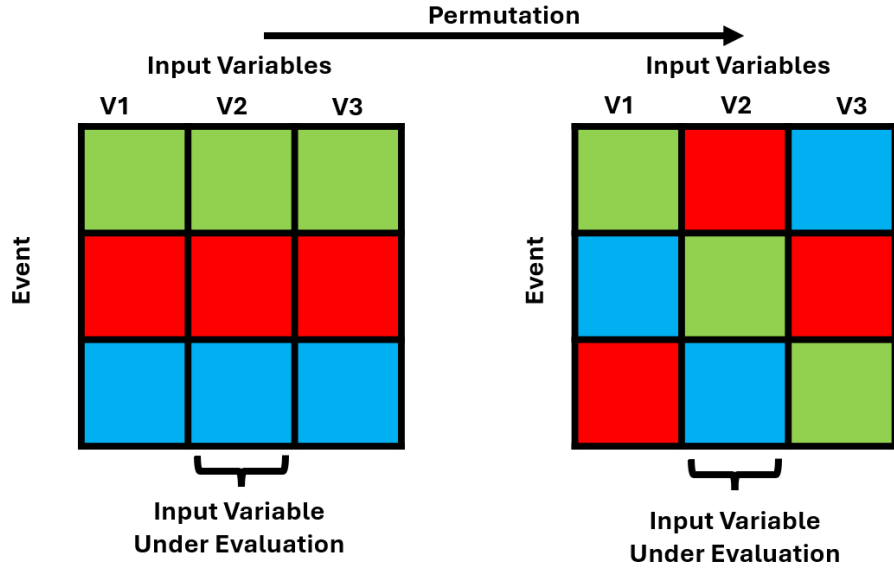


Figure 9.13.: Illustration of Permutation Feature Importance technique for the evaluation of the contribution of a given input variable to the final Neural Network output. The different colors correspond to different values of a given input variable in a given event. For a selected input variable (V_i), these values are randomly shuffled across all events using the permutation procedure. The Neural Network output after the permutation is compared to the one before the permutation in order to determine the impact of the variable on the final result.

9.3.1. Classification in the Run 3 0 – Jet event category

To cope with the decreased amount of data in the Run 3 datasets compared to the Run 2 datasets, the 0 – Jet event categories are merged to a single, inclusive event category containing all final state flavor compositions. In the Run 3 case, these bins were merged into a single p_T -inclusive event category. The distributions of the input variables used for the training in the 0 – Jet category are shown in Appendix (A).

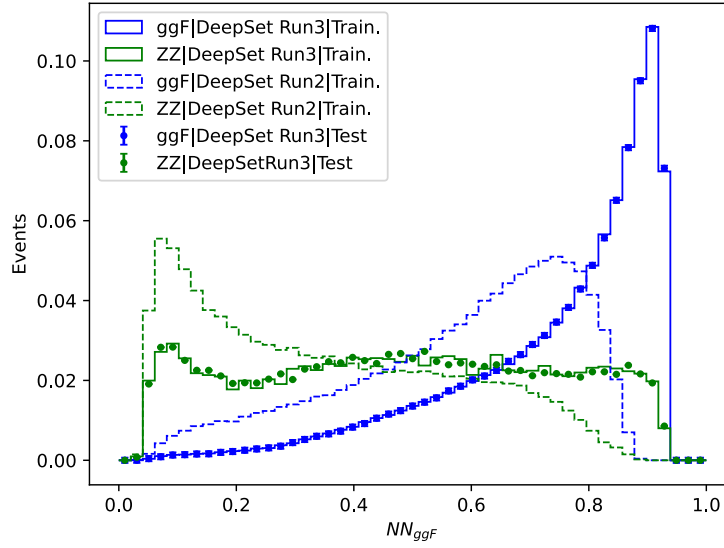


Figure 9.14.: Deep Set Neural Network output distributions for the ggF class in the 0 – Jet event category, after training on the Run 2 (dashed lines) and the Run 3 (solid lines) Monte Carlo samples. The distributions from ggF Higgs signal (blue) and ZZ background (green) are shown separately. The corresponding outputs from the test datasets (full circles) are shown for comparison.

The Neural Network output distributions for the ggF class of events (i.e. ggF signal vs. the ZZ background separation) are shown in Figure 9.14. The Deep Set Neural Network trained on Run 3 samples shows a stronger concentration of ggF events in the region with large output values $NN_{ggF} > 0.8$ when compared to the corresponding results with Run 2 samples. For the ZZ events, the Run 3 output distribution is almost flat with a small peak around $NN_{ggF} < 0.1$. The Run 2 distribution on the other hand has a much more pronounced peak in that region and a noticeable slope in the region with $NN_{ggF} > 0.7$. While the predictive behaviour of the Run 3 trained Deep Set is not confident in the ZZ -background channel compared to the Run 2 trained model, the Run 3 model strongly dominates at the prediction of ggF signal events, showing very confident predictions.

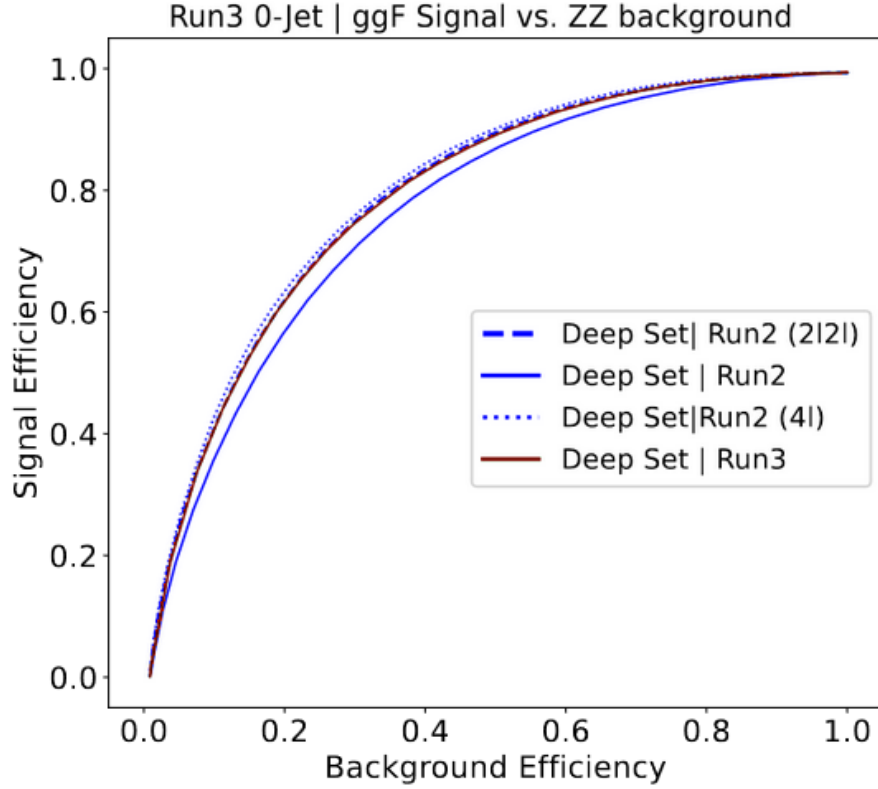


Figure 9.15.: The ROC curves obtained from the Deep Set Neural Network trained in the $0 - Jet$ inclusive event category with the Run 2 data (solid blue line) and Run 3 data (full green line). In addition, the corresponding ROC curves from Run 2 data are also shown for the original Run 2 $0 - Jet$ $4l$ (blue dotted line) and $0 - Jet$ $2e2\mu$ (dashed blue line) event categories for comparison.

The performance of the Deep Set setups trained with Run 2 and Run 3 data is directly compared with the corresponding ROC curves, shown in Figure 9.15. For the inclusive $0 - Jet$ category, it can be seen that the Run 3 data provides a better discrimination between the ggF signal and the ZZ background compared to the Run 2 data. For a signal efficiency of 60%, this yields signal significances of 2.175 and 2.001 for the Run 3 and Run 2 Deep Set setups respectively. This is expected since the higher center-of-mass energy available in the Run 3 dataset compared to the Run 2 dataset yields more pronounced differences in the topology of the final state particles of the $H \rightarrow 4l$ decay, as discussed in Section 7.1.1. The Deep Set model trained on Run 2 data in the inclusive $0 - Jet$ event category performs significantly worse than the Run 2 Deep Set models trained in the $0 - Jet$ $4l$ and $0 - Jet$ $2e2\mu$ categories, with a drop in signal significance from 2.269 and 2.240, for the $0 - Jet$ $4l$ and $0 - Jet$ $2e2\mu$ categories respectively, to 2.001 for the $0 - Jet$ category at the working point. This shows that the separation of the $0 - Jet$ category according to the final state flavor composition yields additional discrimination power during the Deep Set training,

The impact of each input variable on the final result was evaluated using the Permutation Feature Importance technique shown in Figure 9.16. The figure shows the relative change in the Neural Network output performance after a random shuffling is respectively applied to each input variable. In the case of the $0 - Jet$ event category, the Neural Network output is largely dominated by the matrix element information based D_{ZZ} variable with a relative increase of the loss, with regards to the baseline Deep Set model, by a factor of 1.35. An additional substantial impact is also seen to come from the transverse momentum of the combined lepton system p_T^{4l} , with further smaller contributions from the masses m_{12} and m_{34} of the two Z boson candidates and from the single-lepton properties.

The large impact of the D_{ZZ} variable is the variable is specifically defined to distinguish between the ggF signal and the ZZ background (see Section 7.1.1). The strong impact of this observable also explains the additional improvements in Neural Network performance when splitting the inclusive $0 - Jet$ event category into two separate $0 - Jet2\ell\mu$ and $0 - Jet4l$ categories (compare in Figure 9.15), since these two final states result in slightly different matrix-element values due to the interference effects in the case of four same-flavor leptons. The strong discrimination power of D_{ZZ} may, however cause instabilities in the the Deep Set model with the danger of over-emphasizing small changes in D_{ZZ} while disregarding contributions from other input variables. To alleviate this, the probability of the DropOut regularization (see Section 7.2.4) was increased in this event category, allowing for the relative increase of the loss to drop from originally 1.5 to 1.35. At the same time, the relative seen here, while the relative importance of other features input variables increased.

Without the presence of jets in this bin, the final state of the $0 - Jet$ event category is fully described by the kinematics of the four leptons from the $H \rightarrow 4l$ decay. As discussed in Section 7.1.1, the kinematic properties of the leptons resulting from the off-shell Z boson in ggF signal give the largest discrimination from the leptons in the ZZ background process. This also explains the large impact of the Z -boson association indicator $\mathbb{1}_{Z_{off}}$, which allows for the distinction between the leptons from on-shell and off-shell Z bosons.

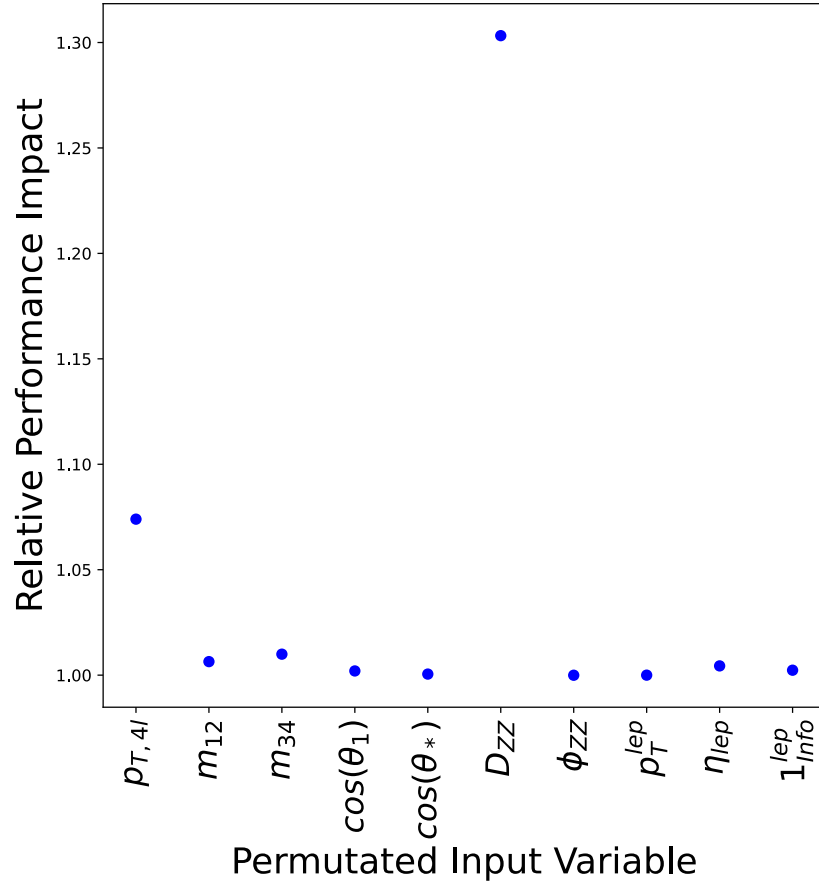


Figure 9.16.: Impact of each input variable in the Run 3 0 – *Jet* event category on the Neural Network output, expressed in terms of the relative change of the loss value after employing the Permutation Feature Importance technique.

9.3.2. Classification in the Run 3 1-Jet event category

While the 1 – Jet event category is split into low- p_T^{4l} and a high- p_T^{4l} category for the analysis of the Run 2 data, a p_T^{4l} -inclusive category is used for the Run 3 data due to the expected smaller size of the Run 3 dataset. The corresponding distributions of the input variables employed in this merged 1 – Jet category are shown in Appendix (A).

The Neural Network is trained for three output classes of events, ggF , VBF and ZZ . The resulting output distributions are shown in Figure 9.17. In the case of the ggF output class, shown in Figure 9.17a, the Run 3 Deep Set output a distribution for the ggF events is much more shifted towards higher NN_{ggF} values compared to the Run 2 Deep Set ggF output class. The VBF and ZZ events are considered as a background in the ggF output class. For these events, the Run 3 Deep Set ggF output has a less pronounced peak at $NN_{ggF} \approx 0.1$ compared to the Run 2 Deep Set, with a larger tail at higher output values. In the case of the VBF output class, shown in Figure 9.17b, the Run 3 Deep Set output for the VBF signal events has a more pronounced peak at $NN_{VBF} \approx 0.8$ and a softer tail in comparison to the Run 2 Deep Set VBF output. The NN_{VBF} distribution for the combined ggF and ZZ background events in this output class is slightly shifted to higher output values compared to the Run 2 Deep Set. In summary, while the Run 3 Deep Set setup is better performing in identifying the signal candidate events, its performance for the background identification is reduced.

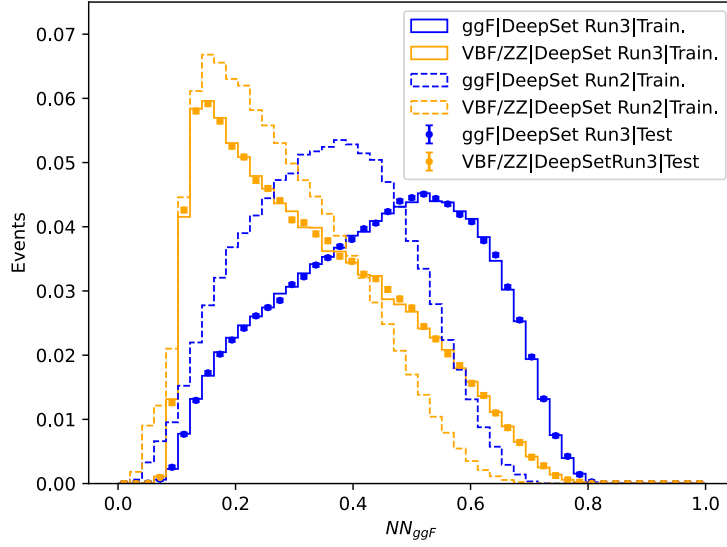
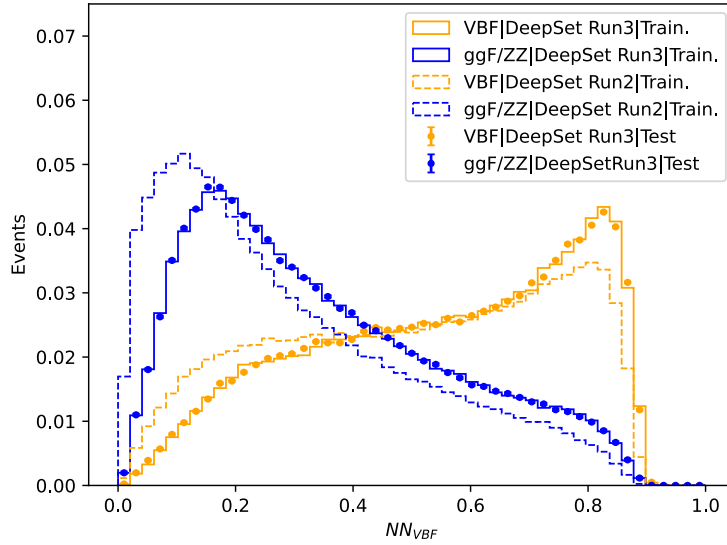
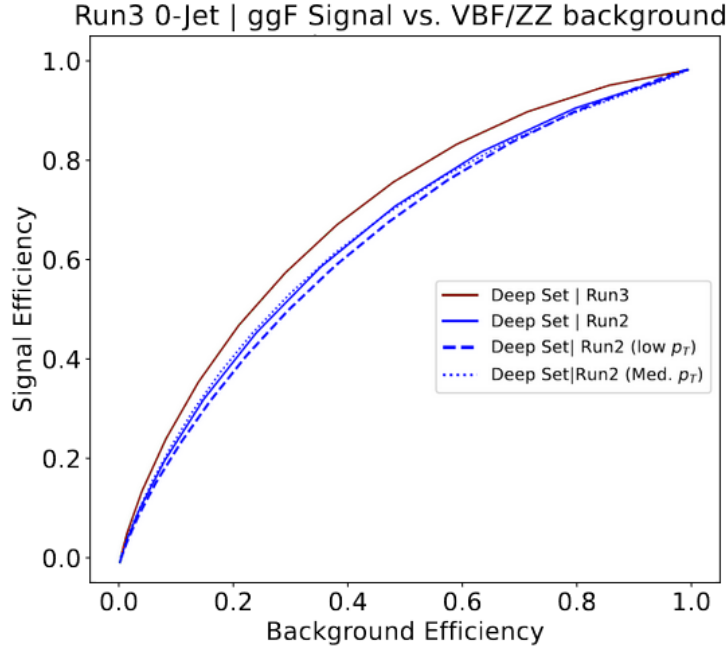

 (a) ggF class probabilities

 (b) VBF class probabilities

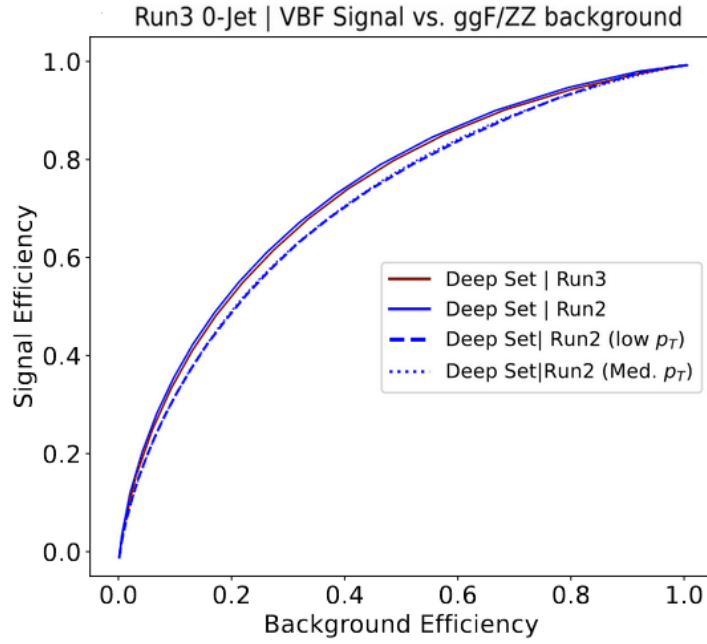
Figure 9.17.: Deep Set Neural Network output distributions for the a) ggF class and b) VBF class in the 1 – Jet event category, after training on the Run 2 (dashed lines) and the Run 3 (solid lines) Monte Carlo samples. The distributions from ggF Higgs signal (blue) and VBF background (yellow) are shown separately. The ZZ process events are respectively included in the VBF and ggF output distribution in a) and b). The corresponding outputs from the test datasets (full circles) are shown for comparison.

The more direct comparison of the network performance is thus provided by the corresponding ROC curves, shown in Figure 9.18. In the case of the ggF output-class (Figure 9.18a), the p_T^{4l} -inclusive Deep Set Neural Network trained on Run 3 samples in the inclusive- $p_T^{4l} 1 - Jet$ event category has a significantly improved performance compared to the Deep Set trained on Run 2 samples in the same event categories, with signal significances of 2.538 and 2.252 respectively for signal efficiencies of 60%. The much stronger signal-to-background separation of the Run 3 Deep Set Neural Network compared to its Run 2 counterpart is due to a much better ggF signal identification, which overshadows the weaker performance in background event rejection. In the case of the VBF class, almost no improvements are observed, with identical signal significances of 1.079 for the Deep Set and RNN setups.

The splitting of the event category into the low- p_T^{4l} and medium- p_T^{4l} event categories is observed to be detrimental to the performance for Run 2 data. In the case of the VBF output-class, shown in Figure 9.18b, both the Run 2 and Run 3 models trained in the p_T^{4l} -inclusive $1 - Jet$ category have a similar performance while the splitting of the category into a low- p_T^{4l} and a medium- p_T^{4l} event category only decreases the performance with the Run 2 data. This is expected, since there are no major differences in distributions of input variables between the two sub-categories while the size of the data sample in each category is reduced. The low- p_T^{4l} category in particular suffers from a low number of events available during the training, as a drop in signal significance of 9% from 1.079 to 0.99 can be observed at the working point compared to the combined $1 - Jet$ category. The improved signal identification together with the weaker background rejection in the Run 3 Deep Set model ultimately cancel each other out. In summary, the higher energy available due to the increase in center-of-mass collision energy from 13 TeV in Run 2 to 13.6 TeV in Run 3 helps to improve the learning capabilities of the Deep Set Neural Network for the identification of ggF events, while the VBF identification of events is barely affected.



(a) ggF signal vs ZZ/VBF background



(b) VBF signal vs ZZ/ggF background

Figure 9.18.: The ROC curves obtained from the Deep Set Neural Network trained in the $1 - Jet$ inclusive event category with the Run 2 data (solid blue line) and Run 3 data (full green line). In addition, the corresponding ROC curves from Run 2 data are also shown for the original Run 2 $1 - Jet$ medium- p_T^{4l} (blue dotted line) and low- p_T^{4l} $1 - Jet$ (dashed blue line) event categories for comparison.

The impact of each input variable on the final output, as evaluated by the Permutation Feature Importance technique is shown in Figure 9.19. The largest impact on the Neural Network output is made by the transverse momentum of the four-lepton system p_T^{4l} , followed by the matrix element-based D_{ZZ} variable and the jet kinematic properties (η^{jet} and p_T^{jet}). The lepton kinematic properties p_T^{lep} and η^{lep} have very little impact compared to the 0 – Jet event category (see Section 9.1.2). Similar to case of the 0 – Jet category, the D_{ZZ} variable is important due to its strong distinction of the ZZ background from the ggF and VBF Higgs events. The additional information from lepton kinematic properties add little to improve this separation. The remaining input variables with strong impact help to separate VBF from ggF events. The VBF production is characterized by jets that are emitted with higher p_T in a more forward direction. This is due to the fact the jet in a ggF process originates from initial state radiation, while the jets produced in a VBF process originate from the hard-scatter itself. In addition, the p_T^{4l} distribution is also shifted towards higher values in the VBF production.

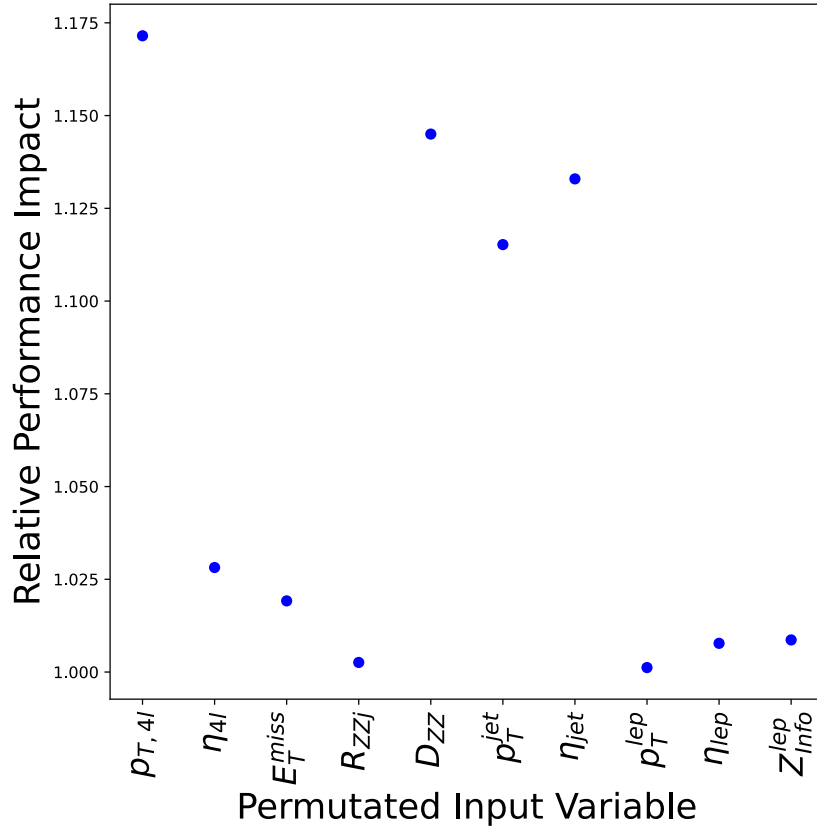


Figure 9.19.: Impact of each input variable in the Run 3 1 – Jet event category on the Neural Network output, expressed in terms of the relative change of the loss value after employing the Permutation Feature Importance technique.

9.3.3. Classification in the Run 3 2-Jet low- p_T^{4l} event category

The Run 3 event categories with two jets in the final state are defined in the same way as for the Run 2 data. The 2 – Jet low- p_T^{4l} category contains the events from the ggF , VBF and VH processes. Instead of a three-class classification of the Run 2 data analysis, these three processes are, for Run 3 data, classified into just two classes: VH signal events and combined VBF/ggF background events. The corresponding distributions of the input variables employed in the 2 – Jet low- p_T^{4l} category are shown in Appendix (A).

The corresponding Neural Network output distributions are shown in Figure 9.20. The Run 3 output distributions for VH events is shifted towards substantially higher NN_{VH} output values compared to the Deep Set Run 2 output distributions. The distribution for VBF and ggF background events, on the other hand, is slightly less shifted towards low NN_{VH} values in the Run 3 data. This is a similar trend as in the case of the 0 – Jet and 1 – Jet event categories, where Run 3 data also yielded better signal identification at the cost of a slightly weaker background rejection.

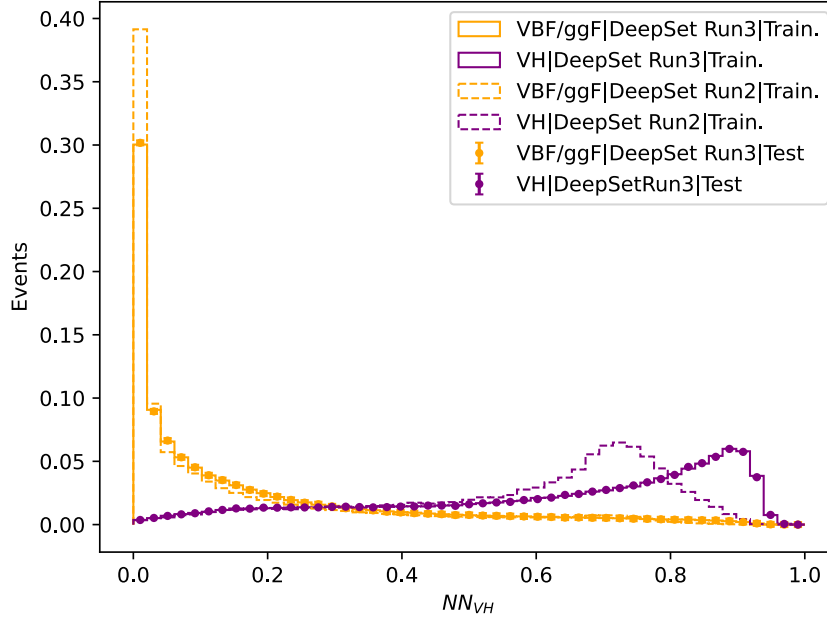


Figure 9.20.: Deep Set Neural Network output distributions for the VH class in the 2 – Jet low- p_T^{4l} event category, after training on the Run 2 (dashed lines) and the Run 3 (solid lines) Monte Carlo samples. The distributions from VH Higgs signal (purple) and a combined ggF and VBF background (yellow) are shown separately. The corresponding outputs from the test datasets (full circles) are shown for comparison.

The ROC curves for the discrimination of the VH signal from the VBF and ggF background are shown in Figure 9.21. In the binary classification case (VH vs. VBF/ggF) substantially higher signal significances are observed with Run 3 data compared to Run 2 with signal significances of 1.876 and 1.823 respectively for signal efficiencies of 60%. It is notable that the original multi-class (VH vs. VBF vs. ggF) neural network setup trained on Run 2 data yields a much better performance compared to the two-class setup, with an increase in signal significance of 26% from 0.209 to 0.263. This may seem surprising at first, since binary classification is in general more robust compared to a multi-class alternative. However, it has to be taken into account, that some input variables may provide strong discrimination between ggF events and VH events, while having no separation between VBF events and VH events, and vice versa. For example, the distribution of the invariant mass of the two leading jets m_{jj} is very similar between the ggF and VH events and differs only for the VBF events where it is shifted towards higher values. Similarly, the distribution of the jet transverse momentum p_T^{jet} tends to be shifted towards lower p_T values for ggF events, while the p_T^{jet} distribution for VBF and VH events are far more similar to each other. As such, it is easier

for the Neural Network to discriminate e.g. VBF and ggF events, without combining ggF and VH contributions together. The large difference in signal-to-background separation between the performance of Run 3 trained and Run 2 Deep Sets in the case of binary classification is due to changed distributions of key input variables with the increased center-of-mass-energy.

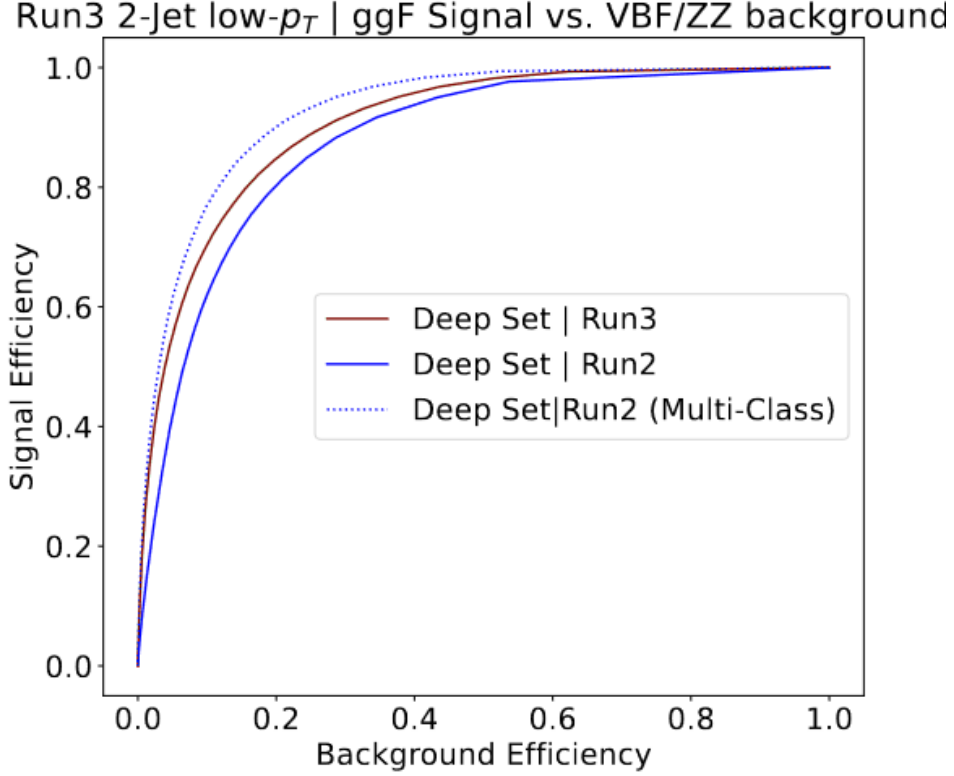


Figure 9.21.: The ROC curves obtained from the Deep Set Neural Network trained in the $2 - Jet$ low- p_T^{4l} event category with the Run 2 data (solid blue line) and Run 3 data (full green line). In addition, the corresponding ROC curves from Run 2 data are also shown for the original Run 2 $2 - Jet$ low- p_T^{4l} , multi-class based Deep Set model (blue dotted line) for comparison.

The impact of each input variable on the Neural Network performance is shown in Figure 9.22 for the case of the two-class classification. The dominant discriminating input variables are the transverse momentum of the combined system of four leptons and the two leading jets, p_T^{4lj} and the leading jet pseudorapidity η^{jet} . Additional impact contributions come from the invariant mass of the dijet system (m_{jj}) and the leading lepton pseudorapidities η^{lep} .

As discussed in Section 7.1.1, the dominant input variables p_T^{4lj} and η^{jet} help to distinguish all three Higgs production processes contributing to the low- p_T^{4l} $2 - Jet$ event category. Other

variables that discriminate between two of the three contributing Higgs production processes (VH vs. VBF and VH vs. ggF) have a smaller impact on the final network performance. This is particularly visible for the case of lepton and jet transverse momenta, p_T^{lep} and p_T^{jet} , which both exert a negligible impact on the neural network output, even though these distributions provide a clear distinction between the ggF process compared to both VBF and VH processes (see Figure A.9 and Figure A.10). While these variables should be helpful to identify ggF events, the neural network model is unable to leverage this information due to the contradicting behavior between the VH and VBF events that are combined into a single background class. As such the multi-class based Deep Set model visibly outperforms the setup with the binary classification Deep Set. It is at first glance surprising that the η^{lep} variable has a comparatively strong impact, since the η^{lep} distributions for all leptons are very similar, as seen in Figure (A.9). While there is a very slight shift towards larger lepton pseudorapidity values in VH events, it seems far too small to explain a relatively big change in loss of. It is likely that this variable contributes to more complex correlations among several input variables related to the geometry of the four-lepton final state.

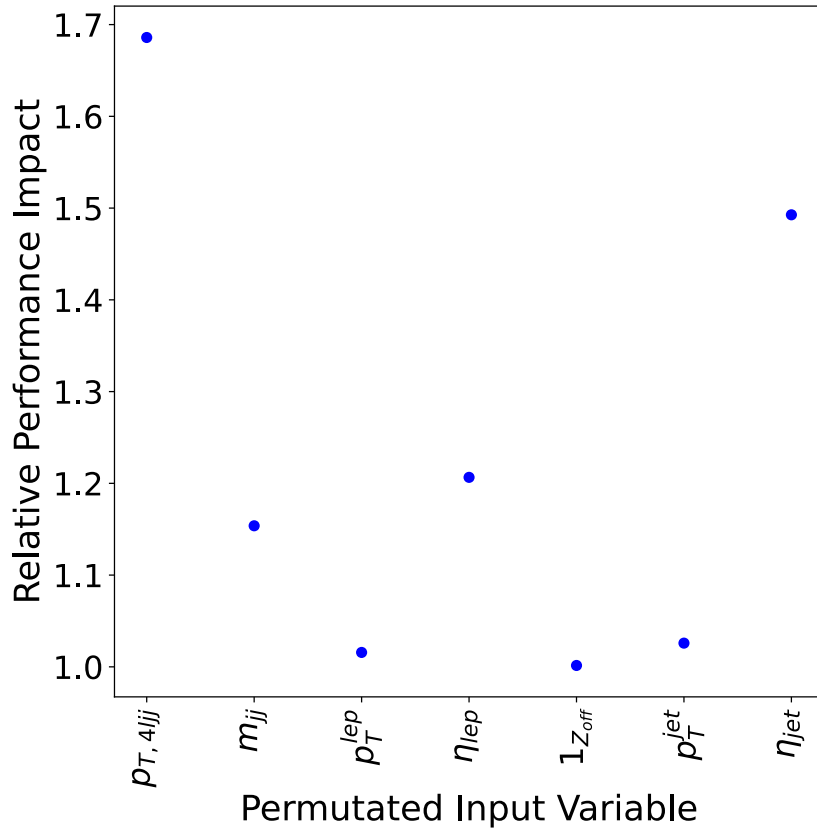


Figure 9.22.: Impact of each input variable in the Run 3 2 – Jet low- p_T^{4l} event category on the Neural Network output, expressed in terms of the relative change of the loss value after employing the Permutation Feature Importance technique.

9.3.4. Classification in the Run 3 2-Jet High- p_T^{4l} event category

Finally the 2 – Jet high- p_T event category for the Run 3 data is also defined in the same way as for Run 2. The full distributions of the input variables used in the 2 – Jet low- p_T^{4l} category are shown in Appendix (A).

The Neural Network output distributions obtained for this event category in a classification between VBF signal and ggF background events are shown in Figure 9.23. The NN_{VBF} distribution for VBF events is shifted slightly towards higher NN_{VBF} values compared to the corresponding Run 2 distribution. The Run 3 Deep Set model also performs better in rejecting the ggF background events with the NN_{VBF} distribution being more shifted towards lower NN_{VBF} values for ggF events.

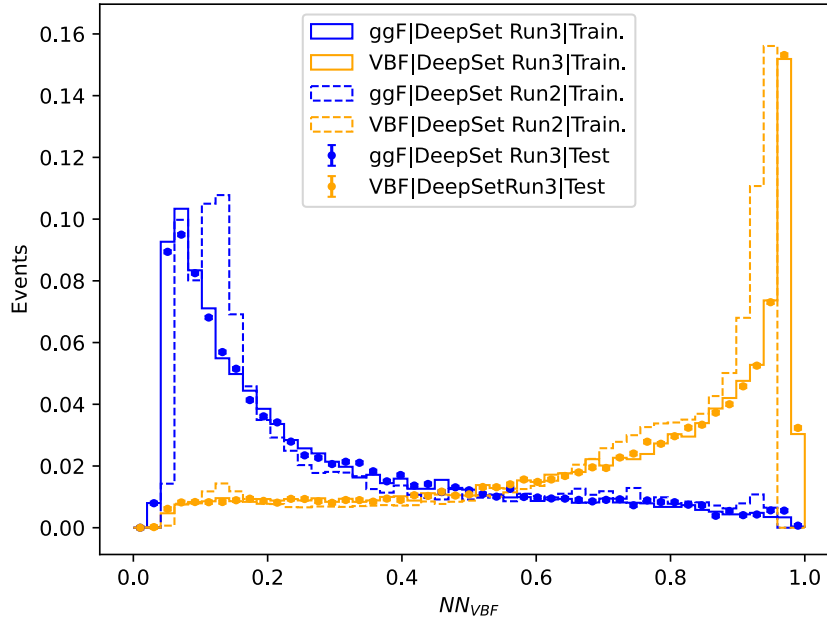


Figure 9.23.: Deep Set Neural Network output distributions for the VBF class in the 2 – Jet high- p_T^{4l} event category, after training on the Run 2 (dashed lines) and the Run 3 (solid lines) Monte Carlo samples. The distributions from VBF Higgs signal (yellow) and ggF background (blue) are shown separately.

The corresponding ROC curves are shown in Figure 9.24. While there is a similar performance for the Run 2 and Run 3 setups in the regions of high signal efficiency, the Run 3 Deep Set has a slightly better performance in case of very low background efficiency (below 0.2). This yields signal significances of 1.422 and 1.357 for the Run 3 and Run 2 Deep Set setups respectively. This can be credited to a more smooth neural network distribution for ggF

events and its shift to lower output values.

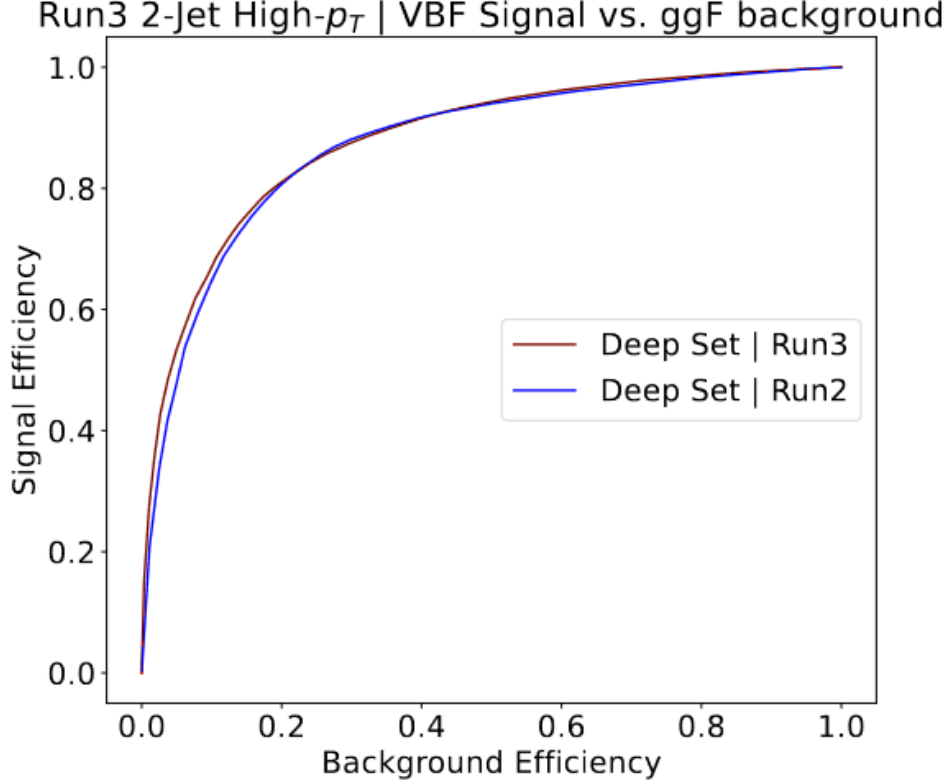


Figure 9.24.: ROC curve comparison of the signal-/background separation of VH signal events against VBF/ggF background. The Deep Set Neural Networks trained in the low- p_T^{4l} 2 – Jet event category on Run 3 (green, solid) and Run 2 (blue, solid) samples in the binary VH vs VBF/ggF case is compared with the Deep Set Neural Network trained Run 2 samples in the multi-class VH vs. VBF vs. ggF case (blue, dotted).

The impact of the employed input variables on the neural network performance in the 2 – Jet high- p_T^{4l} event category is shown in Figure 9.25. The by far largest impact is seen for the jet pseudorapidity η^{jet} . Additional contributions come from the transverse momentum of the combined system of four-lepton and two leading jets system p_T^{4ljj} , the jet transverse momentum p_T^{jet} . It can be seen in Figure A.10 that the η^{jet} distribution has a more pronounced shift to larger η^{jet} values for VBF events in Run 3 compared to the Run 2 data samples. This explains the slightly improved signal-to-background separation for the Run 3 Deep Set model.

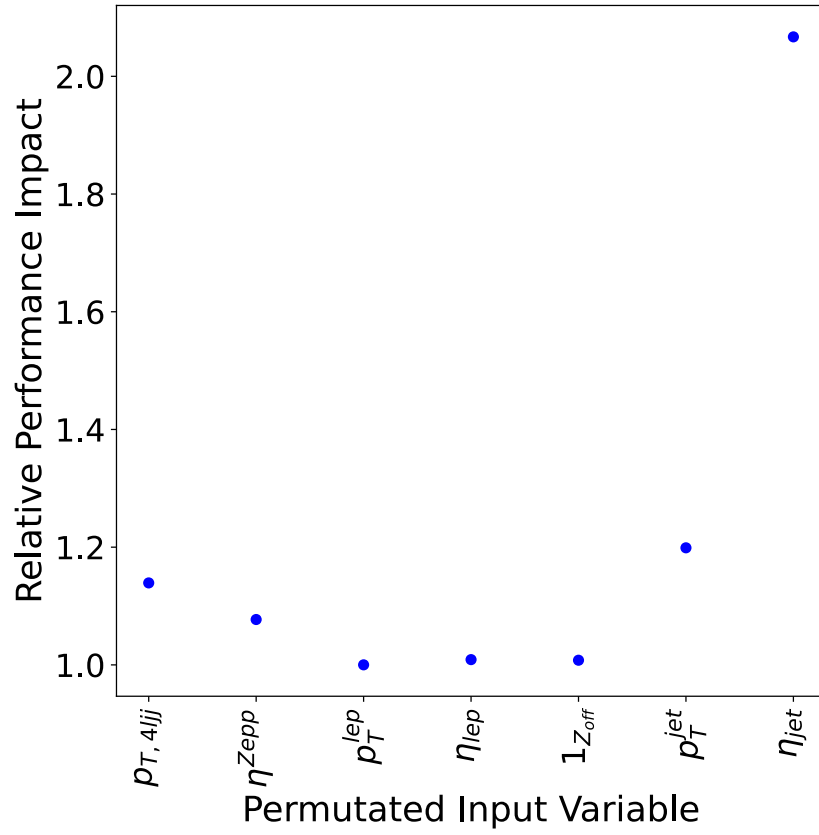


Figure 9.25.: Impact of each input variable in the Run 3 2 – Jet high- p_T^{4l} event category on the Neural Network output, expressed in terms of the relative change of the loss value after employing the Permutation Feature Importance technique.

9.4. Summary

A brief summary of the various studies on the performance of the Deep Set models discussed in this chapter will follow in this section.

The performance of Deep Set models fully trained on Run 2 data was compared to that of Run 2 RNN models in all categories of the Run 2 event categorization scheme. To quantify the difference in signal-to-background separation between the Deep Set and RNN models, the respective signal significances were compared at signal efficiencies of 60%. A full overview of the resulting performance differences between the Deep Set and RNN models for all Run 2 event categories is shown in Table 9.1. The Deep Set is observed to yield improved signal significances in all event categories. Only minor improvements to the signal significance are observed for the Deep Set compared to the RNN for the identification of VBF signal in the $1 - Jet$ (signal significances of 0.99 and 0.98 for the Deep Set and RNN, respectively) and $2 - Jet$ high- p_T^{4l} (signal significances of 0.92 and 0.91 for the Deep Set and RNN, respectively) categories. The identification of ggF signal in the $0 - Jet$ and $1 - Jet$ categories, on the other hand, yields larger improvements for the Deep Set compared to the RNN, ranging from 9% in the $0 - Jet$ $4l$ event category (signal significances of 2.240 and 2.060 for the Deep Set and RNN, respectively), to 15% in the $1 - Jet$ medium- p_T^{4l} event category (signal significances of 2.640 and 2.296 for the Deep Set and RNN, respectively). The Deep Set also leads to improvements to the signal significance of 7% for the identification of VH signal in the $2 - Jet$ low- p_T^{4l} category, with signal significances of 0.263 and 0.245 for the Deep Set and RNN, respectively. This demonstrates, that the Deep Set architecture is a competitive alternative to the RNN architecture for the classification of $H \rightarrow 4l$ production modes.

Similar performance comparisons were made between Deep Set models trained on Run 2 data to Deep Set models trained on Run 3 data in the Run 3 event categorization scheme, a full overview of which can be seen in Table 9.2. The Deep Set models trained on Run 3 data are observed to perform on par or better than the ones trained on Run 2 data. A working point of a signal efficiency of 60% is again chosen to compare the resulting signal significances between Run 3 and Run 2 Deep Sets. Compared to the Run 2 baseline, the identification of VBF signal in Run 3 Deep Set models also showed little or no improvement, with relative changes in signal significance ranging from 0% in the $1 - Jet$ low- p_T^{4l} category (signal significance of 1.079 for both the Run 3 and Run 2) to 4% in the $2 - Jet$ high- p_T^{4l} event category (signal significances of 0.246 and 0.209 for Run 3 and Run 2, respectively). More significant changes in signal significance are once again observed for ggF and VH signal, ranging from 9% in the $0 - Jet$ event category (signal significances of 2.175 and 2.001 for Run 3 and Run 2, respectively) to 18% in the $2 - Jet$ high- p_T^{4l} event category (signal significances of 0.246 and 0.209 for Run 3 and Run 2, respectively). This shows that the increased center-of-mass energy available in the Run 3 data is largely beneficial to the training of the Deep Set neural networks. As the increased center of mass energy in the Run 3 data changes the distributions of the input variables, more pronounced differences between the distributions of signal and

background events were observed in several key kinematic properties, such as the transverse momentum p_T^{4l} of the four lepton system, which is shifted to higher values in the Run 3 data.

The impact of the differences between the Run 3 and Run 2 event category scheme on the neural network performance were also studied. It was observed, that the merging of the $1 - Jet$ low- p_T^{4l} and $1 - Jet$ medium- p_T^{4l} categories into an inclusive $1 - Jet$ category improved the signal significance at the working point by 9% from 0.99 in the $1 - Jet$ low- p_T^{4l} event category to 1.079 in the combined $1 - Jet$ category. The merging of the $0 - Jet$ $4l$ and $0 - Jet$ $2e2\mu$ categories and the switch from a multi-class to a binary-class case in the $2 - Jet$ low- p_T^{4l} case caused significant decreases in performance, on the other hand, with the signal significance at the working point dropping by 9% (signal significances of 2.240 and 2.001 in the $0 - Jet$ $4l$ and combined $0 - Jet$ event categories, respectively) and 24% (signal significances of 0.263 and 0.209 in the $2 - Jet$ p_T^{4l} event category with multi-class and binary classification, respectively), respectively.

The impact of the various input variables on the Deep Set outputs was also studied. Out of the seven to ten input variables, depending on the event category, used in the Deep Set training, it was observed that the most dominant input variables were, for event categories with ZZ process contributions, the matrix element-based D_{ZZ} variable, for event categories with jet multiplicities larger than zero, the transverse momentum p_T^{jet} and pseudorapidity η^{jet} of the jets, and, in $0 - Jet$, $1 - Jet$ and $2 - Jet$ low- p_T^{4l} event categories, the transverse momentum p_T^{4l} of the four lepton system.

Table 9.1.: Summary of performance comparisons between the Deep Set and RNN architectures in all Run 2 event categories. The ratio between Deep Set and RNN signal significance at signal efficiencies of 60% is chosen as performance metric

Event Category	Contributing Processes	Targeted Process	Signal Significance at 60% Signal Efficiency		
			Deep Set	RNN	Deep Set/RNN
$0 - Jet$ $2e2\mu$	ggF, ZZ	ggF	2.269	2.044	1.11
$0 - Jet$ $4l$	ggF, ZZ	ggF	2.240	2.060	1.09
$1 - Jet$ low- p_T^{4l}	VBF, ggF, ZZ	VBF	0.99	0.98	1.01
		ggF	2.430	2.209	1.10
$1 - Jet$ medium- p_T^{4l}	VBF, ggF, ZZ	VBF	0.92	0.91	1.01
		ggF	2.640	2.296	1.15
$2 - Jet$ low- p_T^{4l}	VH, VBF, ggF	VH	0.263	0.245	1.07
$2 - Jet$ high- p_T^{4l}	VBF, ggF	VBF	1.357	1.339	1.01

Table 9.2.: Summary of performance comparisons between Deep Set models trained on Run 2 and Run 3 data in all Run 3 event categories. The ratio between Run 2 and Run 3 signal significance at signal efficiencies of 60% is chosen as performance metric

Event Category	Contributing Processes	Targeted Process	Signal Significance at 60% Signal Efficiency		
			Run 3	Run 2	Run 3/Run 2
0 – Jet	ggF, ZZ	ggF	2.175	2.001	1.09
1 – Jet	VBF, ggF, ZZ	VBF	1.079	1.079	1.00
		ggF	2.538	2.252	1.12
2 – Jet low- p_T^{4l}	VH, VBF, ggF	VH	0.246	0.209	1.18
2 – Jet high- p_T^{4l}	VBF, ggF	VBF	1.422	1.357	1.04

10. Conclusions and Outlook

In this thesis, possible optimizations of machine learning methods for ATLAS cross section measurements in the $H \rightarrow ZZ^* \rightarrow 4l$ decay channel have been investigated. These optimizations were investigated with data from the second data-taking run of the Large Hadron Collider (Run 2) at a centre-of-mass energy of $\sqrt{s} = 13$ TeV and from the currently ongoing third data taking run (Run 3) at a centre-of-mass energy of $\sqrt{s} = 13.6$ TeV. The cross sections are measured for key Higgs production modes, such as gluon-gluon fusion (ggF), vector-boson fusion (VBF) and associated vector boson production (VH), by separate measurements of event yields in dedicated event categories with different jet multiplicities in the final state. Since several Higgs production modes can contribute in each category, dedicated algorithms have to be used to provide discrimination between these modes.

Neural networks are used as classification algorithm to separate a given targeted Higgs production mode from the dominant ZZ background and contamination by other Higgs production modes in each event category. The currently used Recurrent Neural Network-based (RNN) approach for the production mode classification in $H \rightarrow 4l$ events was discussed and a possible, Deep Set Neural Network based alternative introduced. The use of a Deep Set based Neural Network is motivated by its more accurate representation of the final state topology due to its handling of jet and lepton kinematic variables as unordered sets as opposed to the ordered set representation of the jet and lepton variables in the RNN case. Deep Set Neural Networks were optimized for event categories matching those used for the development of the RNN with Run 2 and Run 3 Monte Carlo simulated data. The performance of the Neural Networks in each event category has been evaluated in dependence on various tuning parameters that control the training setup (hyperparameters), such as the algorithm used for the Neural Network optimization, the number of fitting-parameters of the Neural Network or the size of the updates to the Neural Network fitting parameters in each optimizer step. The Deep Set model performance was evaluated with regards to the signal-to-background separation, until ultimately a best performing Deep Set Neural Network was determined and employed for the final analysis results.

The performance of the trained Deep Set Neural Networks was then evaluated in each event category, separately for the training with Run 2 and Run 3 data. First, the performance of the Deep Set Neural Network trained with the Run 2 data was compared to the corresponding performance in the RNN-based approach. This was done for each event category defined by the event categorization scheme used previously for the training of the RNN used in

Run 2, which consists of six categories defined by cuts on the jet-multiplicity, the transverse momentum p_T^{4l} of the four lepton system and the di-jet mass m_{jj} , and the final state flavour composition: two event categories with a jet multiplicity of zero, divided by mixed ($0 - Jet\ 2e2\mu$) and same-lepton flavor ($0 - Jet\ 4l$) final states, two event categories with a jet multiplicity of one, divided by a cut at $p_T^{4l} = 60$ GeV ($1 - Jet\ low-p_T$ and $1 - Jet\ medium-p_T$), and two event categories with jet multiplicities larger than one, divided by a cut at $p_T^{4l} = 200$ GeV and $m_{jj} = 200$ GeV ($2 - Jet\ low-p_T^{4l}$ and $2 - Jet\ high-p_T^{4l}$). Subsequently, the Deep Set Neural Network model was retrained with Run 3 data in an event categorization scheme with a reduced number of four categories, in which the event categories with jet multiplicities of zero and one were respectively merged into one combined $0 - Jet$ and one combined $1 - Jet$ category. In this reduced event categorization scheme, Deep Set models trained on Run 3 and Run 2 data were compared. All performance comparisons were made by using the Neural Network output distributions and the ROC curves from the respective Neural Network models. Additionally, the impact of each input variable used in the Neural Network training on the output performance was determined separately for each event category by comparing the outputs of the Deep Sets in a given test dataset with the outputs in the same dataset after applying a random shuffling to the data of a given input variable, referred to as permutation feature importance technique.

The Deep Set Neural Networks, compared to the RNN based Neural Networks, were observed to provide comparable or improved signal-to-background separation across all event categories. For signal efficiencies of 60%, improvements to the signal significance of up to 15% are observed for the Deep Set compared to the RNN. Thus, the Deep Set architecture appears to be a competitive alternative to the RNN base architecture for the classification of $H \rightarrow 4l$ production modes. The comparison between neural network outputs showed that the Deep Set compared to the RNN yields improved signal-to-background separations, even though the improvements in signal efficiency may come at the cost of a decreased background rejection and vice versa. The ROC curves obtained from the neural network outputs of the Deep Set models in most cases outperform the ones from the RNN-approach.

Deep Sets Neural Networks trained on Run 3 simulated data were shown to perform significantly better compared to the same Deep Set models trained on Run 2 simulated data. For a signal efficiency of 60%, improvements to the signal significance of up to 18% are observed. This can be attributed to changes in the distributions of several key variables with the increase of center-of-mass-energy in Run 3 data. For example, the transverse momentum of leptons from Higgs decays is shifted towards higher values, improving the discrimination against ZZ background. Thus, a higher signal efficiency is seen with the Neural Networks trained on Run 3 simulated data.

The reduced event categorization scheme with four event categories for the analysis of Run 3 data did not always result in better performance compared to the corresponding Run 2 categorization scheme with six event categories. Improvements to the signal significance

of 7% in the signal significance were observed when merging the low- p_T^{4l} and medium- p_T^{4l} $1 - Jet$ event categories. The merging of the $4l$ and $2e2\mu 0 - Jet$ event categories, on the other hand, lead to the worsening of the signal significance by 9%. This is due to the weak end separation power of the matrix-element-based D_{ZZ} input variable .

The impact of additional input variables beyond the ones currently used for the Run 2 RNN training was studied for the Deep Set Neural Networks trained on Run 3 samples. In particular, two additions to the input variables were investigated: The variable $\mathbb{1}_{Z_{off}}$, tagging the leptons originating from the off-shell Z boson, which replaces the less motivated p_T -based ordering of leptons in the RNN based setup, and the addition of the azimuthal angle ϕ^{lep} , ϕ^{jet} and invariant mass m^{lep} , m^{jet} of the leptons and jets, to include the full four-momentum information of the final state particles, was investigated. It was observed that the addition of $\mathbb{1}_{Z_{off}}$ significantly improves the predictive performance of the Deep Set Neural Network models.

The addition of the azimuthal angle the invariant mass for the leptons and jets did not lead to improvements. The utilization of the Lorentz-invariance of the $H \rightarrow 4l$ process in the Neural Network architecture requires further extension of the Neural Network architecture could be further investigated in the future.

The studies of the impact of the individual input variables on the Neural Network performance generally showed that not all variables contribute with the same weight. In the event categories with jet multiplicities larger than zero, jet-related input variables such as η^{jet} and p_T^{jet} were found to greatly impact the neural network output, with relative impact on the Neural Network outputs of up to 2.05 after applying the permutation feature importance technique. The matrix-element-based D_{ZZ} variable had a high impact in event categories where the ZZ process was included, with relative impact on the Neural Network output of up to 1.3. As some variables contributed very little to the neural network output, it might be feasible to reduce the number of input variables while maintaining comparable predictive quality in the future. This would not only reduce the required computational resources during training and inference, but also allow to remove potentially destabilizing correlations between different variables.

Other possible extension of the Neural Network architecture include alternative permutation-invariant Neural Network architectures, such as Graph Neural Networks, or an attentionally based extension to the Deep Set architecture, such as Attentional Sets.

As a result of the analysis in Run 3 presented in this thesis, the Deep Set Neural Network approach is now being implemented into the official common framework for the ATLAS $H \rightarrow ZZ^* \rightarrow 4l$ analysis.

A. Figures

A.1. Kinematic Distributions (0 – Jet)

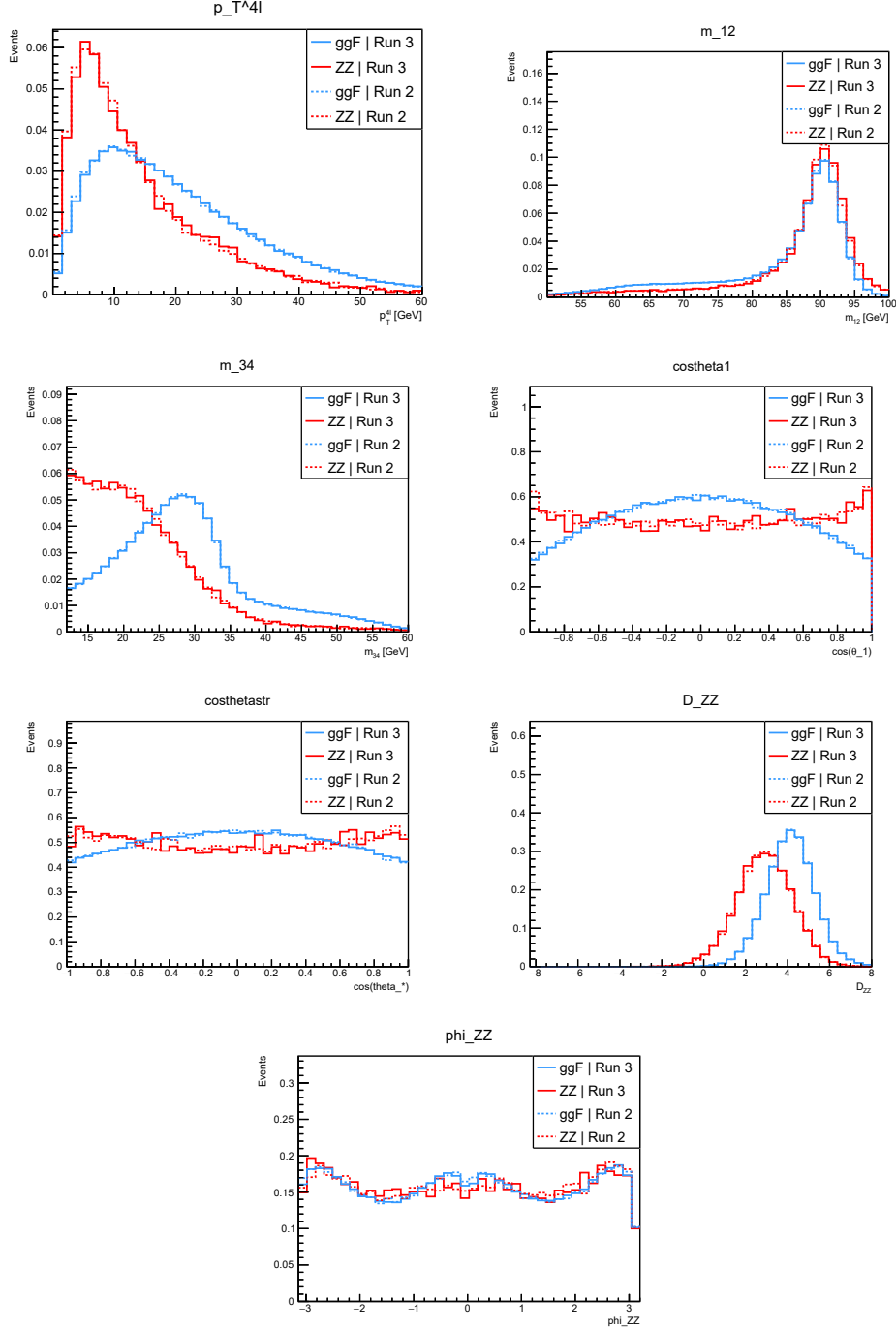


Figure A.1.: Global Input Features in the 0 – Jet case for the Run 3 samples (solid) and Run 2 samples (dashed).

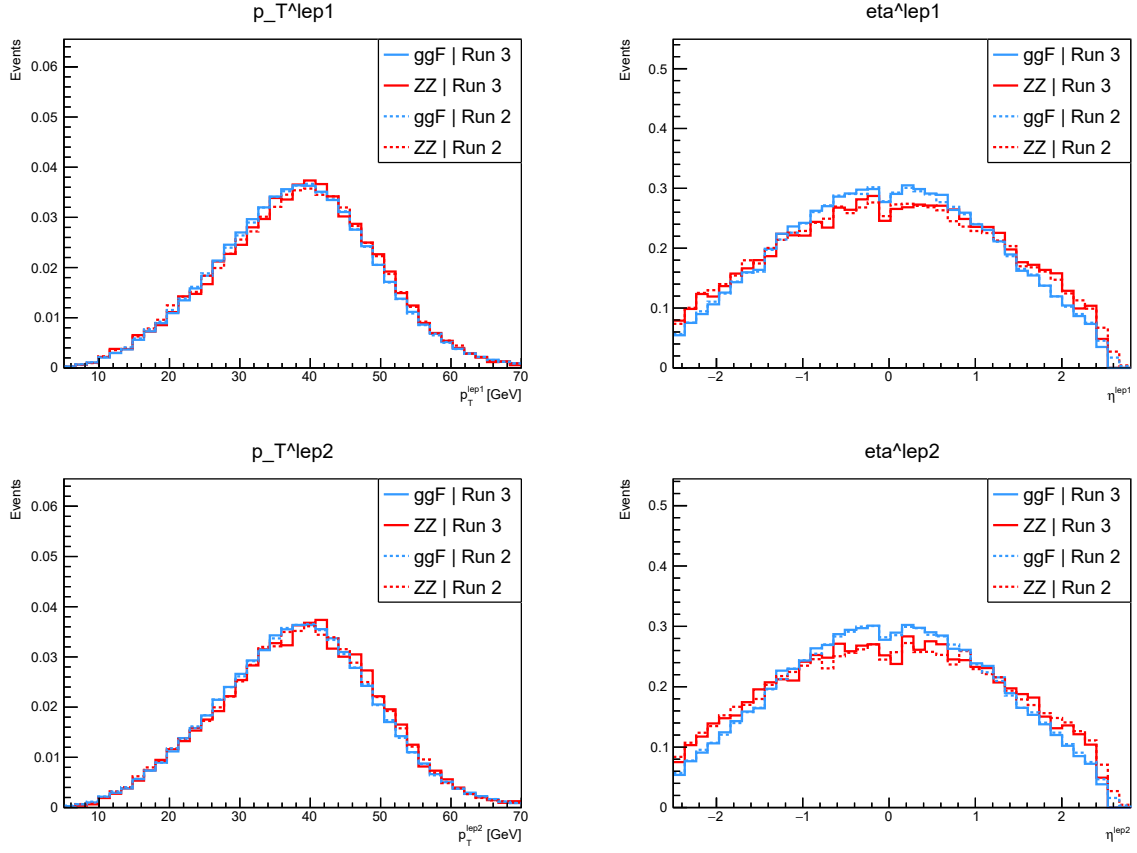


Figure A.2.: Lepton Input Features of the leading lepton pair in the 0 – Jet case for the Run 3 samples (solid) and Run 2 samples (dashed). Leptons are shown in order of descending p_T

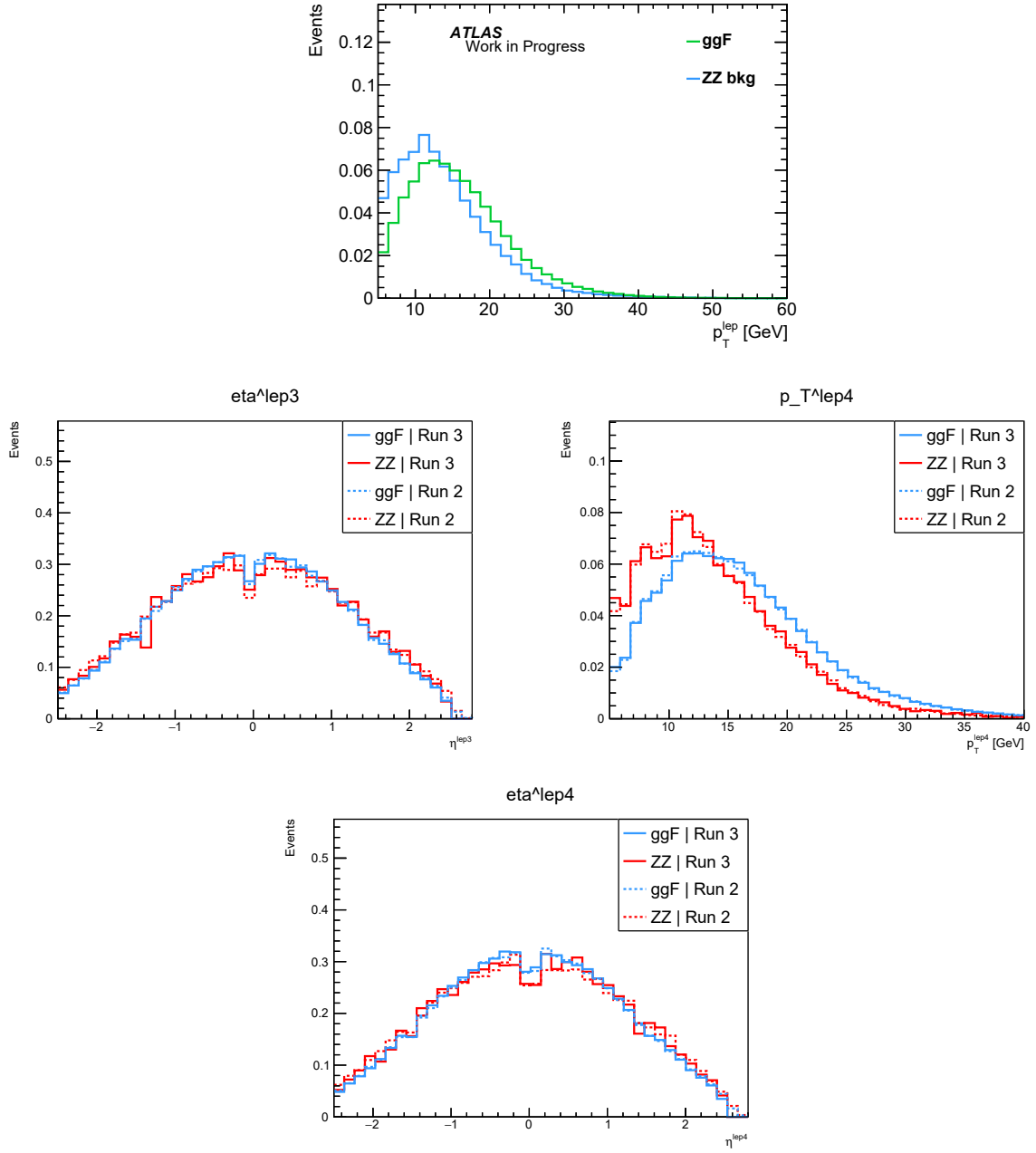


Figure A.3.: Lepton Input Features of the sub-leading lepton pair in the $0 - Jet$ case for the Run 3 samples (solid) and Run 2 samples (dashed). Leptons are shown in order of descending p_T

A.2. Kinematic Distributions (1 – Jet)

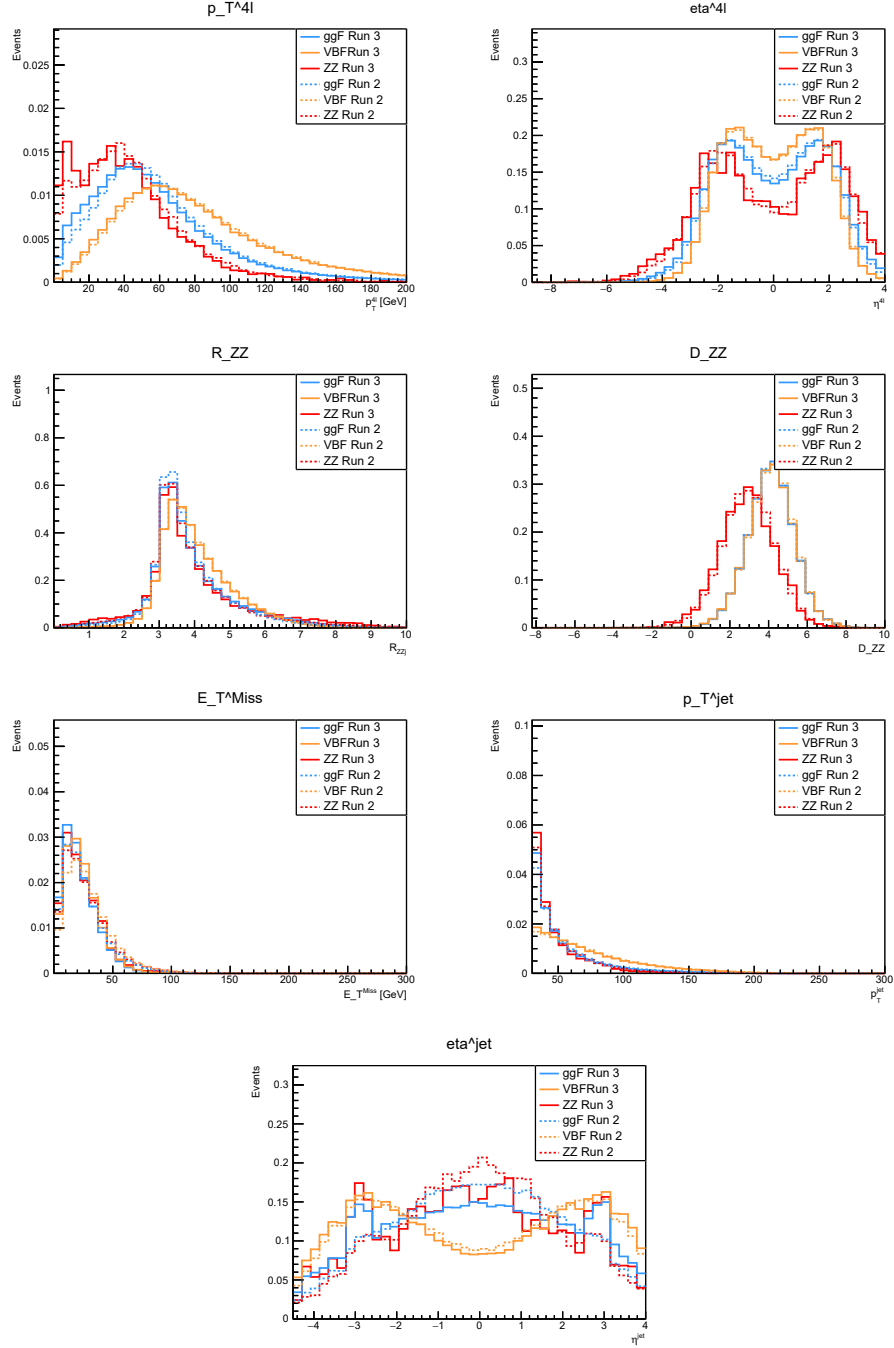


Figure A.4.: Global Input Features in the 1 – Jet case for the Run 3 samples (solid) and Run 2 samples (dashed).

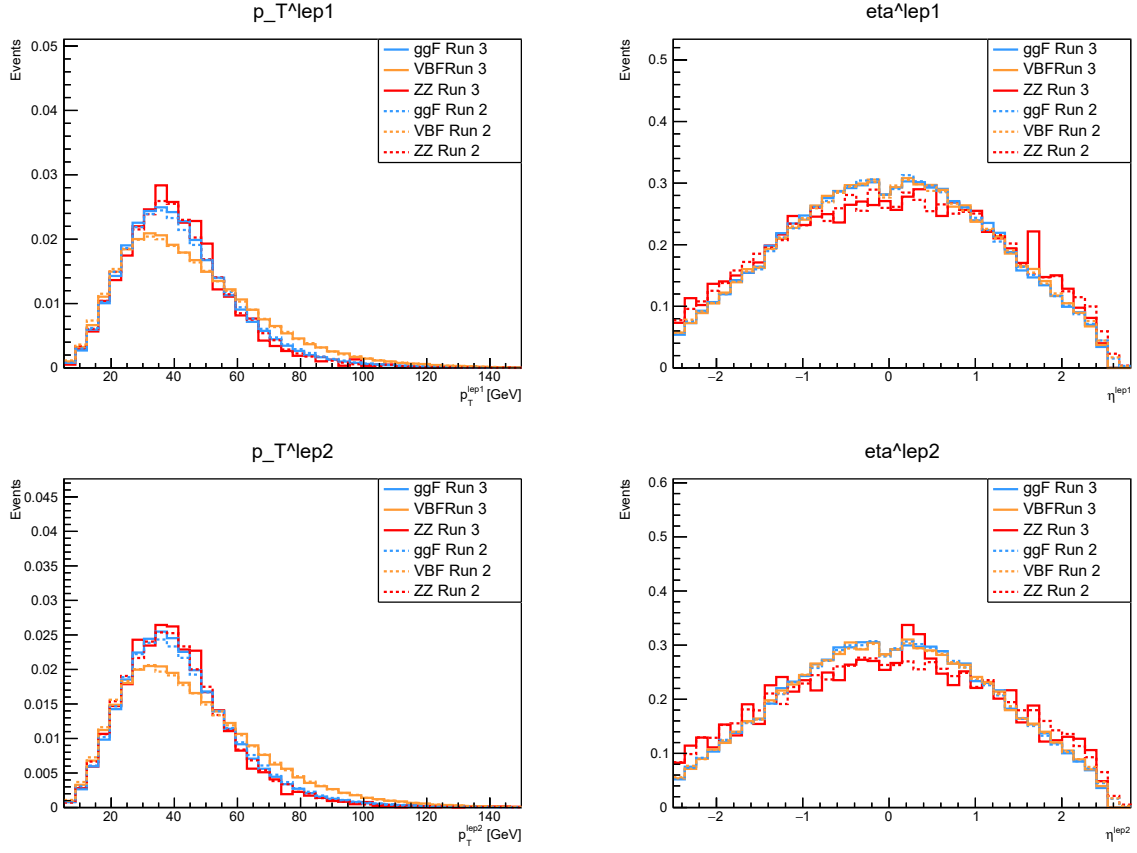


Figure A.5.: Lepton Input Features of the leading lepton pair in the 1 - Jet case for the Run 3 samples (solid) and Run 2 samples (dashed). Leptons are shown in order of descending p_T

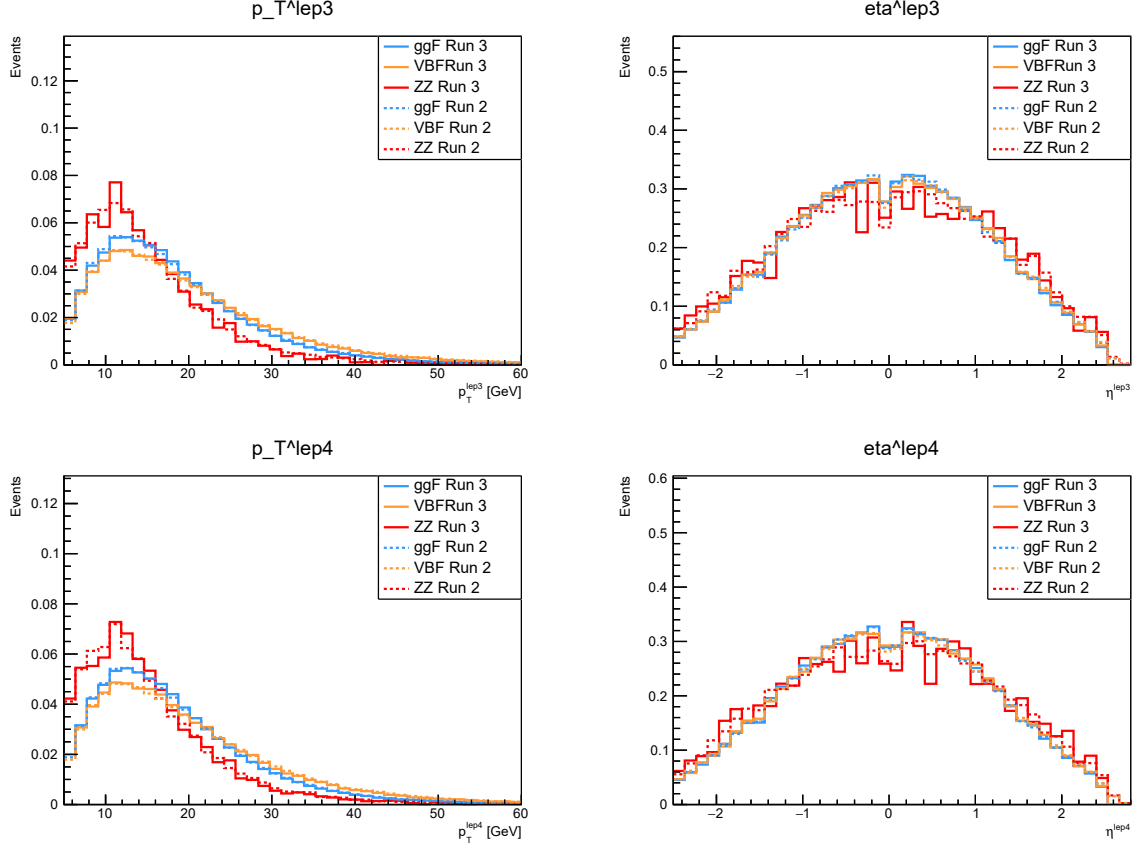
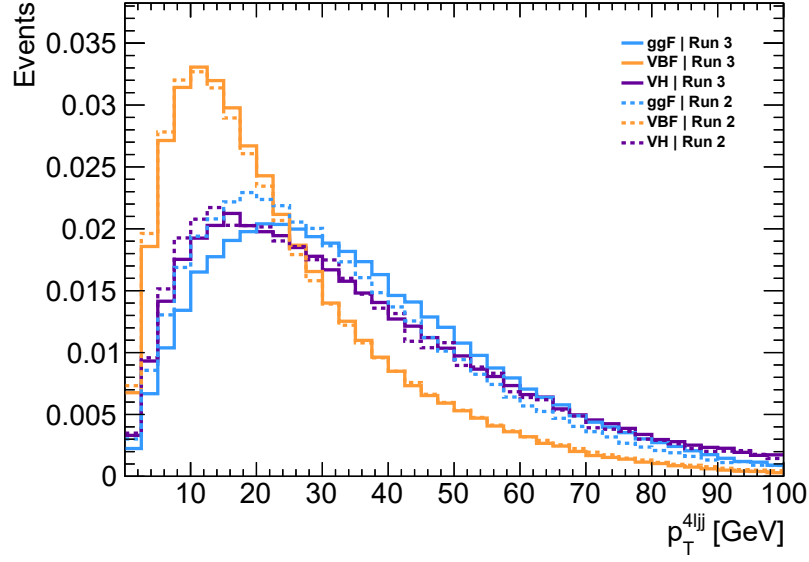


Figure A.6.: Lepton Input Features of the sub-leading lepton pair in the 1 - Jet case for the Run 3 samples (solid) and Run 2 samples (dashed). Leptons are shown in order of descending p_T

A.3. Kinematic Distributions (2 – Jet low- p_T)



(a)

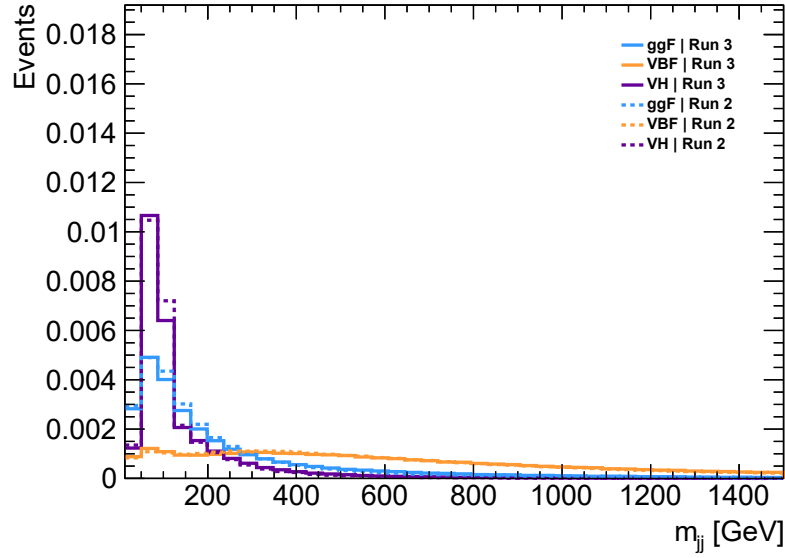


Figure A.7.: Global Input Features in the 2 – Jet low- p_T case for the Run 3 samples (solid) and Run 2 samples (dashed).

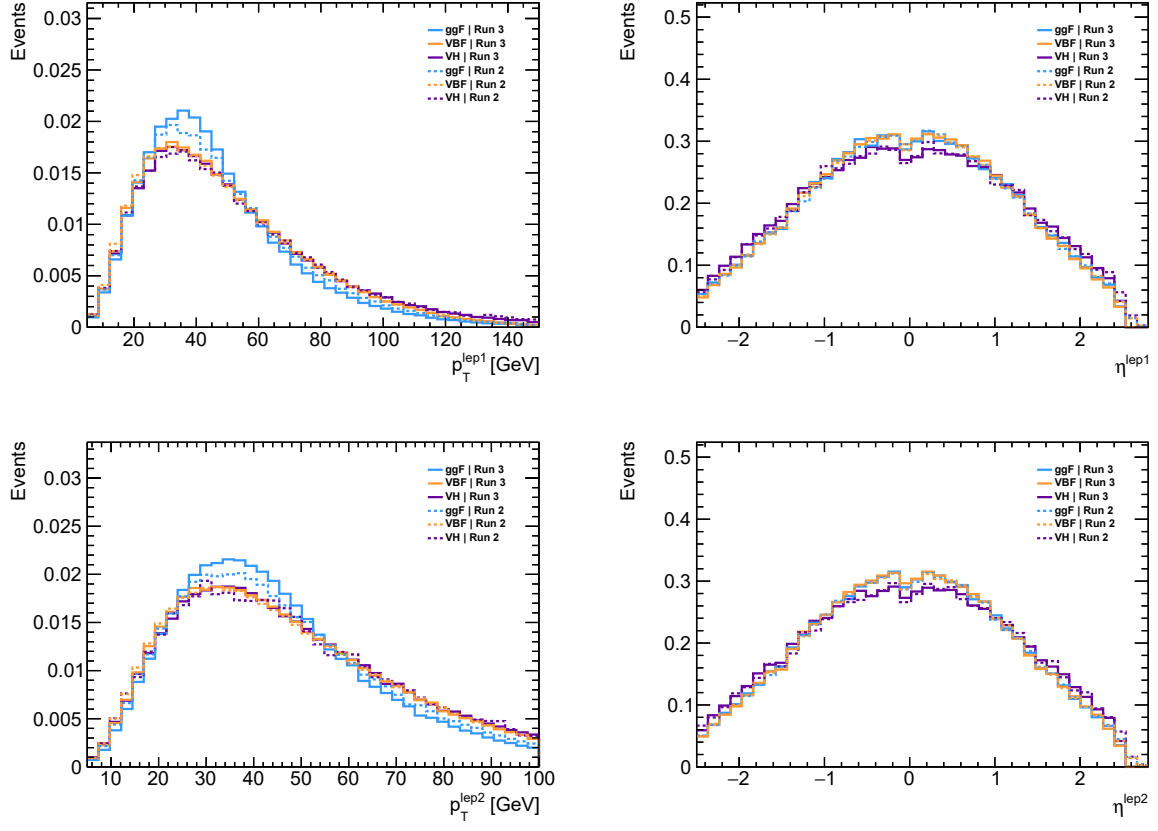


Figure A.8.: Lepton Input Features of the leading lepton pair in the 2 – Jet low- p_T case for the Run 3 samples (solid) and Run 2 samples (dashed). Leptons are shown in order of descending p_T

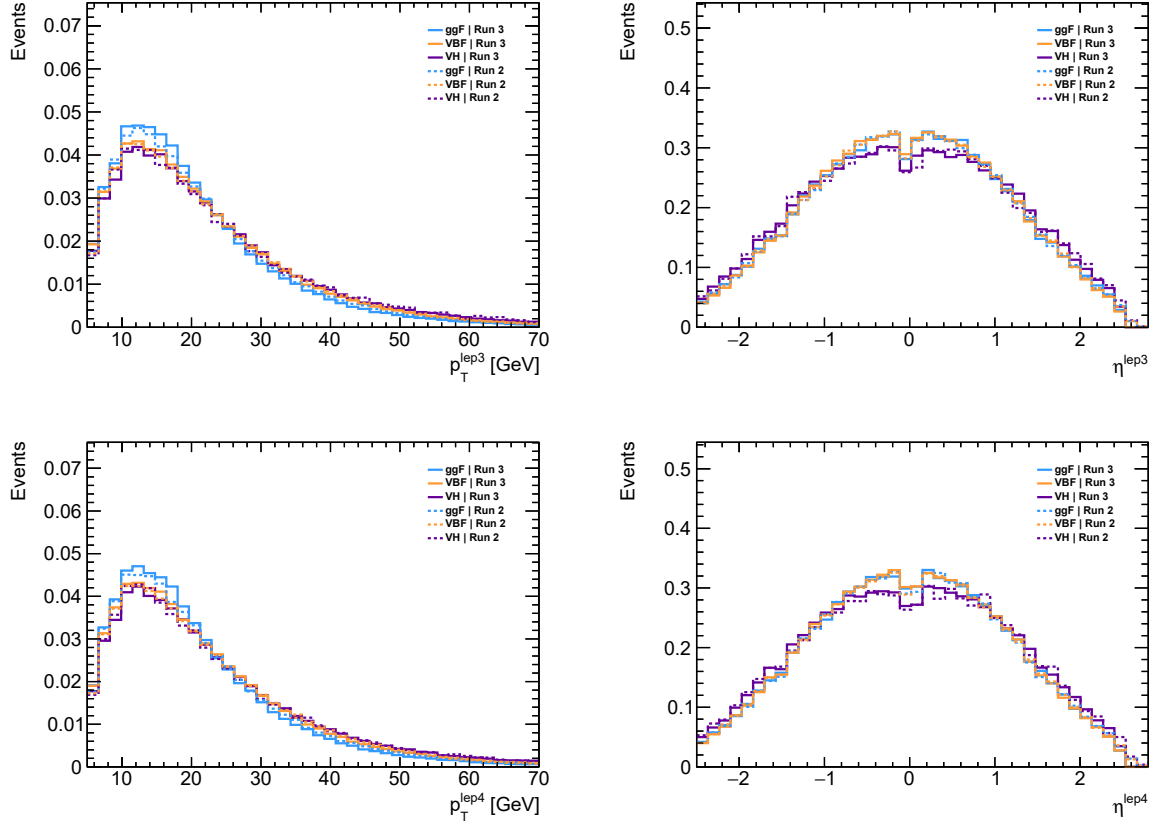


Figure A.9.: Lepton Input Features of the sub-leading lepton pair in the 2 – Jet low- p_T case for the Run 3 samples (solid) and Run 2 samples (dashed). Leptons are shown in order of descending p_T

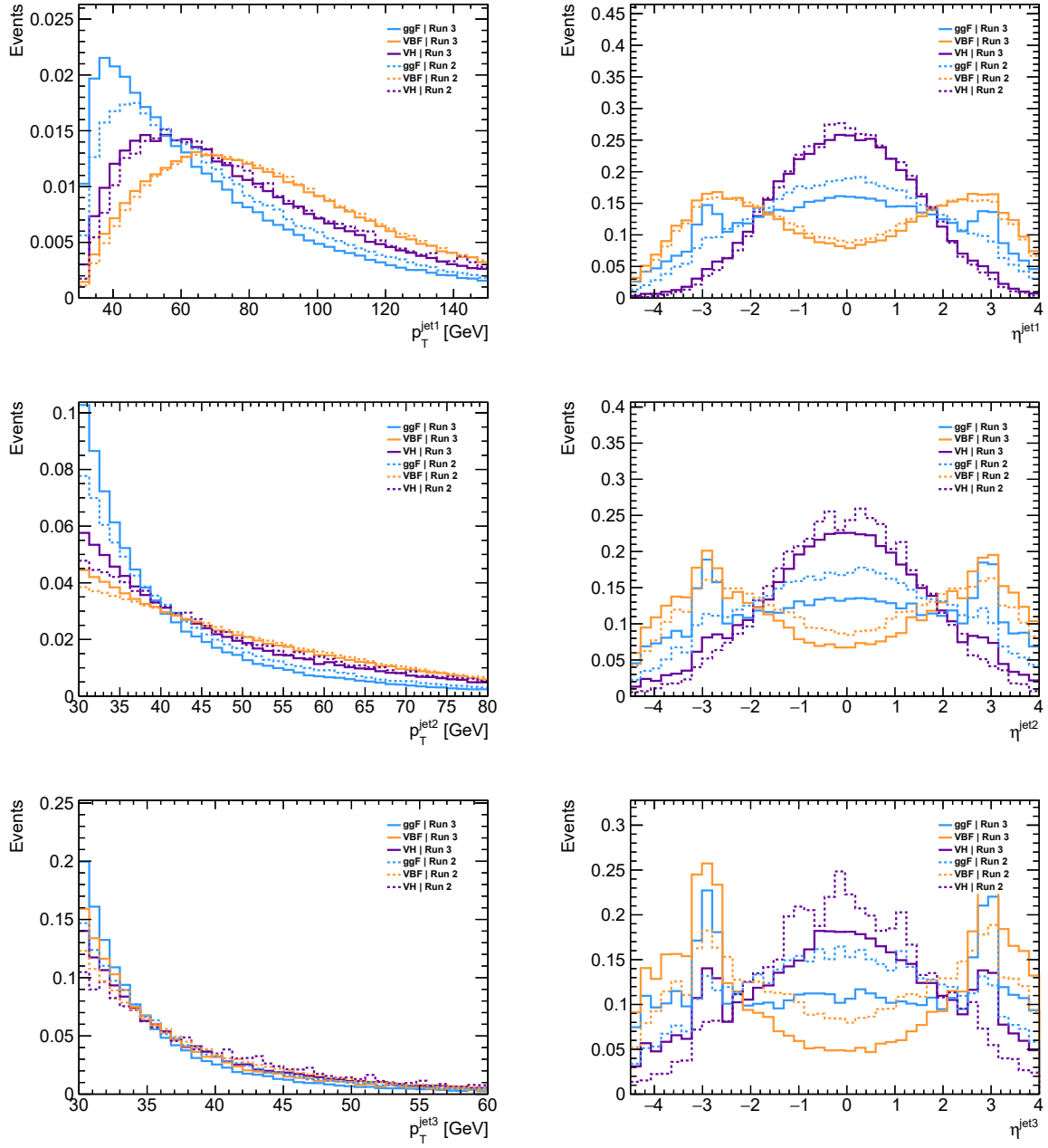
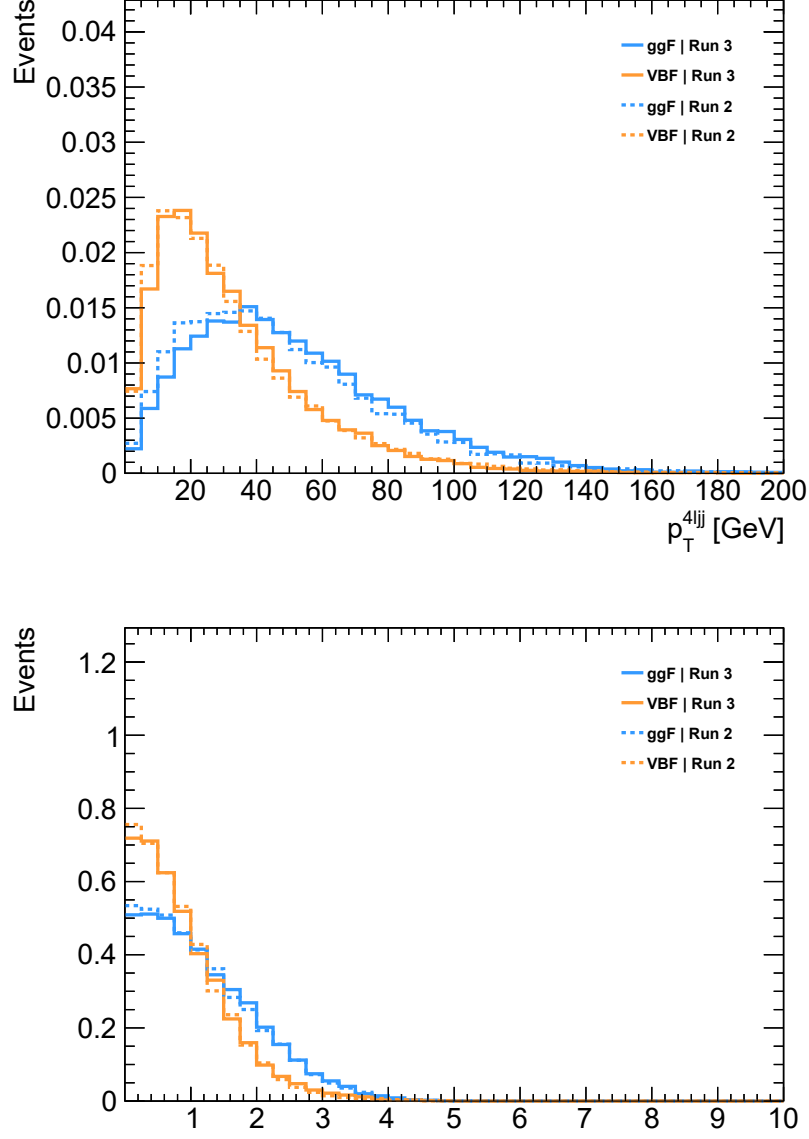


Figure A.10.: Jet Input Features in the 2 – Jet low- p_T case for the Run 3 samples (solid) and Run 2 samples (dashed). p_T and η are shown for all jets in the final state in order of descending p_T

A.4. Kinematic Distributions (2 – Jet high- p_T)



(b)

Figure A.11.: Global Input Features in the 2 – Jet high- p_T case for the Run 3 samples (solid) and Run 2 samples (dashed).

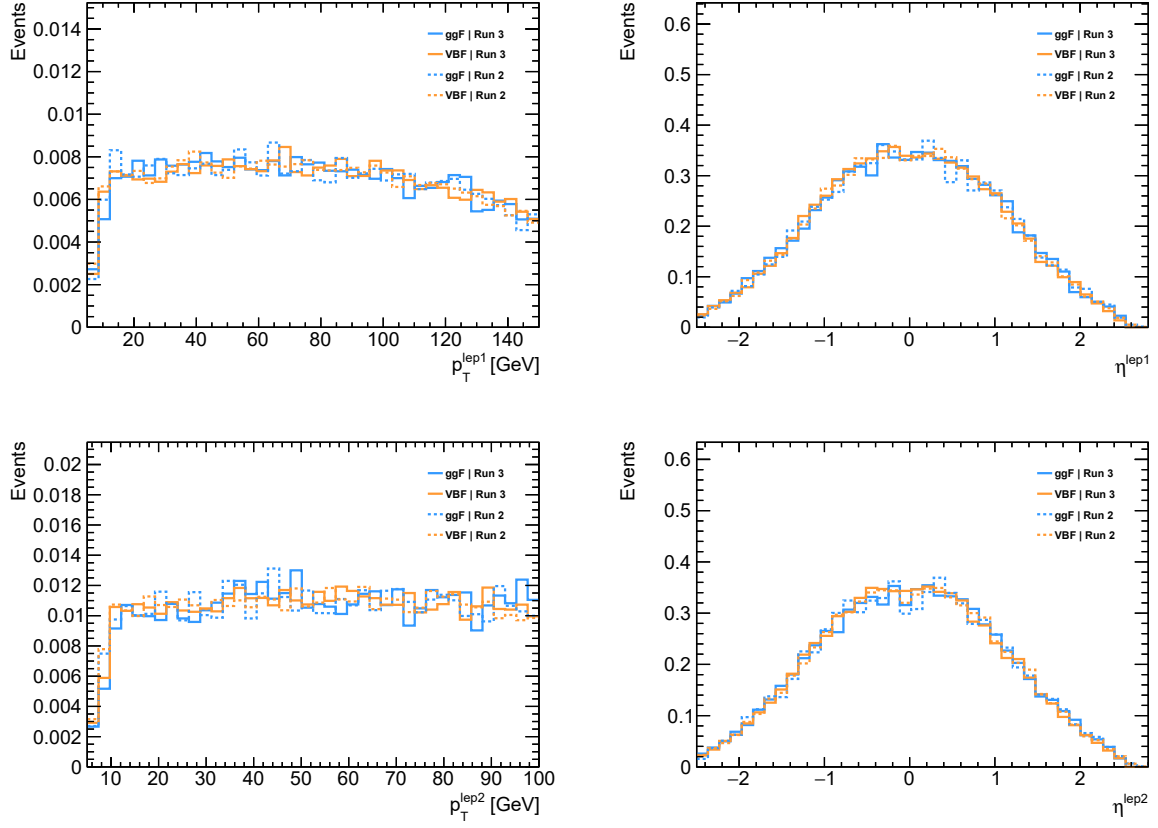


Figure A.12.: Lepton Input Features of the leading lepton pair in the 2-Jet high- p_T case for the Run 3 samples (solid) and Run 2 samples (dashed). Leptons are shown in order of descending p_T

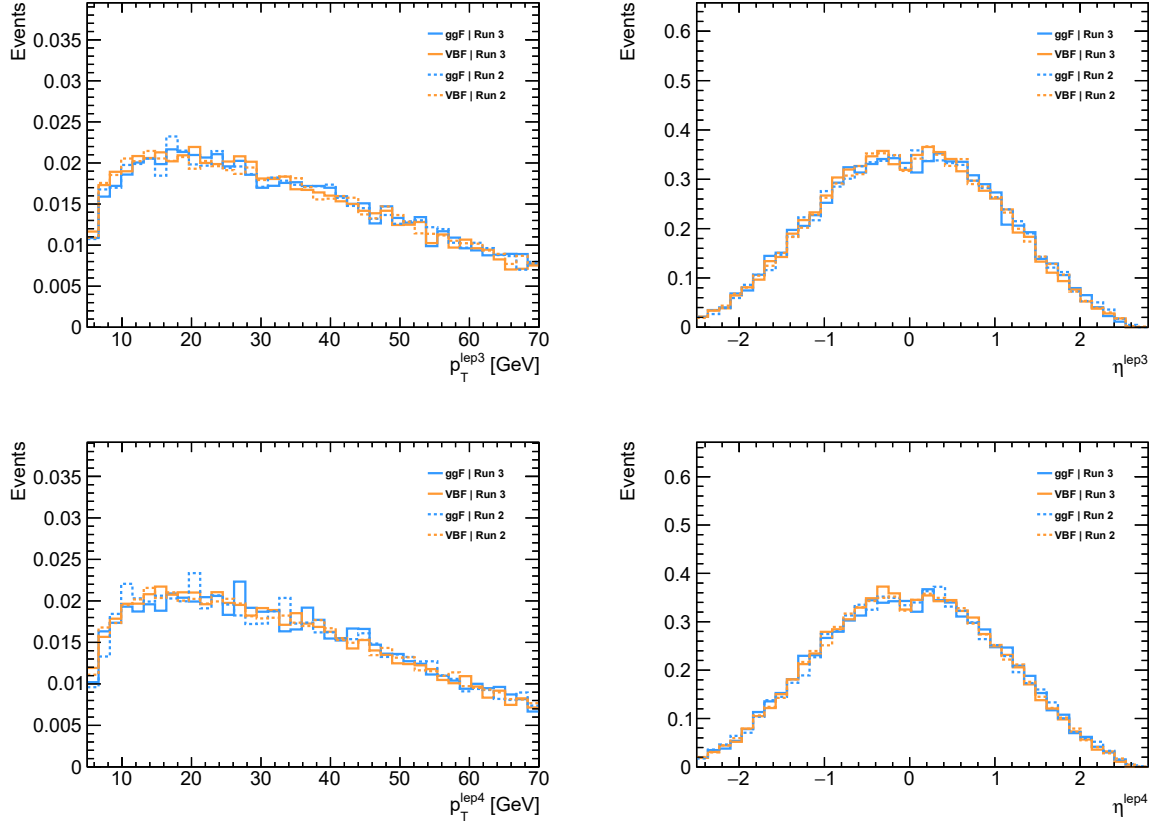


Figure A.13.: Lepton Input Features of the sub-leading lepton pair in the 2 – Jet high- p_T case for the Run 3 samples (solid) and Run 2 samples (dashed). Leptons are shown in order of descending p_T

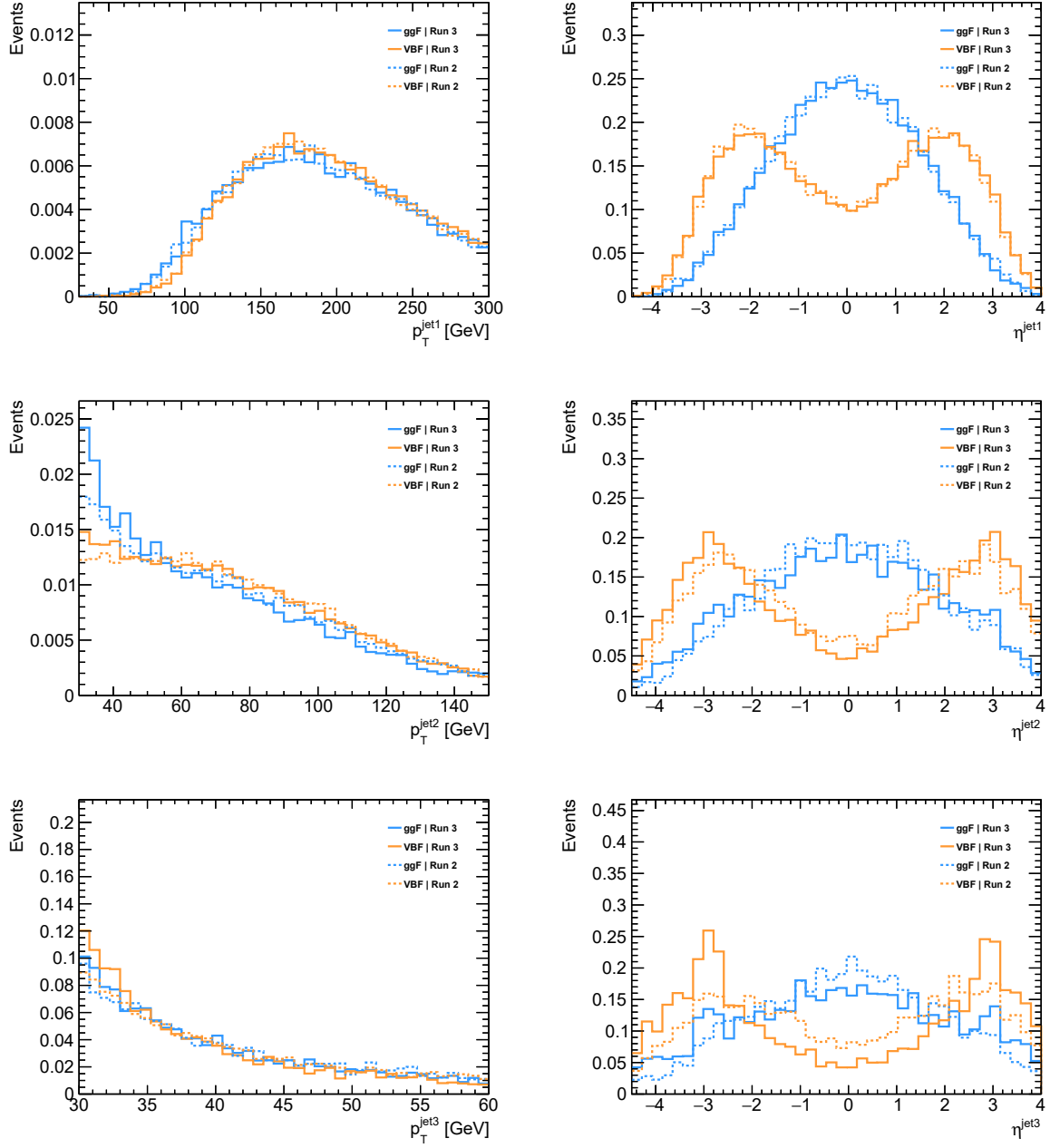


Figure A.14.: Jet Input Features in the 2 – Jet high- p_T case for the Run 3 samples (solid) and Run 2 samples (dashed). p_T and η are shown for all jets in the final stat in order of descending p_T

B. Specifications of Neural Network Training Setups

B.1. Hyperparameters of the Deep Set Neural Networks

Table B.1.: Summary of the Neural Network hyperparameters for all sub-Neural Networks of the combined $0 - Jet$ event category and the $0 - Jet\ 4l$ and $0 - Jet\ 2e2\mu$ event categories.

Sub-Neural Network	Input nodes	Output nodes		Nodes per hidden layer	Hidden Layer activation function	Output Layer activation function
Lepton Deep Set						
Pre-Processing NN	3	20	2	15	ReLU	ReLU
Post-Pooling NN	20	10	2	15	ReLU	ReLU
Global NN						
Dense NN	7	20	2	15	Tanh	Tanh
Final NN						
Dense NN	30	1	2	20	Tanh	Sigmoid

Table B.2.: Summary of the Neural Network hyperparameters for all sub-Neural Networks of the $1 - Jet$ low- p_T^{4l} event category.

Sub-Neural Network	Input nodes	Output nodes		Nodes per hidden layer	Hidden Layer activation function	Output Layer activation function
Lepton Deep Set						
Pre-Processing NN	3	20	2	15	ReLU	ReLU
Post-Pooling NN	20	10	2	15	ReLU	ReLU
Global NN						
Dense NN	5	20	2	15	Tanh	Tanh
Final NN						
Dense NN	30	3	3	20	Tanh	Softmax

Table B.3.: Summary of the Neural Network hyperparameters for all sub-Neural Networks of the 1 – *Jet* medium- p_T^{4l} event category.

Sub-Neural Network	Input nodes	Output nodes		Nodes per hidden layer	Hidden Layer activation function	Output Layer activation function
Lepton Deep Set						
Pre-Processing NN	3	20	2	15	ReLU	ReLU
Post-Pooling NN	20	10	2	15	ReLU	ReLU
Global NN						
Dense NN	7	20	2	15	Tanh	Tanh
Final NN						
Dense NN	30	3	3	20	Tanh	Softmax

Table B.4.: Summary of the Neural Network hyperparameters for all sub-Neural Networks of the 2 – *Jet* low- p_T^{4l} event category in the binary classification case (*VH* vs. *VBF* + *ggF*).

Sub-Neural Network	Input nodes	Output nodes		Nodes per hidden layer	Hidden Layer activation function	Output Layer activation function
Lepton Deep Set						
Pre-Processing NN	3	20	2	15	ReLU	ReLU
Post-Pooling NN	20	10	2	15	ReLU	ReLU
Jet Deep Set						
Pre-Processing NN	2	20	2	15	ReLU	ReLU
Post-Pooling NN	20	10	2	15	ReLU	ReLU
Global NN						
Dense NN	2	20	2	15	Tanh	Tanh
Final NN						
Dense NN	50	1	3	20	Tanh	Sigmoid

Table B.5.: Summary of the Neural Network hyperparameters for all sub-Neural Networks of the 2 – Jet low- p_T^{4l} event category in the multi-class classification case (VH vs. VBF vs. ggF).

Sub-Neural Network	Input nodes	Output nodes		Nodes per hidden layer	Hidden Layer activation function	Output Layer activation function
Lepton Deep Set						
Pre-Processing NN	3	20	2	15	ReLU	ReLU
Post-Pooling NN	20	10	2	15	ReLU	ReLU
Jet Deep Set						
Pre-Processing NN	2	20	2	15	ReLU	ReLU
Post-Pooling NN	20	10	2	15	ReLU	ReLU
Global NN						
Dense NN	2	20	2	15	Tanh	Tanh
Final NN						
Dense NN	50	3	3	20	Tanh	Softmax

Table B.6.: Summary of the Neural Network hyperparameters for all sub-Neural Networks of the 2 – Jet high- p_T^{4l} event category.

Sub-Neural Network	Input nodes	Output nodes		Nodes per hidden layer	Hidden Layer activation function	Output Layer activation function
Lepton Deep Set						
Pre-Processing NN	3	20	2	15	ReLU	ReLU
Post-Pooling NN	20	10	2	15	ReLU	ReLU
Jet Deep Set						
Pre-Processing NN	2	20	2	15	ReLU	ReLU
Post-Pooling NN	20	10	2	15	ReLU	ReLU
Global NN						
Dense NN	2	20	2	15	Tanh	Tanh
Final NN						
Dense NN	50	1	3	20	Tanh	Sigmoid

Table B.7.: Summary of the hyperparameters used in the training setup used for the optimization of the Deep Set Neural Networks in each event category. T_{min} , T_{min} and T_{rest} refer to the minimum and maximum learning rates and epochs until the restart of the learning rate cycle in the Cosine Annealing Warm Restarts learning rate scheduler method (see section 7.2.2). The training is done in four consecutive phases over 200 epochs each. After each phase, the learning rate is doubled. The initial batch size refers to the batch size used in the first training phase.

Event category	T_{min}	T_{min}	T_{rest}	Initial batch size (number of events)
0 – Jet 0 – Jet 4l 0 – Jet 2e2μ	$5 \cdot 10^{-5}$	0	30	16
1 – Jet 1 – Jet low- p_T^{4l} 1 – Jet medium- p_T^{4l}	$5 \cdot 10^{-4}$	0	50	16
2 – Jet low- p_T^{4l}	$5 \cdot 10^{-4}$	0	30	16
2 – Jet high- p_T^{4l}	10^{-5}	0	50	4

B.2. Hyperparameters of the Deep Set Training Setups

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